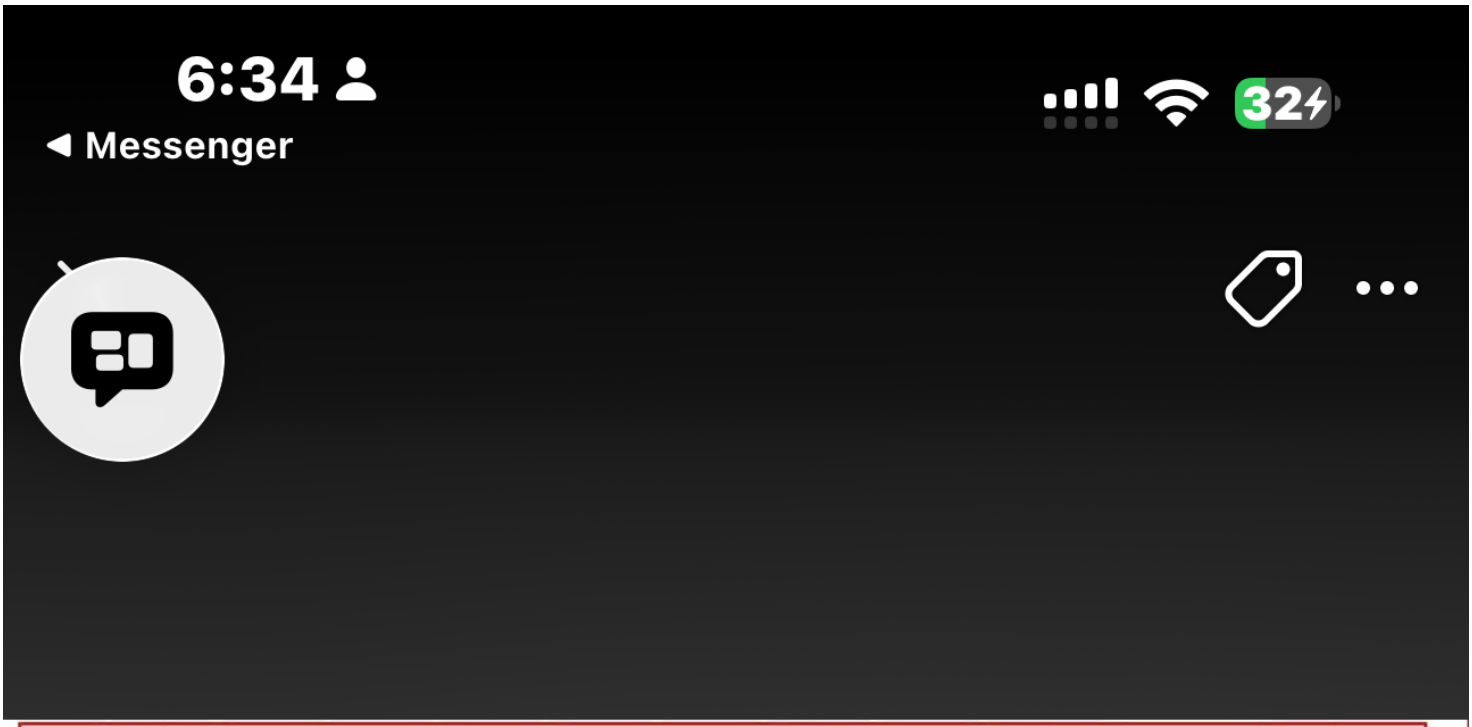
FORENSIC RESEARCH DOSSIER - Patented Technologies, Human Perception, Subliminal Influence, Neuromodulation, Radio-Frequency Auditory Effects, Military Research and the Scientific Evidence for Behavioural Influence

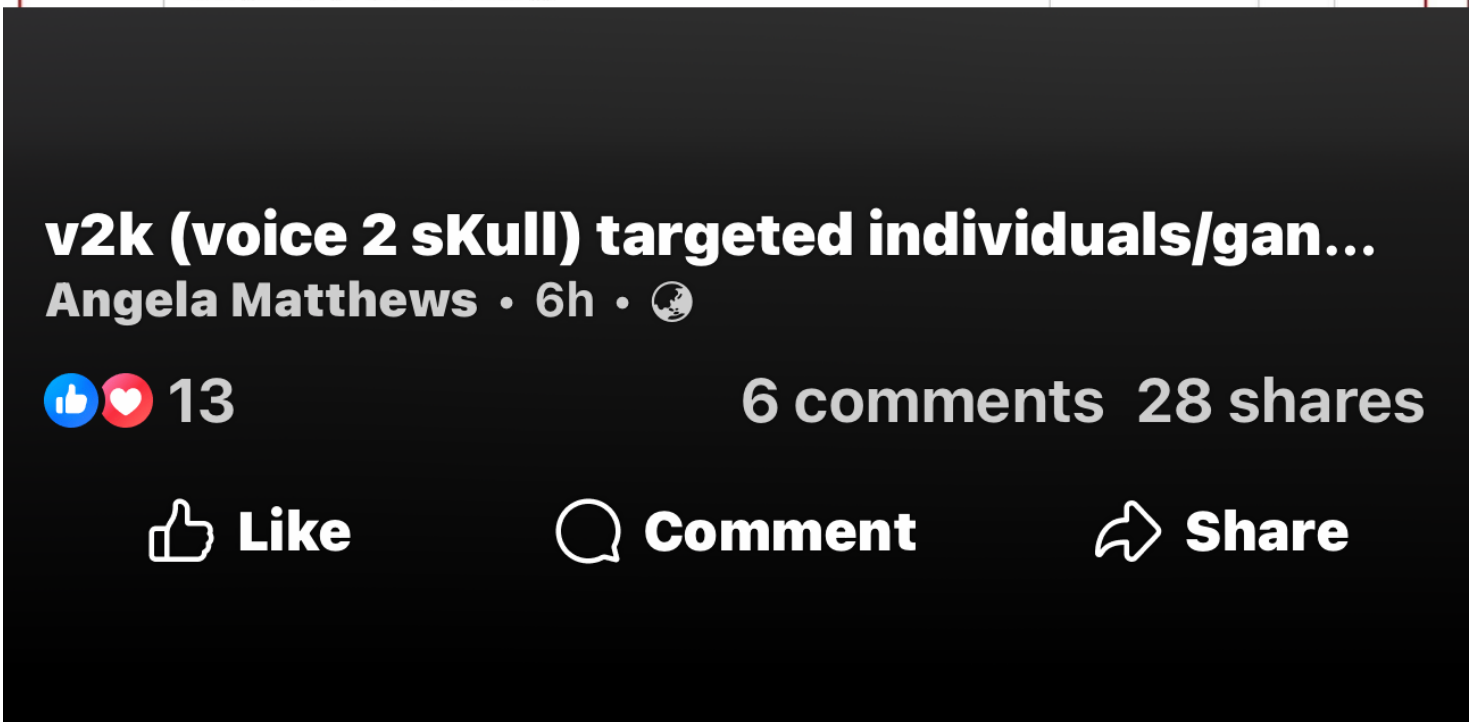
A staged, source-linked documentary investigation

Created and ordained with intentional commands by Barran Dodger



US Patent#	Description	Inventor	Filed	Issued
US3014477	Hypnotic Inducer (Mind Control Machine)	Carlin	1956	**1961**
US3060795	Apparatus For Producing Visual Stimulation (Subconscious Transmission via Movie Film)	Corrigan et al	1958	**1962**
US3278676	Apparatus For Producing Visual and Auditory Stimulation (Subconscious Transmission via TV)	Becker	1958	**1966**
US3393279	Nervous System Excitation Device	Flanagan	1962	**1968**
US3563246	Method and Apparatus For Improving Neural Performance In Human Subjects By Electrotherapy	Puharich	1967	**1971**
US3629521	Hearing Systems (RF/Microwave)	Puharich	1970	**1971**
US3712292	Method Of and Apparatus For Producing Swept Frequency Modulated Audio Signal Patterns For Inducing Sleep (Brain Frequencies Broadcast)	Zentmayer, Jr.	1971	**1973**
US3884218	Method of Inducing and Maintaining Stages Of Sleep in the Human Being (FFR - Frequency Following Response Microwave)	Monroe	1970	**1975**
US3951134	Apparatus and Method For Remotely Monitoring and Altering Brain Waves (RF/Electromagnetic Waves)	Malech	1974	**1976**
US4395600	Auditory Subliminal Message System and Method (Subliminal Brainwash via Music or Other Sound)	Lundy	1980	**1983**
US4717343	Method Of Changing A Person's Behavior (Subconscious Brainwash via Video)	Densky	1986	**1988**
US4777529	Auditory Subliminal Programming System (Silent Brainwash Via Music or Other Sound)	Schultz/Dolejs	1987	**1988**
US4834701	Apparatus For Inducing Frequency Reduction In Brain Wave (FFR - Brain Frequencies Transmission)	Masaki	1985	**1989**
US4858612	Hearing Device (Microwave Hearing)	Stocklin	1983	**1989**
US4877027	Hearing System (Microwave Hearing via open air Broadcast)	Brunkan	1988	**1989**
US5159703	Silent Subliminal Presentation System (aka Silent Sound - Microwave)	Lowery	1989	**1992**
US5356368	Method Of and Apparatus For Inducing Desired States Of Consciousness (FFR/EEG Waveforms By Broadcast)	Monroe	1991	**1994**

US5774088	Method and System For Warning Birds Of Hazards (Microwave Hearing)	Kreithen	1997	**1998**
US5889870	Acoustic Heterodyne Device and Method (Ultrasound. Ventriloquist Effect.)	Norris	1996	**1999**
US6011991	Communication System and Method Including Brain Wave Analysis and/or Use of Brain Activity (Remote Viewing)	Mardirossian	1998	**2000**
US6052336	Apparatus and Method Of Broadcasting Audible Sound Using Ultrasonic Sound As A Carrier (Ultrasound)	Lowery	1998	**2000**
US6470214	Method and device For Implementing The Radio Frequency Hearing Effect (Microwave Hearing)	O'Loughlin/Loree	1992	**2002**
US6587729	Apparatus For Audibly Communicating Speech Using The Radio Frequency Hearing Effect (Microwave Hearing)	O'Loughlin/Loree	2002	**2003**



Prepared for: Dr Richard William McLean — Barran Dodger Purpose: Historical preservation, scientific examination, patent analysis, institutional research and public-interest documentation.

Method: Primary-source patent records, peer-reviewed scientific research, military and government publications, recognised psychological and psychiatric literature, and clearly distinguished assessments of evidence.

Scope: From early hypnosis and subliminal stimulation through audiovisual influence, sleep research, electrotherapy, brain-wave measurement, radio-frequency auditory phenomena and later developments in neurotechnology.

PART I — FORENSIC FRAMEWORK, HISTORICAL SIGNIFICANCE AND THE PRIMARY-SOURCE RECORD

1. Why this investigation matters

The history of attempts to influence human perception and behaviour sits at the intersection of psychology, psychiatry, physiology, communications engineering, military research, intelligence history, medical ethics and the law.

For more than a century, researchers have investigated whether external stimuli can influence attention, perception, emotional response, sleep, memory, decision-making or other aspects of human functioning. Some techniques have established applications in medicine, psychology or communications. Others remain experimental, have limited evidence of effectiveness, or have attracted claims substantially beyond what the available science demonstrates.

The documentary record includes patents for devices intended to induce hypnosis, present subliminal audiovisual material, alter patterns of neural stimulation, influence sleep, investigate electromagnetic auditory effects and propose remote measurement of physiological signals.

These documents are historically significant because they allow researchers to examine what named inventors proposed, what technical methods they described, what applications they claimed, and what questions their work raises about the relationship between technology and human autonomy.

A forensic examination must go further than compiling provocative titles. It must reconstruct the history of each invention, distinguish proposed mechanisms from demonstrated results, identify government or commercial associations where documented, and establish whether independent evidence supports the claimed capabilities.

The objective is not to dismiss unusual technologies simply because they sound extraordinary. Nor is it to accept a patent, a military connection or a suggestive title as proof of every allegation associated with it. The objective is to build the strongest possible documentary record by following the evidence wherever it leads.

2. The five levels of evidence

Every document examined in this dossier should be classified according to what it can actually establish.

1

Primary documentary evidence

The patent specification, patent claims, original government report, laboratory paper, official correspondence, contract, procurement record or other identifiable source. It establishes what the document says and, subject to authentication, when and by whom it was produced.

2

Demonstrated experimental findings

Results obtained under defined conditions, with identifiable methods, measurements and limitations. Replication and independent evaluation strengthen confidence in the findings.

3

Institutional involvement

Evidence that a military service, government department, university, corporation or medical institution funded, conducted, owned or commissioned particular work. Each relationship must be established from the relevant record.

4

Operational deployment

Evidence that a technology was actually built, tested in the relevant environment, acquired, deployed or used against a particular target or population. A patent alone does not establish this level.

5

Causal attribution and individual impact

Evidence connecting a particular technology, intervention or institution to a specific outcome. This requires more than demonstrating that a technology exists or that an institution researched a related subject.

These levels are not interchangeable. A source can establish one level without establishing the next.

For example, an authentic patent may conclusively demonstrate that a named inventor disclosed a proposed method. A separate experiment may establish that one physical effect is possible. Neither fact alone proves that the patented apparatus was operationally successful or that it was used against a particular person.

This framework protects the integrity of the historical record. It also makes the research more useful because it identifies precisely what is established, what remains unresolved, and which records could answer the outstanding questions.

3. Primary-source patent examination: five foundational records

The following records deserve particular attention because their own specifications use language concerning hypnosis, subconscious stimulation, remote monitoring or the transmission of information.

3.1. US3014477A — Hypnotic Inducer

Inventor: Robert L. Carlin | Filed: 16 August 1956 | Granted: 26 December 1961

Primary source:

The patent describes a physical apparatus intended to facilitate hypnosis. Its specification discusses therapeutic applications, recorded suggestions and the possibility of using hypnosis to influence subjective experiences such as anxiety, pain and feelings of well-being.

The record is historically relevant because it shows an inventor attempting to turn the induction of hypnosis into a repeatable apparatus-assisted process.

The scientific context is provided by the , which defines hypnosis in terms of suggestion and changes in sensation, perception, cognition, emotion or motor behaviour. The APA also acknowledges that the nature of hypnotic suggestibility and its therapeutic applications remain subjects of scientific study.

Further reading:

Forensic assessment: The patent supports the historical claim that apparatus-assisted hypnosis was being developed and described in the 1950s. It does not establish that hypnosis produces involuntary obedience, removes personal agency or permits unrestricted control over another person.

3.2. US3278676A — Apparatus for Producing Visual and Auditory Stimulation

Inventor: Hal C. Becker | Original assignee: Precon Process and Equipment Corporation | Granted: 11 October 1966

Primary source:

This is one of the most revealing documents in the collection.

The specification explicitly discusses visual and auditory stimuli presented below the level at which an observer can verbally report the stimulus. It proposes methods for conveying information through audiovisual media, including television and motion pictures, with the intended result of influencing selective perception and subsequent responses.

The patent's description also identifies potential applications in medicine, psychiatry, psychology, education, advertising, propaganda and psychological warfare.

Those statements are historically important. They document the inventor's stated ambitions and the range of applications contemplated in the specification. They are not independent demonstrations that each application was successful.

The patent history also requires precision: the record identifies a priority date in 1958, but the application associated with this patent was filed on 29 October 1962 and granted in 1966. The earlier related patent, , was granted in 1962.

The scientific comparison is the National Research Council's review of subliminal self-help claims in In the Mind's Eye: Enhancing Human Performance (1991). That report distinguished limited effects on simple laboratory tasks from claims of lasting changes to complex behaviour, emotions or cognition.

Forensic assessment: This patent is strong documentary evidence that subliminal audiovisual influence and possible psychological-warfare applications were explicitly contemplated in a patent specification. It is not proof that the proposed method reliably achieved complex behavioural conditioning, or that it was deployed in a particular military or civilian operation.

3.3. US3951134A — Apparatus and Method for Remotely Monitoring and Altering Brain Waves

Inventor: Robert G. Malech | Original assignee: Dorne and Margolin Inc. | Filed: 5 August

1974 | Granted: 20 April 1976

Primary source:

This patent describes a proposed system in which electromagnetic signals of different frequencies are transmitted towards a subject. The inventor proposes that interactions between the signals and the subject's brain activity would produce a modulated signal that could be received, processed and analysed remotely.

The specification further proposes generating a compensating signal intended to affect electrical activity in the brain.

The distinction between the invention's description and established scientific capability is especially important here.

The patent establishes that a technical proposal was documented. To establish that the method works as claimed, investigators would need independent experimental evidence demonstrating that the returned signal actually contains measurable information attributable to brain activity, that alternative explanations have been excluded, and that the feedback process produces the claimed physiological effects.

Modern neuroscience uses a variety of instruments to measure brain activity, including electroencephalography (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI) and implanted recording systems. These technologies have different physical mechanisms, spatial resolutions, limitations and practical requirements.

A useful modern comparison is the . It describes recognised and experimental interventions, including electroconvulsive therapy, repetitive transcranial magnetic stimulation, vagus nerve stimulation and deep brain stimulation.

Forensic assessment: Malech's patent is a primary-source record of a proposed remote sensing and feedback technique. It should be investigated on its own technical merits, rather than treated as proof that remote decoding of private thoughts or reliable remote behavioural control has been achieved.

3.4. US5159703A — Silent Subliminal Presentation System

Inventor: Oliver M. Lowery | Filed: 28 December 1989 | Granted: 27 October 1992

Primary source:

The abstract describes a communications system using carriers in very low or very high audio-frequency ranges, including adjacent ultrasonic frequencies, modulated with information and propagated acoustically or through vibration.

The specification contemplates transmitting or recording signals for subsequent presentation through loudspeakers, earphones or piezoelectric transducers.

Its stated purpose is to present information intended to be perceived by a listener's brain, despite the carrier not being heard as ordinary audible sound.

This document is significant because it describes a particular signal-processing and delivery proposal rather than merely making a general statement about the power of suggestion.

Nevertheless, several questions must be kept separate:

- **Can the proposed signal be generated and transmitted?**
- **Can the relevant carrier or modulation produce a measurable physiological response?**
- **Can information be recovered by a human listener under the proposed conditions?**
- **Does that information measurably affect behaviour?**
- **Are any behavioural effects reliable, lasting and reproducible?**

An affirmative answer to one question does not establish affirmative answers to all the others.

The is an important counterweight to the patent's broad claims about subliminal learning. Its findings caution against extrapolating from limited laboratory effects to claims that recordings can reliably change complex behaviour or emotional states.

Forensic assessment: The patent is evidence of a proposed method for delivering information through modulated, non-ordinary-audible carriers. The degree to which that method produces the claimed perceptual or behavioural effects must be established by independent experiments.

3.5. US6470214B1 — Method and Device for Implementing the Radio Frequency Hearing Effect

Inventors: James P. O'Loughlin and Diana L. Loree | Original assignee: United States Department of the Air Force | Granted: 22 October 2002

Primary source:

This is one of the strongest institutional leads in the collection because the patent identifies the United States Department of the Air Force as the original assignee.

The specification concerns modulation and processing of radio-frequency signals to produce an encoded output from which speech may be perceived through the radio-frequency hearing effect. It describes carrier-suppressed modulation and signal-processing methods intended to improve the recovery of intelligible speech.

The patent's statement of government interest provides direct evidence of a government-associated invention. The assignment record and patent history establish the documented institutional relationship.

The underlying auditory phenomenon has a separate scientific history. Allan H. Frey's 1962 paper, Human Auditory System Response to Modulated Electromagnetic Energy, appeared in the Journal of Applied Physiology.

Frey reported auditory perceptions associated with modulated electromagnetic energy under experimental conditions. Later discussions of the microwave auditory effect have examined mechanisms involving rapid energy absorption, thermoelastic expansion and pressure waves reaching the auditory system.

The existence of the effect must not be confused with the much broader proposition that any radio-frequency transmission can deliver intelligible speech into any person's consciousness. The effect depends on signal and exposure conditions, and the evidence for a specific patent's performance must be assessed separately.

Forensic assessment: This patent provides a concrete basis for investigating government-associated work on the radio-frequency hearing effect. Its existence does not, by itself, establish covert deployment, targeting of a particular individual, or unrestricted remote mind control.

4. The scientific baseline: what subliminal influence research actually establishes

A central research item for this entire dossier is the National Research Council's 1991 report:

Daniel Druckman and Robert A. Bjork, editors — In the Mind's Eye: Enhancing Human Performance.

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Its chapter on subliminal self-help examined claims that recordings containing messages outside ordinary conscious awareness could improve memory, confidence, performance or emotional well-being.

The committee distinguished between two materially different propositions:

- **Some stimuli can influence performance on simple tasks under controlled laboratory conditions, even when participants do not report consciously noticing the stimuli.**
- **Such effects establish that recordings can reliably produce lasting changes in complex behaviour, beliefs, personality or emotional conditions.**

The report did not find adequate theoretical and experimental support for the second proposition in the subliminal self-help products it reviewed.

This distinction is foundational. Subliminal perception is not an all-or-nothing question. A person may be unable to report a stimulus while still showing a small, measurable effect under specific conditions. But the existence of such an effect does not establish that a hidden message can compel a person to act against their intentions, implant complex beliefs or reliably control long-term behaviour.

The appropriate forensic approach is therefore to investigate the precise stimulus, its intensity and duration, the method used to determine awareness, the outcome measured, the effect size, the experimental controls and the degree of independent replication.

Key research terms: subliminal perception; masked priming; subjective awareness; objective detectability; semantic priming; attentional processing; implicit cognition; expectancy effects; placebo effects; selective perception; behavioural conditioning; suggestibility.

Useful starting points:

These sources allow the historical patent claims to be compared with established psychological concepts and evidence without collapsing the two into a single conclusion.

5. Military psychological operations: an essential distinction

Military psychological operations are a documented part of modern military history.

Their existence is not speculative.

The United States Army has published an institutional history describing the development of psychological operations during the First World War, including organised propaganda campaigns and the creation of dedicated military units.

Primary institutional source:

That history establishes a documented tradition of using communication, messaging and psychological tactics in military operations. It does not establish that all military psychological operations involve subliminal signals, neurotechnology or direct manipulation of brain activity.

Forensic researchers should distinguish at least four categories:

- **Information operations and propaganda: communication intended to influence beliefs, perceptions or behaviour.**
- **Psychological operations: organised activities designed to influence the attitudes or behaviour of designated audiences in support of operational objectives.**
- **Human-performance research: studies of attention, fatigue, stress, cognition, sleep, decision-making and related physiological processes.**
- **Neuromodulation and signal-based technologies: interventions or proposed systems intended to interact directly with neural or sensory processes.**

These categories may overlap in particular projects, but the relationship must be demonstrated by evidence rather than assumed from similar terminology.

The relevant documentary questions include whether a particular project was funded by a military organisation, whether its purpose was operational or medical, whether human participants were involved, what consent procedures applied, whether the project produced published findings, and whether a specific technology was actually deployed.

6. Documented intelligence research and human-subject ethics

A serious examination of the history of psychological influence must also address documented intelligence research involving drugs, interrogation, behavioural modification and human experimentation.

One identifiable primary-source collection is the CIA's declassified material on MKULTRA.

Primary source:

This is a specific CIA record collection, not a general allegation. The page identifies the material as a declassified collection associated with MKULTRA and lists subjects including research proposals, neurological topics, animal experiments and human experiments.

It is important to inspect the underlying documents rather than infer the contents of every subproject from the MKULTRA label. The programme encompassed different lines of work, and the purpose, methodology and evidentiary significance of each record must be established individually.

The historical record raises major questions about informed consent, deception, medical ethics, accountability and the treatment of human participants.

For a forensic investigation, the most useful follow-up records are original memoranda, funding records, laboratory reports, participant documentation, correspondence, oversight records and contemporaneous reviews.

The ethical issues are not contingent on proving that every experimental claim was successful. A documented experiment involving people can raise serious ethical questions even if its intended effects were not achieved.

At the same time, evidence that an intelligence agency conducted unethical research in one area does not establish that every later patent or unrelated technology was part of the same programme.

7. Brain stimulation in contemporary medicine: a necessary comparison

Modern medicine provides clear examples of technologies that can influence neural activity under defined conditions.

The describes several interventions used or investigated for particular mental-health conditions.

These include:

- **Electroconvulsive therapy (ECT): an established medical procedure involving electrical stimulation that induces seizure activity under clinical supervision.**
- **Repetitive transcranial magnetic stimulation (rTMS): a technique using magnetic fields to stimulate targeted brain regions.**
- **Vagus nerve stimulation (VNS): electrical stimulation of the vagus nerve for certain clinical indications.**

- **Deep brain stimulation (DBS): an intervention involving implanted electrodes, used for selected neurological conditions and investigated for other indications.**

These methods demonstrate that neural activity can be influenced by external electrical or magnetic stimulation, or by devices that deliver stimulation to particular anatomical structures.

They also illustrate why mechanism and delivery conditions matter. A medically established intervention with specified equipment, placement, dosage and monitoring is not equivalent to an unverified proposal for remote control using a general-purpose transmitter.

The correct comparison is not whether all such methods involve energy. It is whether the relevant physical mechanism, delivery pathway, target, measurable response and claimed outcome have been independently demonstrated.

8. Interim findings from Part I

The documents and research cited above support several conclusions.

First, patents exist that explicitly describe attempts to induce hypnosis, present subliminal audiovisual stimuli, propose remote brain-wave monitoring, develop silent subliminal presentation systems and implement the radio-frequency hearing effect.

Second, at least one patent in the collection, US6470214B1, identifies the United States Department of the Air Force as its original assignee. This is a verifiable institutional connection that merits further examination of the patent's technical claims and related records.

Third, subliminal perception, hypnosis, military psychological operations, brain stimulation and the radio-frequency auditory effect are distinct subjects. They require different scientific methods and should not be treated as interchangeable technologies.

Fourth, historical documentation of a proposed technique is not the same as independent verification of its effectiveness. The strongest forensic account identifies both the documented proposal and the evidence needed to evaluate its performance.

Fifth, the ethical implications of psychological and neurological research deserve examination in their own right. Questions of consent, privacy, autonomy, safety, transparency and accountability remain relevant regardless of whether a particular experimental claim is ultimately substantiated.

These findings provide a defensible starting point for the next stage: a patent-by-patent examination of the wider collection, followed by a chronological review of military research, peer-reviewed neuroscience, psychological and psychiatric literature, radio-frequency auditory research, and the institutional records that could establish whether particular inventions progressed from patent documents to practical use.

End of Part I.

The next part should examine the full patent chronology, beginning with the early visual and auditory stimulation patents and continuing through the 1970s remote-monitoring proposal, the 1980s subliminal audio systems, the later radio-frequency hearing patents and the related patent families. Each record should be linked to its original specification and assessed against independently published evidence.

FORENSIC RESEARCH DOSSIER
Part II — Patent-by-Patent Examination and the Historical Development of Technologies Intended to Influence Human Perception and Neural Activity
Continuation of Part I.
This section examines the patents in the original list individually, with particular attention to original patent specifications, inventors, filing and grant dates, proposed mechanisms, stated applications, scientific context and evidentiary limitations.
The governing question is not simply whether a patent exists. It is what the document actually establishes, how its technical proposal relates to independently documented science, and what further records would be needed to establish practical implementation or operational use.
A patent's official title, abstract, detailed description, claims, drawings, priority history, assignments and cited prior art are different sources of information within the same record. They should be examined separately rather than replaced with a summary label such as "mind control," "brainwashing" or "microwave hearing."
The original list includes several patents whose wording makes them particularly relevant to this inquiry. The analysis below preserves that significance while distinguishing the inventors' proposals from independently established results.
9. The 1950s and 1960s: Hypnosis, Subliminal Audiovisual Stimulation and Nervous-System Excitation
9.1. US3014477A — Hypnotic Inducer
Inventor: Robert L. Carlin Filed: 16 August 1956 Granted: 26 December 1961
Original document:
Research terms: hypnosis; hypnotic suggestion; audiovisual stimulation; suggestibility; altered states of consciousness; apparatus-assisted hypnosis.
The patent documents an attempt to use an apparatus to facilitate hypnosis and the

presentation of suggestions. It belongs to the historical development of techniques intended to make the induction of a psychological state more systematic and reproducible.
The wider scientific question is how attention, expectation, suggestion and individual differences in hypnotic responsiveness affect subjective experience and behaviour. Hypnosis is recognised within psychological research, but its effects must be evaluated according to the particular outcome being investigated. Evidence that a person can experience changes in pain perception or subjective experience under hypnosis does not demonstrate that a device can compel arbitrary behaviour or override personal agency.
Relevant scientific sources
Forensic finding: The patent is a primary historical source for apparatus-assisted hypnosis. Its technical claims should be read in full and compared with modern evidence about hypnotic suggestion, not interpreted as proof of unrestricted psychological control.

9.2. US3060795A — Apparatus for Producing Visual Stimulation

Original document:
This patent is relevant to the early development of apparatus for presenting visual material under conditions intended to limit ordinary conscious awareness.
It also matters because the number is easily confused with the similar-looking identifier in some secondary lists. The original record should be used to verify the official title, inventor details, filing history and relationship to subsequent patents.
Research terms: subliminal visual stimulation; visual masking; stimulus detection thresholds; motion-picture projection; selective attention.
The scientifically important distinction is between presenting a stimulus below a person's reportable detection threshold and demonstrating a reliable change in complex behaviour. These are separate empirical questions.
Modern research into masked visual stimuli uses controlled experiments to examine whether stimuli can influence reaction times, choices or other measurable responses. Results depend on stimulus strength, timing, task design, awareness measurements and replication.
Relevant research starting points
Forensic finding: The patent provides a historical starting point for investigating visual stimulation technology. Its existence does not establish the effectiveness of every proposed application or demonstrate that a particular device was used operationally.

9.3. US3278676A — Apparatus for Producing Visual and Auditory Stimulation

Inventor: Hal C. Becker Original assignee: Precon Process and Equipment Corporation
Priority: 1958 Granted: 11 October 1966
Original document:
This patent deserves a detailed reading because its specification connects audiovisual stimulation with the possibility of influencing responses without ordinary conscious recognition of the stimulus.
The document's importance lies partly in its stated range of contemplated applications. Its description refers to areas including psychology, psychiatry, advertising, propaganda and psychological warfare. Those references are evidence of what the specification

contemplated, not proof that all those applications were experimentally successful or adopted by the institutions mentioned.

The relevant forensic questions are:

- What stimuli did the inventor propose presenting?
- How did the apparatus generate and deliver those stimuli?
- What did the inventor claim about conscious awareness?
- What outcomes were asserted or anticipated?
- Were independent experimental results cited?
- Does the patent identify particular customers, contracts, trials or deployments?

The last question is especially important. A patent specification can discuss a possible military or commercial application without documenting a contract, field trial or operational programme.

Scientific comparison

The National Research Council's In the Mind's Eye reviewed claims about subliminal self-help recordings and the scientific evidence for behavioural improvement.

The modern research literature distinguishes relatively limited effects on experimental tasks from claims of reliable, lasting modification of complex beliefs or behaviour.

Forensic finding: This patent is important primary-source evidence of a proposed subliminal audiovisual apparatus and the psychological applications contemplated by its inventor. It should not be presented as conclusive evidence of operational psychological warfare.

9.4. US3393279A — Nervous System Excitation Device

Original document:

Research terms: nervous-system stimulation; electrical excitation; physiological response; stimulation thresholds; neuromodulation.

This patent belongs to a broader history of attempts to produce physiological responses through engineered stimulation.

Forensic interpretation requires the specification to be examined for its actual energy source, waveform, electrodes or transducers, intended anatomical target and measured endpoint. The title alone does not establish that the device could alter thoughts, control behaviour or stimulate the nervous system at a distance.

Electrical stimulation is a legitimate field of medical research, but the mechanism and effect depend on how energy is delivered, the intensity and duration of stimulation, the anatomical target and the physiological response.

Relevant scientific sources

Forensic finding: The original specification is necessary to establish precisely what kind of stimulation was proposed. The title does not establish remote operation, human experimentation or successful behavioural manipulation.

9.5. US3563246A — Method and Apparatus for Improving Neural Performance in Human Subjects by Electrotherapy

Inventor: Andrija Puharich Original document:

Research terms: electrotherapy; neural performance; electrical stimulation; physiological conditioning; human-subject research.

This patent is relevant to the history of proposed electrical methods for influencing neural functioning or performance.
The term electrotherapy encompasses different techniques and should not be treated as a single mechanism. A proper examination would identify the electrical parameters, the proposed mode of application, the human or animal subjects described, the claimed physiological endpoint and any evidence cited by the inventor.
The title's reference to improving neural performance is a statement of intended purpose. It does not independently establish that the method improved cognition, enhanced memory, produced durable behavioural changes or achieved the proposed results under controlled conditions.
Relevant scientific sources
Forensic finding: The patent is a useful primary source for investigating historical attempts to apply electrical stimulation to human performance. Establishing its practical significance requires comparison with independent research using comparable methods and measurable outcomes.

9.6. US3629521A — Hearing Systems

Inventor: Andrija Puharich Original document:
Research terms: hearing systems; radio-frequency energy; auditory perception; electromagnetic stimulation; signal modulation.
This record belongs to the history of proposed systems involving electromagnetic energy and hearing.
The phrase "RF/microwave hearing" in a secondary list is an interpretive label. The original specification must be examined to determine exactly what energy source and delivery mechanism the inventor proposed and whether the document addresses the radio-frequency auditory phenomenon in the same sense as later patents.
The relevant scientific issue is whether the proposed physical mechanism is compatible with independently established auditory physiology and whether the patent's specific design can produce the claimed result.
Relevant sources
Forensic finding: The record should be examined as a specific hearing-system proposal. It should not automatically be equated with later methods for encoding intelligible speech through the radio-frequency hearing effect.

10. The 1970s: Sleep Induction, Brain-Wave Patterns and Proposed Remote Monitoring

10.1. US3712292A — Method and Apparatus for Producing Swept Frequency-Modulated Audio Signal Patterns for Inducing Sleep

Inventor: John E. Zentmeyer Jr. Filed: 20 July 1971 Granted: 23 January 1973
Original patent:
This patent describes producing a familiar, repetitive sound and modifying it with swept-frequency signals selected to resemble frequency ranges associated with electrical brain-wave patterns during sleep.
The specification describes two modulating frequency ranges, approximately 0.8–1.8 Hz and

5.5–6.75 Hz. The intended result was to produce audio patterns that might facilitate sleep.
The important technical distinction is that the patent describes audible sound modulated by low-frequency signals. It does not describe the direct broadcasting of brain waves through the air.
That distinction matters because the phrase “brain frequencies broadcast” can obscure the actual signal path. The patent’s mechanism involves modifying an audio signal; the brain-wave frequencies are used as a reference for modulation.
Scientific context
Sleep research distinguishes electrical brain activity recorded by EEG from the acoustic signals used to influence a sleeping or awake person. Similarity between a modulation frequency and a frequency observed in EEG does not, by itself, establish that the sound entrains brain activity or induces a particular sleep stage.
Relevant research:
Forensic assessment: This is evidence of a proposed audio-based sleep-induction method using frequency modulation informed by EEG patterns. Its actual effectiveness must be established through controlled sleep studies, not inferred from the similarity of the selected frequencies to brain-wave measurements.

10.2. US3884218A — Method of Inducing and Maintaining Various Stages of Sleep in the Human Being

Inventor: Robert A. Monroe Filed: 10 September 1970 Granted: 20 May 1975 Original assignee: Monroe Industries, Inc.
Original patent:
This patent describes generating a familiar, repetitive audio signal and modulating it with a pattern derived from EEG activity associated with sleep.
The subject could select a preferred repetitive sound, and the signal’s volume could be adjusted to overcome ambient noise.
The proposal is notable because it connects three distinct elements:

- A sound that the listener can hear.
- A modulation pattern informed by recorded electrical brain activity.
- An intended physiological outcome involving sleep.

These elements are technically distinguishable. Generating the audio signal is an engineering question; determining whether the listener enters a particular sleep stage is a physiological question; establishing that the modulation caused the change requires an experimental design capable of separating the effect from ordinary sleep variation, expectation, relaxation and environmental conditions.
The patent therefore provides a useful historical example of the effort to translate observations of brain activity into an external sensory stimulus.
Relevant scientific literature
Forensic assessment: The patent documents a specific proposed relationship between auditory modulation and sleep stages. It does not establish that an external audio signal can reliably impose arbitrary brain states or that the same method can remotely control waking

behaviour.

10.3. US3951134A — Apparatus and Method for Remotely Monitoring and Altering Brain Waves

Inventor: Robert G. Malech Filed: 5 August 1974 Granted: 20 April 1976

Original patent:

This patent is one of the most technically ambitious items in the original collection. It proposes a system that uses electromagnetic transmission and reception to investigate brain-wave activity and generate feedback intended to alter that activity.

The title is direct evidence of the inventor's stated objective. It is not evidence that the apparatus achieved reliable remote EEG measurement or effective control of neural activity.

A rigorous technical examination would investigate:

- The frequency and power of the transmitted signals.
- The geometry and distance between transmitter, subject and receiver.
- The expected signal-to-noise ratio.
- The method used to distinguish a signal attributed to brain activity from other physiological or environmental sources.
- The claimed method of interpreting the returned signal.
- The mechanism by which a compensating signal is intended to change neural activity.
- Any experimental data or independently replicated results.
- Any subsequent patents, technical publications, contracts or operational records linked to the invention.

The distinction between sensing and influencing is crucial. A method that detects a physiological signal does not automatically provide the ability to decode thoughts. A method that stimulates neural tissue under certain conditions does not automatically provide the ability to select a particular thought or compel a complex action.

Modern scientific comparison

These search links identify relevant research areas; they do not constitute evidence that Malech's particular method was independently validated.

Forensic assessment: Malech's patent warrants close examination because it explicitly proposes remote monitoring and alteration of brain-wave activity. The next evidentiary step is to establish whether the proposed system was experimentally demonstrated, whether its results were independently replicated and whether there is evidence of actual use.

11. The 1980s: Subliminal Audio Systems, Behavioural Conditioning and Microwave Hearing

11.1. US4395600A — Auditory Subliminal Message System and Method

Inventors: René R. Lundy and David L. Tyler Priority and filing date: 26 November 1980 Granted: 26 July 1983 Original assignee: Proactive Systems Inc.

Original patent:

This patent proposes a system for presenting auditory messages at a level intended to remain below a listener's conscious recognition threshold. It includes a masking signal whose amplitude can be adjusted in response to ambient sound.

The stated application includes environments such as retail shopping areas, where ambient

audio conditions may vary.
This is a particularly useful document for investigating how subliminal messaging was translated into a commercial engineering proposal. It describes not merely an idea about unconscious influence but a system for combining signals and adjusting masking levels. Its specification also makes a claim about subliminal signals influencing behaviour. That claim must be treated as a claim made in the patent, not as a substitute for independent experimental evidence.
The forensic questions are:

- What was the message signal?
- How was its amplitude controlled?
- What measurements established that it was below conscious recognition?
- How did the system respond to changes in ambient noise?
- What behavioural outcomes were measured?
- Were participants tested under blinded conditions?
- Were any claimed effects independently replicated?

Relevant research
Forensic assessment: The patent demonstrates a documented attempt to engineer an adjustable subliminal audio system. Whether it reliably altered purchasing decisions or other complex behaviour is a separate empirical question.

11.2. US4717343A — Method of Changing a Person's Behavior

Inventor: Alan B. Densky Priority date: 30 June 1986 Granted: 5 January 1988
Original patent:
The patent describes a method in which a person views a programme of pictures presented on a screen. The inventor proposes that the programme could condition unconscious thought patterns and produce a desired change in behaviour.
The document is significant because the intended outcome is stated in behavioural terms rather than merely in terms of presenting an unusual sensory stimulus.
However, the title and abstract describe the intended result of the proposed method. They do not independently demonstrate that the method produced the result.
A forensic assessment should distinguish:

- Exposure to visual material.
- Conscious recognition of the material.
- Processing that occurs without reliable conscious report.
- Short-term priming or changes in task performance.
- Longer-term behavioural conditioning.
- Durable changes to beliefs, habits or complex decision-making.

These outcomes require different measures and cannot be assumed to follow one another automatically.
Relevant research
Forensic assessment: The patent is a primary-source record of an attempted visual-conditioning method. Claims of reliable or lasting behavioural change require independent evidence beyond the patent itself.

11.3. US4777529A — Auditory Subliminal Programming System

Inventors: Richard M. Schultz and Raymond Dolejs Filed: 21 July 1987 Granted: 11 October 1988 Original assignee: R. M. Schultz and Associates, Inc.
Original patent:
This patent describes an encoding and playback system for auditory subliminal messages. Its abstract refers to fixed-frequency security tones, an encoder, a decoder/mixer and a conventional stereo system. The system combines the subliminal message with selected audio signals and outputs the composite through ordinary speakers.
The use of security tones is a technical detail worth examining because it describes a means of distinguishing encoded message material from other audio input.
The patent does not establish that the system was used covertly against unsuspecting listeners, nor does it establish that the messages could reliably control behaviour. Those propositions require separate documentary and experimental support.
The most useful technical investigation would reconstruct the signal path from message encoding to output, identify the amplitude and frequency characteristics of the proposed signal, and determine whether the patent cites test results establishing perceptibility or behavioural effects.
Relevant sources
Forensic assessment: This patent documents a specific system for encoding, mixing and presenting auditory messages. The claimed psychological effect must be evaluated independently of the engineering design.

11.4. US4834701A — Apparatus for Inducing Frequency Reduction in Brain Wave

Inventor: Masaki Original patent:
This item should be treated as a proposed method for influencing brain-wave frequency, with the exact mechanism determined from the full specification rather than inferred from a secondary label.
The phrase “brain frequencies transmission” is not a sufficiently precise technical description. It could conceal important differences between direct stimulation, sensory entrainment, acoustic modulation and the use of an external signal whose characteristics are selected by reference to EEG measurements.
A forensic review should establish the patent’s precise signal source, the intended recipient or anatomical target, the method of stimulation and the measured endpoint.
Relevant scientific sources
Forensic assessment: The title establishes the intended subject of the invention, but the original claims and supporting experimental evidence are necessary to determine what it actually proposes and whether its stated effect was demonstrated.

11.5. US4858612A — Hearing Device

Inventor: Robert W. Stocklin Original patent:
The supplied list categorises this patent as “microwave hearing.” That description should be verified against the original specification, particularly the energy source, transmission method, receiver or physiological mechanism.
The term hearing device can refer to a wide variety of technologies. A rigorous examination

must not assume that the device shares the same mechanism as another patent simply because both are associated with electromagnetic energy or hearing.
Research sources
Forensic assessment: This patent should be analysed in its own technical context. Its inclusion in a broader list does not establish that it enabled remote speech transmission or that it was operationally linked to other inventions.

11.6. US4877027A — Hearing System

Inventor: Wayne B. Brunkan Priority date: 6 June 1988 Original patent:
The original record describes a hearing-system proposal. The technical details need to be read directly to determine the precise sound-generation method, frequency range, signal source and intended auditory response.
The supplied list describes this as "microwave hearing via open-air broadcast." That wording should not be treated as an established finding until the patent's actual disclosure and claims have been checked.
Relevant research
Forensic assessment: The patent is a primary source for investigating a proposed hearing system. Its specific mechanism and any evidence of real-world use must be established from the record and corroborating sources.

12. The 1990s: States of Consciousness, Radio-Frequency Hearing and Ultrasonic Audio

The 1990s are particularly relevant because the patent collection includes several distinct approaches: auditory stimulation intended to influence consciousness, systems involving the radio-frequency hearing effect, ultrasonic sound carriers, and proposals involving analysis or use of brain activity.
These must be separated technically. Ultrasonic audio transmission, subliminal recordings, EEG-based sound modulation and radio-frequency auditory effects are not interchangeable mechanisms.

12.1. US5159703A — Silent Subliminal Presentation System

Inventor: Oliver M. Lowery Filed: 28 December 1989 Granted: 27 October 1992
Primary source:
The patent proposes encoding information on carriers in low or high frequency ranges, including adjacent ultrasonic frequencies, and presenting the resulting signal through an acoustic or vibrational pathway.
Its historical significance lies in the proposed delivery of information through a carrier not ordinarily perceived as conventional audible speech.
The essential forensic questions are whether the system can transmit the intended information, whether the signal can be detected or decoded by a human listener, and whether any measurable behavioural effects follow.
Those are different questions. An engineered signal can be physically generated without necessarily delivering intelligible information to a listener, and information can be perceived without necessarily producing the behavioural effect claimed by the inventor.
Research sources

Finding: The patent documents a proposed silent-subliminal presentation system. It does not independently establish reliable remote mind control.

12.2. US5356368A — Method of and Apparatus for Inducing Desired States of Consciousness

Inventor: Robert A. Monroe Filed: 1 March 1991 Granted: 18 October 1994 Original assignee: Interstate Industries Inc.

Primary source:

This patent describes generating sequences of audio signals with selected frequencies for the stated purpose of inducing desired states of consciousness.

The technical interest lies in the proposed relationship between auditory stimulation and changes in subjective or physiological state. The invention belongs to the broader history of audio-assisted relaxation, sleep induction and attempts to influence conscious experience.

It is necessary, however, to distinguish the generation of a signal associated with a particular frequency from proof that the signal produces a specific brain state. EEG measurements, subjective reports and behavioural outcomes are separate forms of evidence.

The patent's existence does not establish that the proposed signals can reliably produce every state of consciousness described by the inventor.

Relevant research

Finding: This is a primary-source record of a proposed audio-based consciousness-induction method. Its effects require independent evaluation using controlled physiological and psychological measurements.

12.3. US5774088A — Method and System for Warning Birds of Hazards

Inventor: Melvin L. Kreithen Original assignee: University of Pittsburgh Filed: 8 May 1997 Granted: 30 June 1998

Primary source:

This patent is an important example of why a forensic analysis must not equate all radio-frequency auditory proposals with human mind-control technology.

The specification proposes using pulsed microwave energy to alert birds to hazardous areas. The abstract describes pulses in the approximate range of 1–40 GHz, with pulse durations of approximately 5–25 microseconds, and proposes that the birds' auditory systems would detect the radiation.

The intended application is animal hazard warning, not human speech transmission.

The patent's significance is that it documents a proposed application of the radio-frequency auditory effect in a system intended to alter animal responses to hazards. The specification is not, by itself, proof that the system achieved reliable warning or avoidance behaviour in field conditions.

Research sources

Finding: The patent documents an attempt to apply an auditory effect associated with pulsed microwave energy to animal hazard warning. It does not demonstrate a corresponding capability to communicate intelligible speech into a human subject's mind.

12.4. US5889870A — Acoustic Heterodyne Device and Method

Inventor: Mark Norris Filed: 1996 Granted: 30 March 1999
Primary source:
The supplied list associates this patent with ultrasound and the ventriloquist effect. Its actual specification should be used to determine how the system produces or directs sound, what role ultrasonic carriers play, and what perceptual effect is claimed.
The term acoustic heterodyning refers broadly to processes in which interacting signals or nonlinear responses can generate additional frequency components. It is important to establish the precise mechanism described in this particular patent rather than assuming that the title implies a single, universally applicable acoustic effect.
The ventriloquist effect is a separate perceptual phenomenon in which visual information can influence the perceived location of a sound. It should not be treated as proof of remote thought transmission.
Relevant research
Finding: This patent warrants examination as an acoustic-engineering proposal. Its significance depends on the actual design and claims, not merely on the association of its title with ultrasound.

12.5. US6011991A — Communication System and Method Including Brain Wave Analysis and/or Use of Brain Activity

Inventor: Aris Mardirossian Priority date: 7 December 1998 Original patent:
The patent proposes a communication system involving analysis or use of brain activity.
This title requires particular care. The phrase “brain wave analysis” can refer to the measurement and interpretation of physiological signals, while the phrase “use of brain activity” can encompass a variety of possible applications. Neither phrase establishes that a system can read arbitrary private thoughts or transmit thoughts directly between people.
A forensic assessment should identify the precise signal acquisition method, the data being analysed, the proposed communication protocol, the experimental conditions and the claims made about accuracy.
Modern brain-computer interface research provides a useful comparison. Such systems can translate particular neural signals into commands or classifications under specified conditions, but their performance depends on the recording method, training, signal quality, task and decoding algorithm.
Research sources
Finding: The patent is a primary source for examining a proposed brain-activity communication system. It does not establish that the invention permits unrestricted remote access to thoughts.

13. The 1998–2003 ultrasonic and radio-frequency communication patents

13.1. US6052336A — Apparatus and Method of Broadcasting Audible Sound Using Ultrasonic Sound as a Carrier

Inventor: Austin Lowrey III Priority date: 2 May 1997 Filed: 1 May 1998 Granted: 18 April 2000
Primary source:
This patent describes using an ultrasonic signal as a carrier, modulated with an information-

bearing input signal, and projecting the resulting signal into an area where listeners may perceive audible sound.

The abstract explains that the ultrasonic signal is amplitude- or frequency-modulated, with additional processing in the amplitude-modulation implementation. It describes a projector unit that broadcasts the signal and a system intended to make the resulting sound detectable by people in the broadcast region.

This is an important distinction from the silent-subliminal patent family. Here, the stated objective is to broadcast audible sound using an ultrasonic carrier. The mechanism is not simply a hidden message intended to be processed unconsciously.

The patent's background also mentions situations involving a crowd. That establishes an application contemplated in the specification, not evidence that the invention was used against a particular crowd or that it was deployed in an operational incident.

Technical issues for examination

- The ultrasonic carrier's frequency and intensity.
- The modulation and demodulation mechanism.
- Whether audible sound is produced through nonlinear interaction in air or another mechanism.
- The effective range and directionality.
- The acoustic field produced in the target area.
- Exposure and hearing-safety requirements.
- The intelligibility of the reproduced signal under real-world conditions.

Relevant sources

Finding: The patent documents a directional-audio engineering proposal. The existence of a system for transmitting audible sound on an ultrasonic carrier is not proof of inaudible psychological conditioning or direct neurological control.

13.2. US6470214B1 — Method and Device for Implementing the Radio Frequency Hearing Effect

Inventors: James P. O'Loughlin and Diana L. Loree Granted: 22 October 2002 Original assignee: United States Department of the Air Force

Primary source:

This patent deserves separate treatment because it provides a direct, identifiable institutional connection to the US Air Force.

Its stated subject is the implementation of the radio-frequency hearing effect. The technical proposal involves processing radio-frequency signals so that speech may be perceived through the relevant auditory phenomenon.

The documentary facts are important: the patent has named inventors, a technical specification, claims and an identified original assignee. These features make it possible to investigate the invention's history without relying on unsupported summaries.

The questions that remain include whether the patent's method was independently demonstrated, how the claimed signal characteristics relate to the auditory mechanism, what practical constraints apply, and whether later technical records document development or deployment.

Relevant scientific sources

Finding: The patent establishes documented Air Force-associated work on the radio-frequency hearing effect. It does not independently establish that the system was used against particular individuals, that it was deployed covertly, or that it allows unrestricted remote transmission of thoughts.

13.3. US6587729B2 — Apparatus for Audibly Communicating Speech Using the Radio Frequency Hearing Effect

Inventors: James P. O'Loughlin and Diana L. Loree Granted: 3 July 2003

Primary source:

This patent is closely related in subject matter to US6470214B1. The relationship is important because patent families and related applications can reveal how a technical proposal was developed, refined or protected over time.

The proper forensic approach is to compare the claims and specifications rather than count related patent numbers as independent experimental confirmations.

A family of patents may document multiple refinements or different legal claims associated with a technology. It does not necessarily represent multiple independent demonstrations of the underlying physical effect.

Questions for the patent-family review

- Which claims are shared between the related patents?
- Which technical features were added or changed?
- What prior art is cited?
- Are there related applications or continuation records?
- Do the records identify testing, contracts, procurement or actual use?
- Are independent experimental papers available that evaluate the specific method?

Relevant sources

Finding: The related patents should be examined as a technical and legal family. Their existence strengthens the documentary trail for the proposed technology, but patent-family size is not a measure of experimental validation or operational deployment.

14. Cross-patent analysis: what is common, and what is different?

The list contains several superficially similar concepts that need to be separated before any broader historical conclusion is drawn.

Technology family	Principal mechanism proposed	Key question
Hypnosis apparatus	Suggestion and structured stimulation	What effects were measured, and under what

		conditions?
Subliminal audiovisual systems	Presentation of stimuli below ordinary reportable awareness	Can the proposed stimuli reliably influence the intended outcome?
Sleep-induction systems	Audible signals modulated using selected frequency patterns	Does the sound produce measurable changes in sleep architecture?
Electrotherapy and neural stimulation	Electrical stimulation or related physiological intervention	What tissue is stimulated, and what response is demonstrated?
Remote brain-wave proposals	Electromagnetic sensing and feedback	Can the signal be measured and interpreted reliably at the proposed distance?
Silent-subliminal audio	Information encoded onto low-, high- or	Can the information be perceived or

	ultrasonic carriers	decoded, and does it affect behaviour?
Ultrasonic directional audio	Ultrasonic carrier used to generate audible sound	What sound field is produced, and under what conditions?
Radio-frequency auditory systems	Pulsed RF energy and auditory transduction	What auditory effect is produced, and what does it permit in practice?
Brain-activity communication systems	Measurement or interpretation of neural signals	What information can be decoded, with what accuracy and limitations?

The critical finding is that these patents do not describe one uniform technology.
They involve different signal pathways, different physical mechanisms, different intended outcomes and different standards of evidence. Some are directed at ordinary sensory perception; some propose physiological stimulation; others concern signal processing or remote measurement.
A credible investigation must preserve those distinctions.

14.1. What the chronology does establish

Taken together, the records document recurring engineering interest in the relationship between external stimuli and human or animal perception.
The patent trail includes attempts to:

- Present audiovisual information below ordinary conscious recognition.
- Use sound patterns intended to influence sleep.

- Apply electrical stimulation to physiological or neural functioning.
- Propose remote monitoring and alteration of brain-wave activity.
- Encode information in auditory signals.
- Use ultrasonic carriers to produce audible sound.
- Implement the radio-frequency hearing effect.
- Develop communication systems involving brain activity.

That is a meaningful historical finding.

14.2. What the chronology does not establish

The list alone does not demonstrate:

- A single coordinated programme connecting every inventor or patent.
- A common institutional owner for all technologies.
- That every invention worked as described.
- That all patents were built or tested.
- That any particular technology was used against a particular individual.
- That the patents collectively establish remote access to thoughts or reliable behavioural control.

Those conclusions would require additional evidence, including technical demonstrations, independent experimental studies, assignment and procurement records, contracts, operational reports and corroborated documentation of use.

15. Research record for continuation

The following source links provide a working primary-source register for the patents examined in Parts I and II.

Early audiovisual and neural-stimulation patents

- Sleep, remote monitoring and subliminal systems
- Later auditory and consciousness-related patents

Important verification note: The links above identify the patent records to examine. Before publishing a final forensic register, the remaining bibliographic fields—including exact inventor spellings, filing dates, grant dates, assignees and patent-family relationships—should be reconciled against each original record. A secondary list should never override the primary patent record.

End of Part II.

The next stage should expand beyond patents into the independently documented scientific and institutional record: military psychological operations, declassified intelligence research, human-subject experimentation, peer-reviewed radio-frequency auditory research, neurotechnology, psychiatric literature and the ethical frameworks governing research on human participants.

FORENSIC RESEARCH DOSSIER

Part III — Military Psychological Operations, Intelligence Research, Human Experimentation and the Scientific Record

Continuation of Parts I and II

This part moves beyond patent documents to examine a wider body of independently identifiable evidence: military doctrine, declassified intelligence records, government investigations, scientific research, psychiatric literature and research-ethics frameworks.

The central historical question is how governments and military organisations have investigated methods of influencing human perception, attention, behaviour and physiological functioning, and what the surviving records establish about the methods actually used.

The distinction between documented institutional activity and demonstrated technical capability remains essential. A declassified programme may establish that research was funded or performed without proving that every proposed method worked. A scientific paper may establish a physiological effect without demonstrating that the effect was incorporated into a military system. A military manual may establish an operational doctrine without establishing the deployment of a particular patent.

The strongest documentary account connects these different classes of evidence while preserving their differences.

16. Military psychological operations: the documented historical record

16.1. United States Army psychological operations

Primary institutional source:

The US Army's institutional history describes the development of organised psychological operations, including wartime propaganda, information campaigns and dedicated military structures.

The historical significance is straightforward: military organisations have formally studied and practised methods intended to influence how target audiences perceive events, understand information and make decisions.

These activities can involve leaflets, radio broadcasts, loudspeaker messages, printed materials, audiovisual communications and other methods of delivering information. Their effects depend on the message, audience, circumstances, credibility, repetition and operational context.

This history is relevant to the patent collection because some of the patent

specifications explicitly contemplate psychological or propaganda-related applications. It supplies a documented military context in which influence over perception and behaviour has been treated as an operational objective.

However, the connection must be drawn carefully. The Army's history does not establish that every subliminal-message patent was used in military psychological operations, nor that conventional information operations depended on direct neurological stimulation.

Research terms: psychological operations; PSYOP; military propaganda; influence operations; information operations; audience analysis; persuasion; military deception; strategic communication.

16.2. US Department of Defense doctrine

The US Department of Defense publishes doctrine and policy documents defining information-related military activities.

Primary source portal:

Relevant topics include psychological operations, information operations, military deception and the coordination of communication-related activities.

These documents are useful because they establish official terminology and the operational objectives recognised by the institution. They also help prevent anachronistic interpretations in which a modern term is applied indiscriminately to an earlier technology.

Forensic researchers should record the exact edition, publication date, issuing authority and applicable definition for each doctrinal document. Doctrine changes over time, and the existence of a doctrine does not prove that every described capability was used in a particular operation.

Questions for documentary examination

- **What audience was the operation intended to influence?**
- **What communication method was specified?**
- **What behavioural or perceptual outcome was sought?**
- **What evidence exists of actual execution?**
- **What assessment was made of the results?**
- **Were the methods conventional communications, sensory stimulation or another technical intervention?**

The answers determine whether a source documents an operational influence activity, a technical proposal or merely an institutional concept.

16.3. NATO and allied military research

For a cross-national examination, NATO's research and doctrine publications provide an additional source of institutional evidence.

Primary institutional portal:

Research portal:

NATO's scientific and technical research spans human performance, operational medicine, human factors, cognitive performance, communications and other defence-related disciplines. The existence of research in these fields is not equivalent to evidence of a particular mind-control capability.

A defensible multinational investigation should identify individual reports, their authors, publication dates, sponsors and experimental methods. Broad institutional descriptions are useful for discovering records but should not be presented as proof of specific programmes.

16.4. Australian Defence research

For an Australian component, the principal institutional starting points are:

These institutions publish information about defence science, human performance, operational medicine, psychology, communications and related technical subjects.

Their existence provides a route for researching Australian defence activity. It does not establish that any particular technology in the patent list was acquired, tested or deployed in Australia.

The relevant records would be specific reports, procurement documents, research contracts, technical evaluations, parliamentary answers, historical archives or released documents that identify the particular subject under investigation.

Forensic principle: An institutional connection must be documented at the level of the actual project, patent, contract or report. It should not be inferred solely from an organisation's general research interests.

17. Intelligence research and behavioural modification: MKULTRA and related records

17.1. The MKULTRA documentary record

Primary-source portal:

The CIA's declassified records include documents concerning MKULTRA, an intelligence research programme associated with investigations of drugs, human behaviour and methods relevant to interrogation and intelligence operations.

One searchable record is:

The documentary importance of MKULTRA is not dependent on accepting every claim made in later accounts of the programme. Its surviving records can be examined for the actual research proposals, funding arrangements, named investigators, institutional relationships and stated objectives.

The records also belong to a broader history of government investigations into drugs, interrogation, human behaviour and the ethics of experimentation.

For a rigorous examination, the original document should be read in full and its provenance established. A page describing a subproject is not necessarily a complete record of the work performed, and the subject of one subproject should not automatically be attributed to every other project bearing the MKULTRA name.

17.2. The 1977 US Senate hearing

A particularly important public record is the US Senate's 1977 hearing concerning the CIA's behaviour-control research.

Primary source:

This hearing is important because it places intelligence research into a formal public record. It provides a basis for examining testimony, programme administration, research practices and the problems created by the destruction or incompleteness of records.

It is a stronger source than an unsourced online summary because it is an identifiable congressional document with a defined date and institutional context.

The hearing should be examined for:

- **The programmes and subprojects specifically identified.**
- **The methods and substances described.**
- **The roles of participating institutions and researchers.**
- **The documentation available to investigators.**
- **The stated purposes of the work.**
- **The ethical questions raised by human experimentation.**
- **The limitations created by missing or destroyed records.**

What this evidence establishes: The hearing documents official investigation of CIA behaviour-modification research and provides primary material for reconstructing aspects of that history.

What it does not establish: It does not prove that every later patent concerning hypnosis, subliminal audio, electromagnetic signals or brain-wave monitoring was connected to MKULTRA.

Such a connection would require a documentary link—for example, a contract, memorandum, named researcher, funding record or technical report connecting the specific invention to the programme.

17.3. Human experimentation and informed consent

MKULTRA is also significant because it illustrates why the ethical assessment of research cannot be reduced to whether a proposed technology worked.

Research involving deception, psychoactive substances or interventions that affect a person's mental state raises questions about informed consent, foreseeable risks, the right to withdraw and the responsibilities of investigators.

These issues are addressed in modern research-ethics frameworks, including:

The Belmont Report identifies three central principles: respect for persons, beneficence and justice. The Declaration of Helsinki sets out ethical principles for medical research involving human participants.

These modern frameworks should not be treated as if every provision applied identically to every historical experiment. Their value is to identify the ethical questions and standards relevant to evaluating research practices, especially where participants may not have understood the purpose or risks of an intervention.

Forensic conclusion: Documented intelligence research into behaviour modification establishes an important historical context for investigating human autonomy and institutional accountability. It does not eliminate the need to establish a direct evidentiary connection between that history and any later patent or alleged deployment.

18. Radio-frequency auditory effects: the scientific record from 1962 onward

This subject requires particular care because there is a documented scientific literature on auditory sensations associated with pulsed or modulated electromagnetic energy, but that literature must not be extended beyond what its experiments demonstrate.

18.1. Allan H. Frey (1962): Human Auditory System Response to Modulated Electromagnetic Energy

Author: Allan H. Frey Journal: Journal of Applied Physiology Volume: 17, Issue 4, pages 689–692 Publication: July 1962 DOI: 10.1152/jappl.1962.17.4.689 PubMed identifier: PMID 13895081

Primary scientific sources:

Frey's paper is an important early publication in the history of the radio-frequency auditory effect. It reports experiments in which people perceived sounds associated with exposure to modulated electromagnetic energy.

The paper describes sound perceptions in normal-hearing and deaf participants and discusses how the effect varied with carrier frequency and modulation. It also discusses possible locations and mechanisms for the response.

The study is historically significant because it places an unusual auditory phenomenon into a published physiological research record. It provides an identifiable starting point for subsequent investigations into the interaction between electromagnetic energy and auditory perception.

However, a careful reading must distinguish the paper's reported observations from later interpretations. Frey's study does not, by itself, establish that all radio-frequency signals can produce hearing, that the phenomenon permits unrestricted speech transmission, or that it allows a transmitter to control thoughts.

Forensic questions raised by the paper

- **What exposure conditions produced the reported effect?**
- **How were the sounds described and measured?**
- **How did carrier frequency and modulation affect perception?**
- **What was the role of peak power density?**
- **How were alternative explanations evaluated?**
- **What later experiments reproduced or challenged the findings?**
- **How do the conditions in this experiment compare with those specified in later patents?**

These questions provide a pathway from a historical report to a modern evidence review.

18.2. The physical mechanism: the microwave auditory effect

The term microwave auditory effect is commonly used for auditory perceptions associated with particular pulsed radio-frequency exposures.

A leading explanation involves rapid, minute temperature changes in tissue that generate thermoelastic pressure waves. Those pressure waves can interact with the structures of the auditory system, creating an auditory sensation.

The mechanism is not the same as ordinary sound travelling through air from a loudspeaker. Nor does it imply that radio waves carry a person's thoughts or that any arbitrary spoken message can be transmitted into another person's consciousness.

The exact effect depends on exposure parameters, including pulse structure, energy absorption and other physical conditions.

Research starting points

These links are research searches and a primary publication, not a claim that every article returned supports every proposed application.

18.3. Three separate scientific propositions

The following propositions are often conflated in popular accounts. They should be investigated independently.

Proposition A: Electromagnetic exposure can produce an auditory sensation under particular conditions.

Frey's 1962 paper provides a published experimental record relevant to this

proposition.

Proposition B: A technical system can encode information into signals that produce intelligible auditory perceptions.

This requires evidence about signal design, exposure conditions, intelligibility testing and the reliability of the resulting perceptions. A demonstration of an auditory sensation is not, by itself, a demonstration of reliable speech communication.

Proposition C: A system can remotely control thoughts, compel behaviour or reliably alter complex psychological states.

This is a much broader claim. It would require independent evidence of the relevant mechanism, measurable outcomes, reproducibility and causal attribution. The existence of an auditory effect does not establish this proposition.

The distinction is central to the forensic examination of US6470214B1 and US6587729B2. Those patents should be compared with the experimental literature on the precise mechanisms they propose, rather than treated as evidence for every claim made under the label "microwave mind control."

18.4. Radio-frequency exposure and safety

A separate research question concerns exposure limits and biological safety.

The fact that a particular exposure can produce a perceptual effect does not establish that the exposure is harmless, nor does it establish that a particular real-world exposure caused a specific health outcome.

For exposure assessment, investigators should identify the frequency, power density or absorbed energy, pulse characteristics, duration, distance, measurement method and applicable safety standard.

Useful institutional sources include:

These sources address exposure, health and safety questions. They should not be treated as evidence that any particular patent was deployed or that a particular individual experienced a particular exposure.

Interim finding: The radio-frequency auditory effect has a documented scientific history. The scope of the effect, the conditions required to produce it, the intelligibility

of any resulting signal and the practical capabilities of specific systems must be evaluated separately.

19. Subliminal perception: what controlled research supports

19.1. National Research Council (1991): In the Mind's Eye

Editors: Daniel Druckman and Robert A. Bjork Publisher: National Academies Press Publication year: 1991 DOI: 10.17226/1580

Primary source:

The National Research Council's report provides an important evidence-based comparison for the subliminal patents.

Its chapter on subliminal self-help examined claims that audio recordings containing messages outside ordinary conscious awareness could improve self-confidence, memory or other aspects of performance.

The committee concluded that the available evidence did not support claims that subliminal self-help tapes enhanced human performance. It distinguished limited effects observed in some laboratory tasks from the much stronger claim that subliminal messages could produce lasting changes in complex behaviour or emotion.

The chapter also examined whether some commercial recordings contained messages that were actually detectable under controlled conditions.

Direct chapter link:

This is particularly relevant to US4395600A, US4777529A and US5159703A, whose specifications concern systems for presenting or encoding subliminal audio messages.

The comparison does not establish that every subliminal system is identical. It establishes that claims about subliminal influence must be evaluated through controlled measurements rather than accepted solely because a system can generate a hidden or masked signal.

19.2. Subjective awareness versus objective detectability

A critical distinction in subliminal research is the difference between a person reporting that they did not consciously notice a stimulus and the stimulus being objectively

undetectable.

A person may fail to report a stimulus while still being able to discriminate it under a more sensitive test. Conversely, a stimulus may be too weak or masked to produce reliable information processing.

Researchers therefore use methods to test awareness and discrimination, rather than relying exclusively on participants' subjective reports.

This distinction matters when evaluating a patent that claims to present messages below the threshold of consciousness. The engineering system may successfully mask a signal acoustically, but that alone does not establish the signal's perceptual status or behavioural effectiveness.

Relevant research links

19.3. Priming is not equivalent to mind control

Priming is a broad research term for situations in which prior exposure to a stimulus affects subsequent processing or performance.

Some priming effects can occur without a participant accurately reporting the initial stimulus. But the existence of a measurable priming effect does not establish that the stimulus can compel behaviour, implant complex beliefs or reliably overcome a person's intentions.

The effect may depend on the task, context, strength of the stimulus, prior expectations and other variables.

The proper forensic question is not simply whether a stimulus can affect a person. It is whether the particular stimulus produces the particular outcome claimed, with what effect size, for how long, and under what conditions.

Key terms for the research register

- **Subliminal perception**
- **Masked priming**
- **Semantic priming**
- **Objective awareness**
- **Subjective awareness**
- **Signal-detection theory**
- **Implicit cognition**

- **Attentional bias**
- **Expectancy effects**
- **Suggestibility**
- **Behavioural conditioning**
- **Long-term behavioural change**

These terms should be used to identify relevant experiments, not as interchangeable labels for one mechanism.

20. Psychiatry, clinical psychology and the limits of behavioural influence

The patent collection uses terminology associated with hypnosis, subconscious processing, neural performance and altered states of consciousness. Those terms overlap with clinical research, but they do not necessarily mean the same thing in engineering, psychiatry or experimental psychology.

This section establishes the relevant clinical and scientific framework.

20.1. American Psychological Association: hypnosis and suggestion

Institution: American Psychological Association (APA)

Hypnosis is a recognised subject of psychological investigation. Research examines responsiveness to suggestion, subjective experiences, attention, pain perception and clinical applications.

The existence of hypnotic responsiveness does not establish that every person can be hypnotised in the same way, that all suggestions will be accepted, or that hypnosis provides unrestricted control over another person's behaviour.

For the patent examination, this literature is relevant to US3014477A and other inventions that propose to induce or influence altered psychological states.

The scientific questions are whether a method reliably induces the intended state, how responsiveness varies between individuals, what outcomes can be measured and whether the findings are independently replicated.

20.2. American Psychiatric Association: psychiatric diagnosis and assessment

Institution: American Psychiatric Association (different from the American Psychological Association).

Psychiatric diagnosis concerns the assessment of defined symptoms, patterns of functioning, duration, context and clinically relevant impairment. It is not a direct measurement of whether a particular technological exposure occurred.

Likewise, the existence of a psychiatric diagnosis does not establish or disprove a claim about an external event. The underlying event, the person's experience, the possible mechanisms and the clinical assessment are different questions.

This distinction is important in any investigation where claims about unusual auditory or sensory experiences intersect with questions of technology, stress, sleep, environmental exposure or mental health.

A rigorous approach neither assumes that an unusual experience proves the existence of a particular technology nor dismisses an experience merely because its cause is uncertain.

20.3. The National Institute of Mental Health: brain stimulation

Institution: US National Institute of Mental Health (NIMH)

Primary source:

NIMH describes established and emerging brain-stimulation approaches used in particular clinical settings.

These include electroconvulsive therapy, repetitive transcranial magnetic stimulation, vagus nerve stimulation and deep brain stimulation. Their mechanisms, indications and delivery requirements differ.

This literature provides a useful technical comparison for patents that propose neural stimulation or alteration of brain-wave activity.

It demonstrates that neural activity can be influenced through defined medical interventions. It does not establish that a patent involving a different signal pathway can achieve the same effects, or that neural stimulation necessarily produces a specific thought or behaviour.

20.4. The scientific meaning of "brain waves"

The expression brain waves commonly refers to oscillatory patterns of electrical potential recorded using electroencephalography (EEG).

EEG is a measurement technique. It records voltage differences at electrodes placed on the scalp or other recording locations. The resulting signals reflect electrical activity generated by populations of neurons, filtered through biological tissues and affected by the recording environment.

EEG rhythms are commonly described using frequency ranges, including delta, theta, alpha, beta and gamma. These categories are useful descriptions, but they do not correspond to a simple one-to-one mapping between frequency and a particular thought, emotion or intention.

Research sources

This distinction is essential when evaluating patents that describe frequencies resembling EEG patterns. A signal can be generated at a frequency associated with a brain rhythm without necessarily reproducing that rhythm in the brain.

The presence of a frequency in both a stimulus and an EEG recording does not establish causation.

20.5. Clinical and forensic assessment must remain separate

In a documentary investigation, three kinds of information should be recorded separately:

- **The reported experience: what a person says they perceived, heard, felt or observed.**
- **The clinical findings: what a qualified clinician assessed, including symptoms, functioning, history and relevant alternative explanations.**
- **The technical evidence: measurements, exposure data, devices, signals, environmental observations or other evidence capable of testing a proposed physical mechanism.**

These categories can inform one another, but none automatically substitutes for the others.

A person's report is evidence of their reported experience. It does not alone identify the cause. A clinical diagnosis does not alone establish the cause of an external event. A patent does not alone establish that a particular device was present or used.

Forensic conclusion: Clinical literature helps define the limits of what is known about hypnosis, suggestion, brain activity and stimulation. It should be used to evaluate

specific claims, not to prejudge a person's account.

21. Neurotechnology and brain-computer interfaces: what is demonstrably possible?

Modern brain-computer interface research is relevant because it demonstrates that information about certain aspects of brain activity can be recorded and translated into outputs under defined conditions.

It is also an area where public descriptions can easily overstate what the technology does.

21.1. Brain-computer interfaces

A brain-computer interface (BCI) is a system that measures neural signals and uses signal-processing or machine-learning methods to produce an output.

Depending on the system, the output might represent a movement intention, a selection from a limited set of options or a command to a computer. Performance depends on the signal source, recording method, training, task, algorithms and the individual user.

A BCI is not equivalent to unrestricted mind reading. A system trained to classify a limited set of neural patterns is not thereby capable of decoding arbitrary private thoughts.

Research sources

These sources provide routes into peer-reviewed work on neural decoding and assistive technologies.

21.2. Neural decoding and speech research

Research into speech neuroprostheses has demonstrated that neural recordings can be processed to reconstruct aspects of intended speech in carefully defined experimental contexts.

These systems generally require access to neural signals using particular recording technologies, specialised algorithms, extensive calibration or training, and experimental conditions that differ substantially from ordinary remote observation.

Their achievements are significant because they show that some neural information can be decoded under specific circumstances.

Their limitations are equally important: successful decoding of a defined task does not establish universal access to thoughts, unrestricted decoding of memories or emotions, or the ability to transmit arbitrary messages into a person without a suitable stimulation mechanism.

Research searches

21.3. Reading neural activity is not the same as controlling it

The patent list contains proposals involving both monitoring and alteration of brain activity. These functions require separate analysis.

A recording system measures signals. A decoding system interprets signals. A stimulation system delivers energy or another intervention. A closed-loop system combines measurement and intervention through feedback.

Each step introduces additional technical requirements and potential sources of error.

For example, a system that detects a reproducible EEG pattern does not automatically identify the exact thought associated with it. A system that stimulates a brain region does not automatically produce a specific complex mental state. A feedback system requires evidence that its measurements and interventions produce the intended outcome.

Forensic application: Malech's US3951134A should be examined against these distinct requirements rather than treated as a single demonstrated capability because its title mentions both monitoring and altering brain waves.

22. Human research ethics: consent, autonomy and institutional accountability

The historical examination of psychological and neurological research is incomplete without an examination of the standards governing human experimentation.

This is particularly important when a project involves suggestion, altered consciousness, drugs, sensory stimulation, neural intervention or the possibility of undisclosed exposure.

22.1. The Belmont Report

Title: The Belmont Report: Ethical Principles and Guidelines for the Protection of Human

Subjects of Research Institution: US Department of Health and Human Services

Primary source:

The report identifies three central principles:

- **Respect for persons: recognising individual autonomy and providing additional protection where autonomy is limited.**
- **Beneficence: seeking to maximise potential benefits and minimise potential harm.**
- **Justice: considering the fair distribution of the burdens and benefits of research.**

These principles are directly relevant to investigations into historical experiments involving human participants.

They provide a framework for evaluating whether participants were adequately informed, whether risks were assessed, whether vulnerable people received appropriate protection, and whether the burdens of research were distributed fairly.

The Belmont Report is a modern US ethical framework. Its principles are useful for historical analysis, but the legal and regulatory standards applicable to a particular experiment depend on its date, jurisdiction, institutional setting and governing rules.

22.2. The Declaration of Helsinki

Institution: World Medical Association

Primary source:

The Declaration of Helsinki sets out ethical principles for medical research involving human participants, including informed consent, scientific validity, risk assessment, independent ethical review and protection of participants.

For the patent collection, the declaration provides a reference framework for evaluating research involving medical or physiological interventions.

A patent does not normally provide a complete record of the ethics review, consent procedures or participant protections associated with an experiment. If a patent's development involved human trials, the relevant research papers, institutional records and ethics documentation should be sought separately.

22.3. The Nuremberg Code

The Nuremberg Code emerged from the post-war legal and ethical response to human experimentation.

Reference source:

Its principles include the voluntary consent of participants, the scientific necessity of experiments, avoidance of unnecessary suffering and appropriate protections against foreseeable harm.

The historical significance is that ethical questions about experimentation are not dependent on whether the experimental hypothesis proved correct. A study can be scientifically unsuccessful and still involve serious ethical misconduct.

22.4. Research ethics and covert exposure

If a particular allegation concerns exposure to an undisclosed intervention, the forensic questions include:

- **Is there independent evidence that the intervention occurred?**
- **What device, substance or procedure was involved?**
- **Was a human participant involved in a research study?**
- **What consent, oversight and safety procedures applied?**
- **What documentation identifies the responsible organisation?**
- **Is there evidence of a departure from applicable ethical or legal requirements?**

The questions must be answered using the available records. An unethical research history elsewhere does not establish that a particular person was exposed to an intervention, but a documented exposure may raise serious ethical questions even before every aspect of its mechanism is understood.

22.5. Medical ethics and the right to bodily autonomy

The ethical principle of bodily autonomy is broader than the specific history of intelligence research. It concerns a person's ability to make informed decisions about medical interventions and the treatment of their body.

The and the provide identifiable sources for these principles.

In a forensic report, ethical conclusions should be linked to documented facts about the intervention, the participant and the applicable standards. This makes it possible to distinguish a documented breach from an unresolved question or a broader concern.

23. A cross-institutional research register

The following sources provide a foundation for extending the investigation across military, intelligence, scientific and ethical domains.

Military history and doctrine

- **Intelligence and historical investigations**
- **Psychology, psychiatry and neuroscience**
- **Human research ethics**
- **Radio-frequency exposure and auditory effects**

This register is a research starting point, not a claim that every institution or publication supports the same conclusion. Each source must be evaluated according to its authorship, purpose, methodology and relationship to the particular question being investigated.

24. How to establish whether a patented technology progressed to operational use

A central task for the next stage of this dossier is to distinguish patent disclosure from research, development, procurement, testing and deployment.

The following framework can be applied to every patent in the original list.

Stage 1 — Patent disclosure

Obtain the complete specification, claims, diagrams, priority records, inventors, assignees and cited prior art.

Question: What precisely was proposed?

Stage 2 — Experimental demonstration

Identify peer-reviewed experiments, technical reports, laboratory records and independent replication.

Question: Was the claimed physical or psychological effect demonstrated?

Stage 3 — Institutional development

Search for research contracts, funding announcements, technical evaluations, laboratory affiliations and documented transfers of intellectual property.

Question: Did an organisation develop or evaluate the technology?

Stage 4 — Acquisition and testing

Seek procurement notices, equipment records, trial protocols, test reports and documented field evaluations.

Question: Was a working system acquired or tested?

Stage 5 — Operational deployment

Seek contemporaneous operational records, corroborated testimony, technical logs, documented deployments and evidence connecting the system to a particular use.

Question: Was the technology actually used, and can the use be independently established?

24.1. Patent and ownership records

The provides official patent information and resources for investigating patent status and ownership records.

The is a convenient way to locate patent specifications, related applications, citations and family information. For legal status or ownership conclusions, the relevant official records should also be checked.

The investigation should record:

- **The original inventor or inventors.**
- **The original applicant and assignee.**
- **Subsequent assignments where documented.**
- **Priority, filing, publication and grant dates.**
- **Related applications and patent families.**
- **Cited prior art and later citations.**
- **Expiry or other status information where relevant.**
- **Any documentary evidence of licensing, transfer, procurement or development.**

A patent assignment establishes an ownership-related relationship. It does not automatically establish that the assignee manufactured, tested or deployed the invention.

24.2. Government procurement and research records

For US federal government contracts, useful starting points include:

These databases may identify relevant contracts, reports or institutional relationships, depending on the project, date, classification status and available metadata.

A search result that shares a keyword with a patent is only a lead. The underlying record must establish that the project concerns the same technology or a demonstrably related system.

For historical classified programmes, the surviving public record may be incomplete. Missing records should be recorded as a limitation, not treated as proof that the proposed technology was deployed or that it was never deployed.

24.3. Research and publication databases

The following resources can help identify independent research and related publications:

- — biomedical and life-sciences literature.
- — a full-text archive of participating biomedical and life-sciences journals.
- — bibliographic and DOI metadata.
- — scholarly works, authors, institutions and citation relationships.
- — reports from the National Academies.
- — defence-related scientific and technical information.

The strongest scientific record will combine the original patent with independent experimental publications and, where relevant, documented institutional records.

25. Part III — Consolidated findings

The military, intelligence, clinical, scientific and ethical records establish a substantial historical context for investigating technologies intended to influence human perception and behaviour.

The evidence supports the following conclusions.

1. Military psychological operations are historically documented. The US Army’s own history records organised psychological operations dating to the First World War. This establishes an institutional tradition of influence through communications and information, not a universal capability for direct neurological control.

2. Intelligence research into behaviour modification is documented. The CIA Reading Room and the 1977 US Senate hearing provide primary sources for examining MKULTRA and related research. These records raise important questions about human experimentation, institutional accountability and informed consent.

3. Subliminal influence claims require careful experimental evaluation. The National Research Council's 1991 review concluded that the evidence did not support claims that subliminal self-help tapes enhanced human performance. Its analysis distinguishes limited effects on simple experimental tasks from claims of lasting changes in complex behaviour.

4. Radio-frequency auditory effects have a published scientific history. Allan H. Frey's 1962 paper provides an identifiable primary scientific source. Its reported auditory effects should be distinguished from broader claims about arbitrary speech transmission or behavioural control.

5. Modern neuroscience demonstrates specific, limited ways of measuring and influencing neural activity. EEG, brain-computer interfaces and clinical brain-stimulation techniques provide established research contexts. Their capabilities depend on defined mechanisms and conditions; they do not establish unrestricted remote access to thoughts.

6. Human research ethics are independently important. The Belmont Report and the Declaration of Helsinki provide frameworks for examining informed consent, participant protection, risk and accountability. These questions remain relevant even when a research method's effectiveness is uncertain.

7. A direct link between a patent and operational deployment requires additional evidence. Patents, scientific papers, institutional histories, procurement records and operational reports answer different questions. They must be linked through identifiable documentary evidence rather than inference alone.

Final assessment of Part III

The historical record supports a serious investigation of military influence operations, intelligence behaviour-modification research, sensory stimulation, neural technologies and the ethical treatment of human participants.

It also supports a disciplined approach to claims about technological capability. A proposal, an experiment, an institutional programme and an operational deployment are

distinct evidentiary events. The investigation becomes stronger when those events are reconstructed individually and linked through original records.

Part III establishes the institutional and scientific framework for the next stage of the dossier.

The next section should focus on the remaining patent families and their technical relationships, followed by a deeper review of military-funded human-performance research, relevant psychiatric and psychology textbooks, peer-reviewed electromagnetic auditory studies, and the documentary standards needed to assess claims of real-world use.

FORENSIC RESEARCH DOSSIER

Part IV — Technical Cross-Examination of the Patent Record, Scientific Literature and Military Human-Performance Research

Continuation of Parts I–III

This part examines the evidentiary connections between the patents, their underlying physical mechanisms, the history of military human-performance research, and the scientific literature on sensory perception, neural stimulation and electromagnetic exposure.

The objective is to move beyond a chronological list of inventions and construct a traceable research record: what each technology was intended to do, what physical process it relied upon, what independent science says about that process, and what documentary evidence would be required to establish practical use.

One principle governs the entire examination: a patent can be important historical evidence even when the efficacy of its proposed mechanism remains unproven. The existence of a patent is not meaningless, but neither is it a substitute for experimental validation.

26. The patent record as evidence: what the documents can establish

26.1. Why the original specification matters

A US patent ordinarily contains several distinct evidentiary components:

- Title and abstract: a brief description of the proposed invention.
- Background: the problem the inventor says the invention addresses.
- Summary: an overview of the proposed solution.
- Detailed description: the implementation described by the inventor.
- Drawings: representations of components and their relationships.
- Claims: the legal scope of the protection sought or granted.
- References and citations: earlier documents considered or cited in the patent record.
- Assignment history: recorded transfers of patent ownership or associated rights.

These components should not be treated as interchangeable.

The title of US3951134A, for example, refers to remotely monitoring and altering brain waves. That title is historically significant because it identifies the inventor's stated objective. To determine what the inventor actually proposed, however, the full description and claims must be examined.

The same approach applies to US6470214B1 and US6587729B2, which concern the radio-frequency hearing effect. Their specifications can establish the signal-processing approach proposed by the inventors. The claims alone cannot establish that the technology was successfully demonstrated under independent experimental conditions.

Primary research resources

26.2. The difference between a patent claim and an experimental result

A patent application may describe an intended result, an expected benefit or a proposed explanation for a physical phenomenon. The presence of such language establishes that the proposition appeared in the document.

It does not, by itself, establish that the result was achieved.

To establish an experimental result, the investigator should seek the actual protocol, equipment specifications, measurements, controls, statistical analysis, reported failures and independent replication where available.

This distinction is particularly important for inventions whose intended effects concern human consciousness or behaviour. The generation of a signal can be demonstrated through electrical or acoustic measurements, whereas a claimed psychological effect requires a suitable human-subject or behavioural experiment.

A system may therefore be technically functional in one sense while failing to demonstrate its intended psychological outcome.

26.3. A patent is not necessarily a record of deployment

The following propositions require different evidence:

Proposition	Most relevant evidence
An invention was proposed	Original patent specification
An application was granted	Official patent record
Rights were assigned to an organisation	Assignment records
A prototype was constructed	Engineering records, laboratory reports or credible technical publications
A prototype was tested on participants	Trial protocols, research papers and participant

	records, subject to privacy protections
A government agency funded the work	Contract, grant or funding documentation
A system was acquired	Procurement and acceptance records
A system was operationally deployed	Contemporaneous operational records and corroboration
A system caused a particular effect	Measurements, causal analysis and independently assessable evidence

This framework is intended to prevent two opposite errors: dismissing a documented invention simply because operational use has not been established, and treating the invention's existence as proof that operational use occurred.

27. Military human-performance research: documented programmes and their actual objectives

Military research into cognition and human performance is a relevant area of investigation because it demonstrates that defence institutions have formally studied how people learn, maintain attention, make decisions, cope with stress and interact with technology.

The evidence must nevertheless be classified correctly. Research intended to improve a soldier's own performance is not the same as research intended to manipulate an adversary, and neither automatically establishes a capability for covert neurological intervention.

27.1. US Army Research Institute for the Behavioral and Social Sciences

Institution: US Army Research Institute (ARI).

Official research portal:

The Institute describes its role as scientific research and development supporting personnel assessment, leader development and military readiness. Its published research areas include personnel assessment, team assignment, performance, leadership and the development of military personnel.

This is relevant to the historical examination because it provides direct evidence of an institutional commitment to studying the relationship between human cognition, individual characteristics, training and performance.

It does not establish that ARI developed or deployed the specific patents in the original list. A connection would require a report, contract, named investigator, technical reference or other record linking a particular project to the relevant invention.

Key research terms

- Military cognitive performance
- Human factors engineering
- Cognitive workload
- Situational awareness
- Decision-making under uncertainty
- Human performance assessment
- Training transfer
- Behavioural measurement
- Stress and performance
- Team cognition

These terms are useful for locating actual research publications rather than treating the broad phrase “military mind control” as a technical category.

27.2. US Army research into cognitive performance under complex conditions

Official source:

The Army’s description identifies a research objective involving scientifically based training strategies and assessment tools to improve individual and collaborative performance in complex, stressful and ambiguous operational environments.

The significance is institutional and methodological. The programme describes research into how military personnel acquire knowledge, adapt to demanding conditions, make decisions and retain skills.

A forensic researcher should examine the actual assessment instruments, published studies and validation methods. The fact that an institution seeks to improve performance does not establish that its methods work equally well for all people or that they can be used to produce arbitrary changes in another person’s behaviour.

Evidence required for stronger conclusions

- Published research reports and study protocols.
- The measured cognitive outcomes.
- Participant numbers and selection criteria.
- Experimental and comparison conditions.
- Validation and replication results.
- Any documented links to particular technologies or contractors.

27.3. US Army Research Institute of Environmental Medicine

Institution: US Army Research Institute of Environmental Medicine (USARIEM).

Official source:

USARIEM describes research intended to prevent or reduce physical and cognitive performance decrements associated with military training and operational environments. Its stated areas include human performance, readiness and occupational stressors.

This research is relevant to questions about sleep, fatigue, attention, workload and the physiological demands of military operations.

It should not be confused with evidence of a covert system for controlling another person's mind. Research into maintaining alertness or reducing fatigue has a different purpose and mechanism from research proposing to transmit information directly through an electromagnetic auditory effect.

The distinction matters because the same general vocabulary—brain activity, performance enhancement, neural response or behavioural change—can describe very different scientific activities.

28. DARPA and neural-interface research: what official records establish

The Defense Advanced Research Projects Agency (DARPA) is a particularly relevant institution for this dossier because its published records document defence-sponsored research into neural interfaces and advanced neurotechnology.

The correct method is to examine individual programmes and their stated goals rather than infer a single overarching capability from the agency's involvement in neuroscience.

28.1. RE-NET — Reliable Neural-Interface Technology

Institution: DARPA Programme: Reliable Neural-Interface Technology (RE-NET) Official source:

DARPA states that it launched RE-NET in 2010 to address the need for high-performance neural interfaces capable of extracting information from the nervous system at sufficient resolution and speed to control advanced prosthetic limbs. The programme examined central and peripheral neural interfaces, signal quality and the durability of recorded neural information.

This is a documented example of defence-sponsored research into reading neural signals and translating them into machine control.

Its significance is substantial but specific: it establishes an institutional programme concerned with neural interfaces for assistive and prosthetic technology.

It does not establish unrestricted decoding of thoughts, remote reading of a person's private mental activity or the ability to control arbitrary human behaviour.

Relevant technical terms

- Neural signal acquisition
- Peripheral nerve interfaces
- Cortical microelectrodes
- Neural decoding
- Signal-to-noise ratio
- Prosthetic control
- Interface stability
- Long-term neural recording

Forensic comparison with the patent list: RE-NET provides a modern comparator for proposals involving brain-wave monitoring. Unlike a patent title alone, the programme description identifies a concrete engineering objective and explains why earlier methods were inadequate for the intended prosthetic-control applications.

28.2. N3 — Next-Generation Nonsurgical Neurotechnology

Institution: DARPA Programme: Next-Generation Nonsurgical Neurotechnology (N3) Official

source:
DARPA describes N3 as a programme intended to develop high-performance, bidirectional brain-machine interfaces that do not require surgical implantation. Its stated objectives include reading neural signals and delivering targeted interactions with neural tissue for human-machine applications. The programme description also discusses technical challenges involving signal scattering through skin, skull and brain tissue, as well as neural encoding and decoding.
This programme is particularly important for a forensic review because it expressly acknowledges the technical challenges of interacting with neural tissue noninvasively.
Its existence establishes that DARPA pursued research toward advanced nonsurgical neural interfaces. The programme's ambitious target specifications must not be mistaken for proof that every target was achieved, that the proposed performance was available in an operational system, or that the technology could be used remotely without the equipment and conditions required by its design.
A complete assessment would examine the programme's final technical reports, funded performers, publications, demonstrators and independent evaluations.

28.3. RAM — Restoring Active Memory

Institution: DARPA Programme: Restoring Active Memory (RAM) Official source:
DARPA describes RAM as a programme intended to investigate neurotechnology for improving memory formation and recall in people with traumatic brain injury. Its stated objective included developing and testing a wireless, fully implantable neural interface for human clinical use.
This record documents research into memory-related neural intervention. It does not establish a general capability to implant memories, erase arbitrary memories, extract every memory from an individual or alter a person's identity at will.
The distinction between restoring impaired function and controlling normal cognition is essential.
Questions for further examination

- Which neural signals were recorded?
- What intervention was delivered?
- Which memory outcome was measured?
- What were the participant-selection criteria?
- What clinical and ethical oversight applied?
- What outcomes were published, and which remained research objectives?

These questions help establish what the programme actually demonstrated rather than treating its broad objective as a completed capability.

28.4. Comparative assessment of DARPA programmes

Programme	Documented research objective	What the programme description does

		not establish
RE-NET	Extract neural information for high-performance prosthetic control	Unrestricted thought reading or remote behavioural control
N3	Develop advanced nonsurgical brain-machine interfaces	That every stated performance target was achieved or operationalised
RAM	Investigate neurotechnology to support memory function after brain injury	Arbitrary memory implantation, erasure or coercive control

The common thread is the investigation of neural interfaces and the translation of physiological information into defined technological outputs.

The differences in target, method and application are just as important. These are not interchangeable programmes, and they should not be treated as evidence of a single system.

29. National Academies reports: military behaviour, cognitive neuroscience and national security

Two National Research Council reports are especially valuable because they examine military human performance and emerging cognitive technologies from a broader scientific perspective.

29.1. Human Behavior in Military Contexts (2008)

Publisher: National Academies Press Institution: National Research Council DOI:

Full report:

This consensus study was commissioned by the US Army Research Institute for the Behavioral and Social Sciences. It examined areas of behavioural and social research relevant to military personnel, leadership, training and organisational performance. The research

agenda included intercultural competence, teams operating in complex environments, technology-based training, nonverbal behaviour, emotion and behavioural neurophysiology.
The report is significant because it documents an institutional demand for systematic scientific knowledge about human behaviour in military settings.
Its scope also helps clarify an important issue: military behavioural research encompasses a wide range of disciplines, many of which have no connection to covert neurological intervention.
Recommended research topics within the report

- Behavioural neurophysiology.
- Emotion and performance under stress.
- Nonverbal behaviour.
- Team interaction and group decision-making.
- Technology-based training.
- Human performance measurement.
- Organisational behaviour and leadership.

Forensic interpretation: This report supports the conclusion that military organisations have pursued formal scientific research into human behaviour and performance. It does not establish that every proposed method of influence is effective, or that this work is connected to the patents in the original list.

29.2. Emerging Cognitive Neuroscience and Related Technologies (2008)

Publisher: National Academies Press Institution: National Research Council DOI:
Full report:
This report is particularly relevant to the national-security dimension of the dossier.
It identifies several areas of emerging research that could have national-security implications, including methods for measuring indicators of psychological states and intentions, technologies or drugs that could alter physical or cognitive abilities, real-time brain imaging and computational models of brain functions.
The report demonstrates that the potential implications of cognitive neuroscience have been considered at the level of a formal scientific assessment for national-security purposes.
That is a meaningful documentary finding. It establishes institutional interest in the implications of emerging cognitive technologies, not the successful development of every capability discussed.
The report should be examined for the specific technologies it identifies, the state of the science at the time, the limitations it records and the recommendations it makes.
Questions for a detailed review

- Which technologies were already demonstrated in 2008?
- Which were emerging research directions rather than working capabilities?
- What limitations did the authors identify?
- Which applications were relevant to military or intelligence analysis?
- What subsequent publications document progress in the specific fields?
- Which predictions have been supported, revised or superseded by later evidence?

29.3. Measuring Human Capabilities (2015)

Institution: National Research Council DOI:

Full report:
This report addresses the assessment of individual and group performance potential for military accession. It examines topics including working memory, executive attention, cognitive biases, spatial abilities, emotional regulation, performance under stress and the possible contribution of neuroscience measures to assessment.
Its significance lies in the distinction between measuring human capability and altering human capability.
An assessment tool may estimate a person's performance on a defined task without providing a means of changing that person's underlying abilities. A neural measurement may correlate with a task outcome without uniquely identifying a thought, intention or personality trait.
These distinctions are important when interpreting patent descriptions that use terms such as neural performance, brain-wave analysis or behavioural change.
Consolidated finding: National Academies reports establish a substantial research interest in human performance, cognitive neuroscience and national security. They should be used as scientific context, not as substitute evidence for a particular device's operation.

30. Radio-frequency hearing: a deeper review of the peer-reviewed literature

30.1. James C. Elder and colleagues: Auditory Response to Pulsed Radiofrequency Energy (2003)

Journal: Bioelectromagnetics Publication: 2003 DOI:
Primary scientific source:
This paper reviews research on auditory responses to pulsed radio-frequency energy and the physical mechanism proposed to explain them.
The authors describe the radio-frequency auditory effect as a documented phenomenon in which suitable pulsed RF exposure can produce auditory sensations. The review discusses the characteristics of the perceived sounds, relevant exposure parameters and evidence supporting a thermoelastic explanation.
This is an important follow-up to Frey's 1962 publication because it evaluates a broader body of research rather than relying on a single early experiment.
The physical explanation is particularly relevant. According to the thermoelastic expansion theory, absorption of a sufficiently intense, brief RF pulse can produce a very small, rapid temperature change in tissue. The resulting expansion generates a pressure transient that can stimulate the cochlea.
This is different from radio waves carrying an ordinary sound wave through the air. The radio-frequency pulse initiates a physical process that can result in an auditory sensation.
Primary and review sources
Forensic interpretation: The research supports examining RF-induced auditory perception as a real scientific phenomenon. It does not justify treating all claims about microwave hearing, speech transmission, remote surveillance or psychological manipulation as established facts.

30.2. Lubner and colleagues (2020): review of auditory and vestibular symptoms

Article: Review of Audiovestibular Symptoms Following Exposure to Acoustic and

Electromagnetic Energy Outside Conventional Human Hearing Journal: Frontiers in Neurology Publication date: 28 April 2020 DOI: PubMed:
Full text:
This review examines literature concerning auditory and vestibular effects associated with infrasound, ultrasound, radio-frequency energy and magnetic resonance imaging.
The authors identify documented symptoms in some exposure settings while also emphasising substantial gaps in knowledge about the mechanisms and conditions responsible for particular adverse effects. They conclude that evidence for adverse human audiovestibular symptoms from acoustic and electromagnetic exposures outside conventional hearing is largely based on case series or small cohorts.
That qualification is essential. Reports of symptoms, controlled evidence of a physiological response, and proof that a particular source caused a particular person's symptoms are different evidentiary categories.
A retrospective report may be useful for identifying a potential problem, but it may lack calibrated exposure measurements or a suitable comparison group. A controlled experiment can provide stronger causal evidence for a defined mechanism, but its findings may not generalise to different equipment, exposure conditions or individuals.
Research questions arising from this review

- Which exposure types have been studied?
- What were the measured frequencies and intensities?
- Were exposures quantified or retrospectively inferred?
- Were outcomes objectively measured or primarily self-reported?
- Were comparison groups used?
- Which proposed mechanisms have experimental support?
- What uncertainties remain?

The review provides a useful framework for evaluating claims involving unusual sound perception and electromagnetic exposure without assuming either that every proposed mechanism is valid or that every reported symptom is imaginary.

30.3. The distinction between auditory perception and intelligible speech

An auditory sensation may resemble a click, buzz, hiss or chirp. That does not automatically mean that a system can transmit intelligible speech.
A speech-transmission claim requires evidence that the signal encodes speech information and that the recipient can recover that information with measurable accuracy under defined conditions.
The appropriate tests include controlled listening trials, predefined intelligibility measures, blinded or otherwise bias-controlled experimental conditions, repeatability and a description of the physical exposure.
For a patent such as US6587729B2, the relevant question is therefore not merely whether an RF-induced sound can occur. It is whether the particular apparatus described in the patent can reliably produce intelligible speech under the conditions claimed.
The further claim that such a system could remotely control thoughts or compel complex behaviour would require additional and substantially different evidence.

31. Forensic synthesis: what the combined evidence establishes

The documentary and scientific sources examined in this part support several distinct conclusions.

Military research into human performance is documented. The US Army Research Institute, USARIEM, DARPA and National Research Council publications provide identifiable records of research into cognition, stress, neural interfaces, behavioural assessment and human-machine interaction.

Intelligence and national-security institutions have examined emerging cognitive technologies. The 2008 National Research Council report Emerging Cognitive Neuroscience and Related Technologies explicitly examines scientific developments relevant to intelligence and national security. Its stated purpose and research scope are documented in the report itself.

Neural-interface technologies have defined, technically demanding applications. DARPA's RE-NET, N3 and RAM records illustrate different objectives involving neural signal acquisition, brain-machine interfaces and memory-related interventions. The individual programme descriptions establish research aims, not unrestricted operational capabilities.

Radio-frequency auditory effects have a substantial scientific literature. Frey's original paper, Elder's review and the 2020 audiovestibular review provide a traceable research pathway. They also identify important limitations in interpreting the effects and the evidence surrounding particular exposures.

Claims about subliminal influence must be matched to the relevant experimental evidence. The National Research Council's review found insufficient support for claims that subliminal self-help tapes enhanced human performance. It distinguished limited short-term effects on simple laboratory tasks from unsupported claims about long-term changes in complex behaviour.

A patent, a scientific result and an operational deployment are different forms of evidence. The forensic record becomes stronger when the relationship between them is demonstrated through original documents rather than assumed.

31.1. Recommended evidentiary notation for the master dossier

For the continuing investigation, each proposition should be assigned one of the following descriptions:

- Primary document located: the original patent, report, correspondence or official record has been identified.
- Institutional activity documented: an official record establishes that a programme, project or research activity existed.
- Scientific effect reported: a study reports an experimental observation or measured outcome.
- Independently replicated: independent studies reproduce the relevant result under comparable conditions.
- Technical capability demonstrated: a working system has been shown to achieve the specified result under documented conditions.
- Operational use documented: records establish actual use of the technology in a

defined setting.
- Causal attribution established: evidence supports the conclusion that the intervention caused the specific outcome under examination.
- Unresolved: the available evidence is insufficient to reach a reliable conclusion.

These descriptions should not be treated as automatic stages that every invention necessarily passes through. They are categories for identifying exactly what the evidence supports.

32. Research bibliography for Part IV

The following sources are especially useful for preserving the documentary trail established in this section.

Military human performance

- Defence neurotechnology
- Subliminal perception
- Radio-frequency auditory effects

End of Part IV.

The next part should examine the ethical, legal and evidentiary implications in greater depth, including human-subject protections, historical intelligence investigations, research governance and the standards required to connect a documented technology to a specific alleged incident. It should also begin a consolidated patent register that separates verified bibliographic facts from inventor claims, later interpretations and unresolved questions.

FORENSIC RESEARCH DOSSIER

Part IV — Technical Cross-Examination of the Patent Record, Scientific Literature and Military Human-Performance Research

Continuation of Parts I–III

This part examines the evidentiary connections between the patents, their underlying physical mechanisms, the history of military human-performance research, and the scientific literature on sensory perception, neural stimulation and electromagnetic exposure.

The objective is to move beyond a chronological list of inventions and construct a traceable research record: what each technology was intended to do, what physical process it relied upon, what independent science says about that process, and what documentary evidence would be required to establish practical use.

One principle governs the entire examination: a patent can be important historical evidence even when the efficacy of its proposed mechanism remains unproven. The existence of a patent is not meaningless, but neither is it a substitute for experimental validation.

26. The patent record as evidence: what the documents can establish

26.1. Why the original specification matters

A US patent ordinarily contains several distinct evidentiary components:

- Title and abstract: a brief description of the proposed invention.
- Background: the problem the inventor says the invention addresses.
- Summary: an overview of the proposed solution.
- Detailed description: the implementation described by the inventor.

- Drawings: representations of components and their relationships.
- Claims: the legal scope of the protection sought or granted.
- References and citations: earlier documents considered or cited in the patent record.
- Assignment history: recorded transfers of patent ownership or associated rights.

These components should not be treated as interchangeable.

The title of US3951134A, for example, refers to remotely monitoring and altering brain waves. That title is historically significant because it identifies the inventor's stated objective. To determine what the inventor actually proposed, however, the full description and claims must be examined.

The same approach applies to US6470214B1 and US6587729B2, which concern the radio-frequency hearing effect. Their specifications can establish the signal-processing approach proposed by the inventors. The claims alone cannot establish that the technology was successfully demonstrated under independent experimental conditions.

Primary research resources

26.2. The difference between a patent claim and an experimental result

A patent application may describe an intended result, an expected benefit or a proposed explanation for a physical phenomenon. The presence of such language establishes that the proposition appeared in the document.

It does not, by itself, establish that the result was achieved.

To establish an experimental result, the investigator should seek the actual protocol, equipment specifications, measurements, controls, statistical analysis, reported failures and independent replication where available.

This distinction is particularly important for inventions whose intended effects concern human consciousness or behaviour. The generation of a signal can be demonstrated through electrical or acoustic measurements, whereas a claimed psychological effect requires a suitable human-subject or behavioural experiment.

A system may therefore be technically functional in one sense while failing to demonstrate its intended psychological outcome.

26.3. A patent is not necessarily a record of deployment

The following propositions require different evidence:

Proposition	Most relevant evidence
An invention was proposed	Original patent specification
An application was granted	Official patent record
Rights were assigned to an organisation	Assignment records
A prototype was	Engineering records,

constructed	laboratory reports or credible technical publications
A prototype was tested on participants	Trial protocols, research papers and participant records, subject to privacy protections
A government agency funded the work	Contract, grant or funding documentation
A system was acquired	Procurement and acceptance records
A system was operationally deployed	Contemporaneous operational records and corroboration
A system caused a particular effect	Measurements, causal analysis and independently assessable evidence

This framework is intended to prevent two opposite errors: dismissing a documented invention simply because operational use has not been established, and treating the invention's existence as proof that operational use occurred.

27. Military human-performance research: documented programmes and their actual objectives

Military research into cognition and human performance is a relevant area of investigation because it demonstrates that defence institutions have formally studied how people learn, maintain attention, make decisions, cope with stress and interact with technology.

The evidence must nevertheless be classified correctly. Research intended to improve a soldier's own performance is not the same as research intended to manipulate an adversary, and neither automatically establishes a capability for covert neurological intervention.

27.1. US Army Research Institute for the Behavioral and Social Sciences

Institution: US Army Research Institute (ARI).

Official research portal:

The Institute describes its role as scientific research and development supporting personnel assessment, leader development and military readiness. Its published research areas include personnel assessment, team assignment, performance, leadership and the development of military personnel.

This is relevant to the historical examination because it provides direct evidence of an institutional commitment to studying the relationship between human cognition, individual characteristics, training and performance.

It does not establish that ARI developed or deployed the specific patents in the original list. A connection would require a report, contract, named investigator, technical reference or other record linking a particular project to the relevant invention.

Key research terms

- Military cognitive performance
- Human factors engineering
- Cognitive workload
- Situational awareness
- Decision-making under uncertainty
- Human performance assessment
- Training transfer
- Behavioural measurement
- Stress and performance
- Team cognition

These terms are useful for locating actual research publications rather than treating the broad phrase “military mind control” as a technical category.

27.2. US Army research into cognitive performance under complex conditions

Official source:

The Army’s description identifies a research objective involving scientifically based training strategies and assessment tools to improve individual and collaborative performance in complex, stressful and ambiguous operational environments.

The significance is institutional and methodological. The programme describes research into how military personnel acquire knowledge, adapt to demanding conditions, make decisions and retain skills.

A forensic researcher should examine the actual assessment instruments, published studies and validation methods. The fact that an institution seeks to improve performance does not establish that its methods work equally well for all people or that they can be used to produce arbitrary changes in another person’s behaviour.

Evidence required for stronger conclusions

- Published research reports and study protocols.
- The measured cognitive outcomes.
- Participant numbers and selection criteria.
- Experimental and comparison conditions.
- Validation and replication results.
- Any documented links to particular technologies or contractors.

27.3. US Army Research Institute of Environmental Medicine

Institution: US Army Research Institute of Environmental Medicine (USARIEM).
Official source:
USARIEM describes research intended to prevent or reduce physical and cognitive performance decrements associated with military training and operational environments. Its stated areas include human performance, readiness and occupational stressors.
This research is relevant to questions about sleep, fatigue, attention, workload and the physiological demands of military operations.
It should not be confused with evidence of a covert system for controlling another person's mind. Research into maintaining alertness or reducing fatigue has a different purpose and mechanism from research proposing to transmit information directly through an electromagnetic auditory effect.
The distinction matters because the same general vocabulary—brain activity, performance enhancement, neural response or behavioural change—can describe very different scientific activities.

28. DARPA and neural-interface research: what official records establish

The Defense Advanced Research Projects Agency (DARPA) is a particularly relevant institution for this dossier because its published records document defence-sponsored research into neural interfaces and advanced neurotechnology.
The correct method is to examine individual programmes and their stated goals rather than infer a single overarching capability from the agency's involvement in neuroscience.

28.1. RE-NET — Reliable Neural-Interface Technology

Institution: DARPA Programme: Reliable Neural-Interface Technology (RE-NET) Official source:
DARPA states that it launched RE-NET in 2010 to address the need for high-performance neural interfaces capable of extracting information from the nervous system at sufficient resolution and speed to control advanced prosthetic limbs. The programme examined central and peripheral neural interfaces, signal quality and the durability of recorded neural information.
This is a documented example of defence-sponsored research into reading neural signals and translating them into machine control.
Its significance is substantial but specific: it establishes an institutional programme concerned with neural interfaces for assistive and prosthetic technology.
It does not establish unrestricted decoding of thoughts, remote reading of a person's private mental activity or the ability to control arbitrary human behaviour.
Relevant technical terms

- Neural signal acquisition
- Peripheral nerve interfaces
- Cortical microelectrodes
- Neural decoding
- Signal-to-noise ratio
- Prosthetic control

- Interface stability
- Long-term neural recording

Forensic comparison with the patent list: RE-NET provides a modern comparator for proposals involving brain-wave monitoring. Unlike a patent title alone, the programme description identifies a concrete engineering objective and explains why earlier methods were inadequate for the intended prosthetic-control applications.

28.2. N3 — Next-Generation Nonsurgical Neurotechnology

Institution: DARPA Programme: Next-Generation Nonsurgical Neurotechnology (N3) Official source:

DARPA describes N3 as a programme intended to develop high-performance, bidirectional brain-machine interfaces that do not require surgical implantation. Its stated objectives include reading neural signals and delivering targeted interactions with neural tissue for human-machine applications. The programme description also discusses technical challenges involving signal scattering through skin, skull and brain tissue, as well as neural encoding and decoding.

This programme is particularly important for a forensic review because it expressly acknowledges the technical challenges of interacting with neural tissue noninvasively.

Its existence establishes that DARPA pursued research toward advanced nonsurgical neural interfaces. The programme's ambitious target specifications must not be mistaken for proof that every target was achieved, that the proposed performance was available in an operational system, or that the technology could be used remotely without the equipment and conditions required by its design.

A complete assessment would examine the programme's final technical reports, funded performers, publications, demonstrators and independent evaluations.

28.3. RAM — Restoring Active Memory

Institution: DARPA Programme: Restoring Active Memory (RAM) Official source:

DARPA describes RAM as a programme intended to investigate neurotechnology for improving memory formation and recall in people with traumatic brain injury. Its stated objective included developing and testing a wireless, fully implantable neural interface for human clinical use.

This record documents research into memory-related neural intervention. It does not establish a general capability to implant memories, erase arbitrary memories, extract every memory from an individual or alter a person's identity at will.

The distinction between restoring impaired function and controlling normal cognition is essential.

Questions for further examination

- Which neural signals were recorded?
- What intervention was delivered?
- Which memory outcome was measured?
- What were the participant-selection criteria?
- What clinical and ethical oversight applied?
- What outcomes were published, and which remained research objectives?

These questions help establish what the programme actually demonstrated rather than treating its broad objective as a completed capability.

28.4. Comparative assessment of DARPA programmes

Programme	Documented research objective	What the programme description does not establish
RE-NET	Extract neural information for high-performance prosthetic control	Unrestricted thought reading or remote behavioural control
N3	Develop advanced nonsurgical brain-machine interfaces	That every stated performance target was achieved or operationalised
RAM	Investigate neurotechnology to support memory function after brain injury	Arbitrary memory implantation, erasure or coercive control

The common thread is the investigation of neural interfaces and the translation of physiological information into defined technological outputs.

The differences in target, method and application are just as important. These are not interchangeable programmes, and they should not be treated as evidence of a single system.

29. National Academies reports: military behaviour, cognitive neuroscience and national security

Two National Research Council reports are especially valuable because they examine military

human performance and emerging cognitive technologies from a broader scientific perspective.

29.1. Human Behavior in Military Contexts (2008)

Publisher: National Academies Press Institution: National Research Council DOI:

Full report:

This consensus study was commissioned by the US Army Research Institute for the Behavioral and Social Sciences. It examined areas of behavioural and social research relevant to military personnel, leadership, training and organisational performance. The research agenda included intercultural competence, teams operating in complex environments, technology-based training, nonverbal behaviour, emotion and behavioural neurophysiology.

The report is significant because it documents an institutional demand for systematic scientific knowledge about human behaviour in military settings.

Its scope also helps clarify an important issue: military behavioural research encompasses a wide range of disciplines, many of which have no connection to covert neurological intervention.

Recommended research topics within the report

- Behavioural neurophysiology.
- Emotion and performance under stress.
- Nonverbal behaviour.
- Team interaction and group decision-making.
- Technology-based training.
- Human performance measurement.
- Organisational behaviour and leadership.

Forensic interpretation: This report supports the conclusion that military organisations have pursued formal scientific research into human behaviour and performance. It does not establish that every proposed method of influence is effective, or that this work is connected to the patents in the original list.

29.2. Emerging Cognitive Neuroscience and Related Technologies (2008)

Publisher: National Academies Press Institution: National Research Council DOI:

Full report:

This report is particularly relevant to the national-security dimension of the dossier.

It identifies several areas of emerging research that could have national-security implications, including methods for measuring indicators of psychological states and intentions, technologies or drugs that could alter physical or cognitive abilities, real-time brain imaging and computational models of brain functions.

The report demonstrates that the potential implications of cognitive neuroscience have been considered at the level of a formal scientific assessment for national-security purposes.

That is a meaningful documentary finding. It establishes institutional interest in the implications of emerging cognitive technologies, not the successful development of every capability discussed.

The report should be examined for the specific technologies it identifies, the state of the science at the time, the limitations it records and the recommendations it makes.

Questions for a detailed review

- Which technologies were already demonstrated in 2008?
- Which were emerging research directions rather than working capabilities?
- What limitations did the authors identify?
- Which applications were relevant to military or intelligence analysis?
- What subsequent publications document progress in the specific fields?
- Which predictions have been supported, revised or superseded by later evidence?

29.3. Measuring Human Capabilities (2015)

Institution: National Research Council DOI:

Full report:

This report addresses the assessment of individual and group performance potential for military accession. It examines topics including working memory, executive attention, cognitive biases, spatial abilities, emotional regulation, performance under stress and the possible contribution of neuroscience measures to assessment.

Its significance lies in the distinction between measuring human capability and altering human capability.

An assessment tool may estimate a person's performance on a defined task without providing a means of changing that person's underlying abilities. A neural measurement may correlate with a task outcome without uniquely identifying a thought, intention or personality trait.

These distinctions are important when interpreting patent descriptions that use terms such as neural performance, brain-wave analysis or behavioural change.

Consolidated finding: National Academies reports establish a substantial research interest in human performance, cognitive neuroscience and national security. They should be used as scientific context, not as substitute evidence for a particular device's operation.

30. Radio-frequency hearing: a deeper review of the peer-reviewed literature

30.1. James C. Elder and colleagues: Auditory Response to Pulsed Radiofrequency Energy (2003)

Journal: Bioelectromagnetics Publication: 2003 DOI:

Primary scientific source:

This paper reviews research on auditory responses to pulsed radio-frequency energy and the physical mechanism proposed to explain them.

The authors describe the radio-frequency auditory effect as a documented phenomenon in which suitable pulsed RF exposure can produce auditory sensations. The review discusses the characteristics of the perceived sounds, relevant exposure parameters and evidence supporting a thermoelastic explanation.

This is an important follow-up to Frey's 1962 publication because it evaluates a broader body of research rather than relying on a single early experiment.

The physical explanation is particularly relevant. According to the thermoelastic expansion theory, absorption of a sufficiently intense, brief RF pulse can produce a very small, rapid temperature change in tissue. The resulting expansion generates a pressure transient that can stimulate the cochlea.

This is different from radio waves carrying an ordinary sound wave through the air. The radio-frequency pulse initiates a physical process that can result in an auditory sensation.
Primary and review sources
Forensic interpretation: The research supports examining RF-induced auditory perception as a real scientific phenomenon. It does not justify treating all claims about microwave hearing, speech transmission, remote surveillance or psychological manipulation as established facts.

30.2. Lubner and colleagues (2020): review of auditory and vestibular symptoms

Article: Review of Audiovestibular Symptoms Following Exposure to Acoustic and Electromagnetic Energy Outside Conventional Human Hearing Journal: Frontiers in Neurology Publication date: 28 April 2020 DOI: PubMed:
Full text:
This review examines literature concerning auditory and vestibular effects associated with infrasound, ultrasound, radio-frequency energy and magnetic resonance imaging.
The authors identify documented symptoms in some exposure settings while also emphasising substantial gaps in knowledge about the mechanisms and conditions responsible for particular adverse effects. They conclude that evidence for adverse human audiovestibular symptoms from acoustic and electromagnetic exposures outside conventional hearing is largely based on case series or small cohorts.
That qualification is essential. Reports of symptoms, controlled evidence of a physiological response, and proof that a particular source caused a particular person's symptoms are different evidentiary categories.
A retrospective report may be useful for identifying a potential problem, but it may lack calibrated exposure measurements or a suitable comparison group. A controlled experiment can provide stronger causal evidence for a defined mechanism, but its findings may not generalise to different equipment, exposure conditions or individuals.
Research questions arising from this review

- Which exposure types have been studied?
- What were the measured frequencies and intensities?
- Were exposures quantified or retrospectively inferred?
- Were outcomes objectively measured or primarily self-reported?
- Were comparison groups used?
- Which proposed mechanisms have experimental support?
- What uncertainties remain?

The review provides a useful framework for evaluating claims involving unusual sound perception and electromagnetic exposure without assuming either that every proposed mechanism is valid or that every reported symptom is imaginary.

30.3. The distinction between auditory perception and intelligible speech

An auditory sensation may resemble a click, buzz, hiss or chirp. That does not automatically mean that a system can transmit intelligible speech.
A speech-transmission claim requires evidence that the signal encodes speech information and that the recipient can recover that information with measurable accuracy under defined conditions.

The appropriate tests include controlled listening trials, predefined intelligibility measures, blinded or otherwise bias-controlled experimental conditions, repeatability and a description of the physical exposure.

For a patent such as US6587729B2, the relevant question is therefore not merely whether an RF-induced sound can occur. It is whether the particular apparatus described in the patent can reliably produce intelligible speech under the conditions claimed.

The further claim that such a system could remotely control thoughts or compel complex behaviour would require additional and substantially different evidence.

31. Forensic synthesis: what the combined evidence establishes

The documentary and scientific sources examined in this part support several distinct conclusions.

Military research into human performance is documented. The US Army Research Institute, USARIEM, DARPA and National Research Council publications provide identifiable records of research into cognition, stress, neural interfaces, behavioural assessment and human-machine interaction.

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- Operational use documented: records establish actual use of the technology in a defined setting.
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32. Research bibliography for Part IV

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Military human performance

- Defence neurotechnology
- Subliminal perception
- Radio-frequency auditory effects

End of Part IV.

The next part should examine the ethical, legal and evidentiary implications in greater depth, including human-subject protections, historical intelligence investigations, research governance and the standards required to connect a documented technology to a specific alleged incident. It should also begin a consolidated patent register that separates verified bibliographic facts from inventor claims, later interpretations and unresolved questions.

Now I want you to create, going back into time and technology, only with verified, scientific, confirmed observations and experiments that confirm the technology and the level of significance and the advancement of this kind of technology. And go back to, say, the 1940s and give me a bit of an essay about how it came about, like with the Frey effect and how that came about early on, and only report on verified research that is confirmed on government databases or secret services or other psychological papers or psychiatric papers, police, telecommunications, anything like that. Scan the entire internet and give me a kind of a cumulative timeline about how the technology's developed over the decades and an assessment of where it might be now.

A Scientific History of Electromagnetic Auditory Effects, Neural Technology and Behavioural Influence, 1940–2026

A cumulative historical essay and evidence-based assessment of the technology's development, scientific significance and publicly documented state today.

Introduction: how radar, neuroscience and human-performance research converged

The modern history of technologies intended to measure, stimulate or influence human perception did not begin with a single invention. It emerged from several research traditions that developed at different times: wartime radar engineering, radio-frequency physics, experimental psychology, electroencephalography, military human-performance research, auditory science and, much later, computer-assisted neural interfaces.

These traditions sometimes overlap, but they are not the same technology. A radar transmitter can produce electromagnetic energy; an experimental exposure can produce a measurable biological response; a neural interface can translate recorded brain activity into a computer command; and a psychological experiment can demonstrate a limited change in perception or behaviour. Each represents a different scientific achievement.

One particularly important discovery is the phenomenon now commonly called the Frey effect, or microwave auditory effect. It concerns the perception of sounds associated with appropriately configured pulsed radio-frequency energy. Its history provides a useful starting point because it connects early radar observations with later controlled experiments and modern questions about the interaction between electromagnetic energy and biological tissue.

The evidence reviewed here comes from peer-reviewed scientific papers, systematic reviews, official government research programmes and publicly accessible institutional records. It is not literally possible to scan every item on the internet or every classified archive, so this is a cumulative review of identifiable sources rather than a claim to have exhausted all existing records. Where historical observations, experimental results and later interpretations differ in strength, they are identified separately.

The central question is: what has actually been demonstrated, how has the technology advanced, and what can the public evidence reasonably tell us about its present capabilities?

1. The 1940s: radar technology and the first reported microwave-hearing observations

During the Second World War, radar engineering advanced rapidly. Radar systems emitted radio-frequency energy and used returning signals to detect aircraft, ships and

other objects. High-power transmitters, large antennas and pulsed signals became increasingly important components of military technology.

In this environment, an unusual observation was reported in 1947 at the Airborne Instruments Laboratory in Mineola, New York. Personnel near a large radar antenna reportedly experienced auditory sensations that appeared to occur without a conventional external sound source. The accounts described sounds perceived within the head.

A later historical review in Frontiers in Neurology identifies these observations as an early stage in the history of the microwave auditory effect. It reports that the experience was subsequently described publicly and helped stimulate further investigation of the auditory properties of radio-frequency energy.

The distinction between observation and confirmation is important here. The 1947 accounts were historically significant, but they were not equivalent to a modern controlled experiment. An observation near a radar installation does not by itself exclude acoustic noise, equipment vibration, other environmental sources or errors in attributing the sensation to the transmitter.

Nevertheless, the reports raised a legitimate scientific question: could a person perceive an effect associated with electromagnetic exposure even when there was no ordinary airborne sound corresponding to the sensation?

The question was technically interesting because electromagnetic energy is not itself sound. Sound is a mechanical pressure wave, whereas radio-frequency energy is electromagnetic radiation. If RF exposure could reliably produce an auditory sensation, a physical process would have to connect the exposure to the auditory system.

That question would become the focus of the research later associated with Allan H. Frey.

Historical significance: The 1947 observations provide an early documentary starting point for the research history. They should be described as reported observations that preceded the better-known experimental work, not as proof that a complete auditory-transmission technology had already been developed.

Research source: .

2. The 1950s: from reported experience to a scientific problem

During the 1950s, the possibility of unusual auditory sensations associated with radar exposure began to attract more systematic attention. The scientific challenge was to determine whether the phenomenon could be reproduced, under what conditions it occurred, and which physical mechanism could explain it.

Researchers needed to distinguish several possibilities. The sound might have been produced by ordinary acoustic energy from the equipment; it might have been associated with electrical or mechanical artefacts; or the electromagnetic exposure might have initiated a physical process within the head that ultimately stimulated hearing.

Those explanations have different implications. An ordinary acoustic artefact would not establish a direct relationship between RF exposure and auditory perception. A reproducible auditory response associated with defined RF pulse parameters, by contrast, would establish a more specific physical effect.

The later historical review describes a technical account published in the 1950s that collated observations involving radar transmitters and helped frame the subsequent experiments.

This was the transition from an unusual report to a research question that could be investigated experimentally.

Why the distinction mattered

The relevant variables included the RF carrier frequency, pulse duration, pulse-repetition rate, peak power density, average power density, distance from the transmitter, the exposure geometry and the ambient acoustic environment.

A rigorous experiment would need to control or measure those variables, establish when a participant experienced an auditory sensation, and assess whether the response changed predictably when the signal parameters changed.

The distinction between peak and average power is particularly important for pulsed signals. A transmitter can emit brief, high-intensity pulses separated by intervals during which little or no energy is emitted. Its peak power and time-averaged power therefore describe different aspects of the exposure.

These technical details later became central to interpreting the Frey experiments. They also explain why a general statement such as “radio waves affect hearing” is inadequate: the response depends on the characteristics of the exposure and the

conditions under which it occurs.

Level of advancement by the end of the 1950s: the phenomenon had a history of reported observation and was becoming a defined scientific problem. That is a meaningful step in research development, but it is not evidence of a mature, reliable communication system.

3. 1961–1962: Allan H. Frey and the microwave auditory effect

The pivotal early publication was Allan H. Frey's 1962 paper, Human Auditory System Response to Modulated Electromagnetic Energy, published in the Journal of Applied Physiology.

Frey reported experiments in which normal-hearing and deaf participants perceived sounds associated with modulated electromagnetic energy. The article described sounds perceived without the usual external acoustic source and reported that the response depended on carrier frequency and modulation characteristics. The experiments also examined the influence of peak power density and acoustic background noise.

The paper was important because it moved the subject beyond the initial radar observations. It documented a laboratory investigation of the phenomenon, described relevant exposure parameters and considered possible locations and mechanisms of the sensory response.

Frey's participants reportedly described auditory sensations such as clicking, buzzing, hissing and knocking. The sounds were not necessarily experienced as ordinary sounds arriving through the air from a loudspeaker. Rather, the reported sensation appeared to originate within the head.

3.1. What Frey's experiment established

The strongest historically defensible conclusion is that Frey reported a reproducible auditory response associated with appropriately configured RF exposure under the experimental conditions he described.

The experiment established a serious scientific question and provided observations that subsequent researchers could investigate. It did not establish every later claim associated with the phenomenon.

In particular, it did not show that arbitrary speech, sentences or private thoughts could

be transmitted into a person's mind. Nor did it demonstrate a technology for unrestricted behavioural control.

Those propositions require additional experiments with distinct outcomes and substantially different evidence.

3.2. Why the phenomenon was technically significant

The result challenged a simple assumption that electromagnetic exposure could affect the human body only through familiar sensations such as warmth.

It suggested that, under particular conditions, electromagnetic energy could initiate a physical process that resulted in an auditory percept. This made the phenomenon relevant to auditory physiology, exposure science and the investigation of biological responses to pulsed electromagnetic fields.

The significance lay in the connection between the physical properties of the exposure and the resulting sensory experience.

3.3. What the experiment did not establish

A scientific observation is not a blank cheque for every application that might be imagined from it.

Frey's research did not by itself establish:

- **Reliable transmission of intelligible speech through RF exposure.**
- **A means of extracting a person's private thoughts.**
- **A general ability to write arbitrary information into the brain.**
- **A capacity to compel complex decisions or behaviour.**
- **The deployment of such a system by an intelligence agency or military unit against a particular person.**

These are separate propositions. Each requires direct supporting evidence rather than extrapolation from the existence of the auditory effect.

Assessment: The Frey experiments are a genuine milestone in the history of RF auditory research. Their importance rests on the reported phenomenon and the experimental questions they opened, not on unverified claims about what the phenomenon could ultimately be used to do.

4. The 1970s: investigating the mechanism and expanding the patent record

The decade after Frey's work brought a more detailed examination of the physical mechanism. Researchers were no longer concerned only with whether an auditory sensation could occur; they were also investigating how electromagnetic energy could produce it.

One influential hypothesis proposed that absorbed RF energy could cause a rapid, extremely small change in tissue temperature. The resulting thermoelastic expansion could generate a pressure wave that travelled through the head and stimulated the cochlea, the sensory organ responsible for hearing.

This explanation offered a bridge between electromagnetic energy and a familiar biological process: mechanical stimulation of the auditory system.

However, the mechanism was not settled by a single experiment.

4.1. The 1979 Science paper: testing the proposed mechanism

Paper: Allan H. Frey and Edward Coren, Holographic Assessment of a Hypothesized Microwave Hearing Mechanism.

Published: 12 October 1979, Science, volume 206, pages 232–234.

The authors examined a proposed mechanism in which pulse-modulated microwaves generated sound through mechanical motion of head tissue. Their holographic measurements did not show the predicted tissue motion, and they suggested an alternative location for the hearing effect.

This is an important scientific detail. A paper can support the existence of a phenomenon while challenging an explanation previously proposed for it.

The research history therefore should not be reduced to a simple progression in which every subsequent publication confirmed every earlier hypothesis. Scientific progress often involves reproducing an observation, refining its conditions, testing a proposed mechanism and rejecting or modifying parts of the original explanation.

4.2. Patent development and the difference between invention and proof

The 1970s and subsequent decades also saw patents describing proposed methods of influencing or measuring human sensory and neural activity.

These documents are useful for tracing the history of technical ideas. They can reveal how inventors described the problem, which physical mechanisms they proposed, which components they intended to use and which earlier inventions they cited.

But patent publication is not equivalent to an independent experimental confirmation.

A patent may describe a proposed device whose performance has not been independently replicated. It may also describe a genuine physical effect but propose applications that go beyond what has been demonstrated.

For the patents in the earlier register, the relevant investigation is therefore to compare the original specification with contemporaneous experiments and later reviews. The date of a patent can establish that an idea had been documented by that time; it cannot alone establish when a working prototype first existed or whether the invention was used outside a laboratory.

Assessment for the 1970s: The field advanced from the initial observation toward mechanistic investigation, while the patent record expanded the range of proposed applications. The strongest evidence concerns the scientific investigation of RF auditory perception. Broader claims require separate validation.

5. The 1980s: separate technological paths become clearer

By the 1980s, several different areas of research were developing in parallel. They sometimes shared language about perception, stimulation or behaviour, but their technical methods and evidentiary requirements differed substantially.

5.1. RF auditory research

Research continued to examine the conditions under which pulsed RF energy could produce auditory sensations and the physical process responsible.

The key questions were the exposure thresholds, pulse parameters, tissue interactions and auditory pathways involved. This was fundamentally a question about physical exposure and sensory physiology.

5.2. Subliminal perception and behavioural influence

Experimental psychology also investigated whether information presented below the threshold of conscious awareness could affect subsequent responses.

Subliminal priming and microwave auditory perception are different phenomena. Subliminal priming concerns how information is processed when awareness is limited or absent. The microwave auditory effect concerns auditory perception associated with pulsed electromagnetic exposure.

Research demonstrating a short-lived effect on a simple laboratory task does not establish a means of controlling complex behaviour. That distinction was later emphasised in the National Research Council's assessment of subliminal self-help claims.

The committee concluded that the evidence available to it did not support claims that subliminal self-help tapes enhanced human performance, while acknowledging research on limited short-term effects under controlled conditions.

5.3. Neural measurement and direct brain stimulation

Electroencephalography (EEG), developed much earlier, had by this period become an established means of recording electrical activity from the scalp. Other methods investigated ways of stimulating neural tissue for clinical or research purposes.

These technologies differ from RF auditory systems. EEG measures electrical potentials associated with brain activity; a neural-stimulation device attempts to alter neural activity through a specified physical or electrical intervention.

Neither category should be treated as synonymous with remote mind reading or unrestricted thought control.

Assessment for the 1980s: The period is best understood as one of parallel development. The science of sensory effects, psychological priming and neural measurement each advanced according to different experimental methods. Their shared interest in human perception does not establish a common operational system.

6. The 1990s: scientific review, human-performance research and growing scrutiny

The 1990s were important because the scientific and institutional record increasingly included formal reviews of claims about human performance, emerging technology and electromagnetic exposure.

6.1. National Research Council: evaluating claims about human enhancement

In 1991, the US National Research Council published In the Mind's Eye: Enhancing

Human Performance. The report considered a range of methods intended to improve performance, including psychological techniques and claims about subliminal self-help.

The committee's findings are useful because they demonstrate the difference between a research topic being worthy of investigation and a claimed effect being scientifically supported.

The report distinguished evidence of short-term changes on relatively simple tasks from claims that exposure to subliminal suggestions could produce lasting changes in complex actions, cognition or emotions. It did not consider the available evidence sufficient to support the commercial claims made for subliminal self-help tapes.

This was a significant methodological development: a systematic review could acknowledge limited experimental findings while rejecting claims that went beyond the evidence.

6.2. Government and public-health attention to electromagnetic fields

In 1996, the World Health Organization launched its International EMF Project to coordinate scientific assessment of possible health effects from electromagnetic-field exposure. Its work brought together research and expertise from national and international institutions.

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This research programme is relevant because widespread radio and telecommunications technologies made questions about exposure, biological effects and health increasingly important.

However, the existence of a government or international research programme does not establish that a particular harmful effect was found. The results of the investigations must be examined individually.

6.3. Why the 1990s matter to the broader history

The significance of the period lies in the growing distinction between:

- **Exposure to electromagnetic energy.**
- **Demonstrable biological effects.**
- **Adverse health outcomes.**
- **Effects on cognition or behaviour.**
- **Practical technological applications.**

- **Claims that exceed the available experimental evidence.**

Those categories remain essential to assessing the technology today.

7. The 2000s: a more developed physical explanation of microwave hearing

A major review published in 2003 by J. A. Elder and C. K. Chou examined the auditory response to pulsed radio-frequency energy.

Paper: Auditory Response to Pulsed Radiofrequency Energy.

The authors described RF hearing as an established phenomenon and reviewed the evidence concerning its physical mechanism and exposure characteristics. The review emphasised that the auditory response generally involves low-intensity perceived sounds and that a quiet environment is often necessary for the effect to be detected.

7.1. Thermoelastic expansion

The thermoelastic explanation became an important part of the scientific account.

In simplified terms, a brief pulse of RF energy is absorbed by tissue. The rapid energy deposition produces a tiny temperature change and corresponding expansion. That expansion generates a mechanical pressure wave, which can travel through the head and stimulate the cochlea.

The process does not require radio waves to behave as ordinary sound waves. Rather, electromagnetic energy initiates a mechanical event that the auditory system can detect.

This explanation also makes clear why the pulse structure matters. A continuous RF exposure and a train of brief pulses do not necessarily produce the same mechanical response, even if their time-averaged energy levels are comparable.

7.2. What the 2003 review added

The review brought together earlier work and examined the conditions associated with RF hearing. It helped place the phenomenon within a more coherent biophysical framework.

That is a meaningful advancement in scientific understanding. A proposed effect becomes more useful scientifically when researchers can describe its physical

mechanism, identify the variables that govern it and specify the conditions under which it occurs.

But the existence of a plausible physical mechanism does not automatically establish a useful communication device. A communication system would need to demonstrate that information could be encoded and reliably recovered, not merely that a listener could perceive a sound.

7.3. The wider neurotechnology context

During the same broad period, neural engineering increasingly combined sensors, signal processing and computer systems to translate measured physiological signals into defined outputs.

This work provided a foundation for modern brain-computer interfaces. It was a separate technological pathway from the RF auditory effect: the former generally depends on measuring and decoding neural signals, while the latter concerns a sensory response initiated by electromagnetic energy.

Assessment for the 2000s: The RF auditory effect had become a recognised topic of scientific review. Its physical explanation was more developed, but its implications still depended on specific exposure conditions and on the separate question of whether any proposed application could reliably perform its intended task.

8. The 2010s: defence-funded neural interfaces and the limits of non-invasive technology

During the 2010s, defence research programmes began publicly describing ambitious efforts to improve the speed, resolution and practicality of neural interfaces. These programmes are relevant to the question of technological advancement because they identify concrete engineering goals and acknowledge the obstacles that must be overcome.

They should not be confused with evidence that all the envisioned capabilities were achieved.

8.1. DARPA's RE-NET programme

The Defense Advanced Research Projects Agency (DARPA) developed the Reliable Neural-Interface Technology (RE-NET) programme to investigate high-performance neural interfaces, particularly for prosthetic control.

The intended function was to obtain useful information from the nervous system and translate it into control signals for a device. The challenge included acquiring sufficiently reliable neural signals and maintaining the performance of the interface.

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This is a concrete example of defence-funded work involving neural signal acquisition and machine control. Its objective was defined and measurable: improve the interface between a user's nervous system and a technological device.

The programme does not establish that neural activity can be interpreted without limits. A neural interface is designed for particular signals, tasks and outputs, and its performance depends on the recording method, training, signal quality and decoding algorithms.

8.2. DARPA's N3 programme

In March 2018, DARPA announced the Next-Generation Nonsurgical Neurotechnology programme. In May 2019, it announced awards to six research teams working on high-resolution, bidirectional brain-machine interfaces intended for use by able-bodied service members.

The programme's stated objectives included the ability to read neural signals and deliver targeted interactions with neural tissue without surgical implantation.

The official programme description also acknowledged the technical challenges of sending or receiving signals through the skin, skull and brain tissue. It described an ambitious target for reading from and writing to small regions of neural tissue with high temporal and spatial specificity.

This is a significant record of institutional intent and investment. It shows that researchers were attempting to move beyond existing non-invasive methods towards more capable interfaces.

The programme's stated performance goals, however, are not themselves proof that a finished device achieved every target. To assess actual outcomes, one must examine the final programme reports, published experiments, demonstrated prototypes and independent assessments.

8.3. Why the non-invasive problem is difficult

Signals measured at the scalp are affected by the intervening tissues, electrode placement, noise, movement and other sources of interference. Non-invasive methods generally have less direct access to neural activity than implanted electrodes.

The problem is not solved simply by applying more computing power. A system must first acquire a signal that contains enough reliable information about the intended task. Signal processing cannot recover arbitrary information that was never captured with adequate specificity.

Similarly, a method that can stimulate a broad region of tissue does not necessarily have the precision to select an individual thought, memory or decision.

This is why the N3 programme's own description of the physical challenges is important: it distinguishes a research objective from a mature capability.

Assessment for the 2010s: The field made progress in neural-interface engineering, and defence programmes publicly pursued increasingly ambitious goals. The evidence supports the existence of advanced research, not an assumption that unrestricted non-invasive access to human thought had been achieved.

9. The 2020s: systematic reviews, stronger evidence standards and the current technological picture

The 2020s have brought a useful combination of updated scientific reviews, more explicit evaluations of research quality and continued progress in neural-interface engineering.

This period is particularly important because it allows older claims about electromagnetic exposure and cognition to be compared with systematic reviews of many experiments rather than relying on a few historically significant papers.

9.1. The 2020 review of audiovestibular effects

Paper: Review of Audiovestibular Symptoms Following Exposure to Acoustic and Electromagnetic Energy Outside Conventional Human Hearing.

Published: 2020, Frontiers in Neurology.

This review examined literature concerning auditory and vestibular effects associated with infrasound, ultrasound, radio-frequency energy and magnetic resonance imaging.

Its historical section identifies the 1947 radar-related observations and the subsequent research by Frey. It also discusses the reported sounds and the proposed physical mechanisms.

The review's broader conclusion is cautious: evidence for adverse human audiovestibular symptoms from some exposures outside conventional hearing is limited, with much of the literature based on case series or small cohorts. It identifies substantial gaps in knowledge about how particular exposure frequencies, intensities and durations produce particular symptoms.

This matters because the existence of an auditory effect and the cause of a particular adverse symptom are different questions.

9.2. The 2024 systematic review of RF exposure and reported symptoms

Paper: Bosch-Capblanch and colleagues, The Effects of Radiofrequency Electromagnetic Fields Exposure on Human Self-Reported Symptoms: A Systematic Review of Human Experimental Studies.

Journal: Environment International, 2024.

The review included 41 human experimental studies with 2,874 participants. It assessed symptoms such as headache and sleep disturbance and examined whether participants could accurately identify whether RF exposure was present.

The authors reported no or small, statistically non-significant effects for the examined symptom outcomes, with evidence certainty varying across comparisons. They also reported that participants could not identify exposure status better than would be expected by chance in the studies assessed. The authors noted that the experiments were generally short-term and that the findings did not settle every possible question about longer-term or different exposure conditions.

The review concerns symptoms associated with RF exposure below regulatory limits. It does not test every conceivable exposure, every RF pulse structure or the particular technical parameters used in the original Frey experiments.

The appropriate interpretation is therefore specific: the reviewed experiments do not support claims that ordinary low-level RF exposure reliably produces the assessed symptoms. The result should not be incorrectly presented as a test of every technology described in historical patents.

9.3. The 2024 systematic review of RF exposure and cognitive performance

Paper: Pophof and colleagues, The Effect of Exposure to Radiofrequency Electromagnetic Fields on Cognitive Performance in Human Experimental Studies: Systematic Review and Meta-Analyses.

Journal: Environment International, 2024.

The review included 76 studies involving 3,846 participants, with quantitative meta-analyses using data from 50 studies and 2,433 participants. The studies examined several domains of cognitive performance following short-term RF exposure.

The meta-analyses did not find statistically significant effects of RF exposure compared with sham exposure across the assessed cognitive-performance outcomes. The authors nevertheless identified varying degrees of uncertainty in the evidence, depending on the outcome and the quality and consistency of the underlying studies.

This is one of the most relevant modern checks against overextending the historical RF auditory finding.

It indicates that the known existence of an RF-induced auditory phenomenon does not mean that ordinary RF exposure has been shown to reliably reduce cognitive performance or produce general changes in cognition.

It also illustrates why an experiment must be matched to the claim being made. A cognitive-performance review is not a direct test of RF hearing, and a study of RF hearing is not proof of a general cognitive effect.

9.4. The current state of brain-computer interfaces

Modern brain-computer interfaces (BCIs) are systems that acquire neural activity, process the recorded signals and translate them into outputs such as cursor movement, text selection or control of an assistive device.

A 2025 review of non-invasive EEG-based BCIs describes advances in hardware, machine-learning methods and multimodal approaches, while identifying signal degradation and real-world artefacts as continuing limitations.

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This is a genuine technological advancement: recorded brain activity can be translated into defined commands in appropriately designed systems. In certain medical research settings, neural interfaces have also supported communication and assistive control for people with severe disabilities.

But such systems do not read all thoughts indiscriminately. Their performance is tied to the signals recorded, the task being performed, the decoding model, the equipment and the conditions of use.

The ability to decode an intended movement or a constrained communication task is not equivalent to unrestricted access to private mental content. Nor does it establish a means of transmitting arbitrary thoughts into another person's brain through ordinary wireless exposure.

9.5. What public records establish about defence neurotechnology today

DARPA's published N3 programme description states that the programme is complete and remains available as a reference. Its public materials document the programme's objectives and research challenges, while the 2019 announcement records the participating research teams.

Those records establish that the US defence research community invested in advanced non-surgical brain-machine interface research.

To establish the exact capability achieved by the completed programme, the relevant evidence would be the final technical results, peer-reviewed publications, demonstrated prototypes and independent assessments. A programme's original target specification should not be presented as its confirmed final performance.

The same rule applies to other defence or intelligence programmes: a publicly documented research objective is evidence of research activity, but it is not by itself proof of a deployable capability.

10. Cumulative timeline: the verified development of the technology

The following timeline brings the main milestones together. The dates refer to documented observations, publications or programme announcements, not necessarily the date on which an underlying idea was first conceived.

1947

Early radar-related auditory observations

Reports from personnel near radar equipment describe unusual auditory sensations. These are historical observations, not yet equivalent to controlled experimental proof.

1950s

The phenomenon becomes a research question

Reports are collated and investigators begin examining the conditions under which auditory sensations appear in association with radar and RF exposure.

1961–62

Frey's experimental research

Allan H. Frey publishes experimental observations of auditory perception associated with modulated electromagnetic energy, helping establish the modern scientific literature on RF hearing.

1970s

Mechanism testing and patent development

Researchers test hypotheses about how the effect arises, including the proposed role of tissue motion. The 1979 Science paper challenges a predicted motion pattern, demonstrating that the mechanism remained open to investigation.

1980s

Parallel research fields develop

RF auditory studies, subliminal-perception experiments, EEG research and neural-stimulation research pursue different questions. They should not be treated as evidence of a single integrated technology.

1990s

Systematic assessment and exposure research

National Research Council reviews evaluate claims about subliminal enhancement,

while WHO launches its International EMF Project in 1996 to coordinate research into electromagnetic-field exposure.

2003

Major review of pulsed RF auditory responses

Elder and Chou review the auditory response to pulsed RF energy and its proposed mechanisms, consolidating an important body of experimental research.

2010s

Advanced neural-interface programmes

Defence research develops defined programmes for neural signal acquisition and brain-machine interfaces. DARPA announces N3 funding in 2018–19 to pursue high-performance nonsurgical interfaces.

2020

Audiovestibular evidence reviewed

A peer-reviewed review examines the evidence for symptoms associated with acoustic and electromagnetic energy outside conventional human hearing and identifies major gaps in the evidence for some exposure conditions.

2024

Modern systematic reviews

Systematic reviews assess RF exposure and reported symptoms, as well as RF exposure and cognitive performance. The reviewed experimental evidence does not show consistent harmful effects on the assessed outcomes from short-term exposure below guideline limits.

2025–26

Modern BCI development

Reviews describe ongoing improvements in non-invasive brain-computer interfaces, alongside persistent limitations in signal quality, spatial resolution and real-world

reliability.

11. Where might the technology be now?

A useful assessment needs to separate four technological categories.

A. Pulsed RF and the auditory effect

Documented phenomenon

The RF auditory effect has a published scientific history and a recognised proposed physical mechanism involving rapid energy absorption and thermoelastic processes. This is the strongest directly relevant connection between the 1940s radar observations and later experimental work.

The demonstrated phenomenon is narrower than many popular claims. It does not establish unrestricted speech transmission, arbitrary thought induction or remote behavioural control.

B. Brain-computer interfaces

Demonstrated for defined tasks

Modern BCIs can use measured neural activity to control particular computer functions or assistive devices. The extent of control depends on the interface, the task, the quality of the signals and the processing methods.

These systems represent genuine progress in neural engineering, but they are not equivalent to unrestricted access to another person's thoughts.

C. Neural stimulation and neuromodulation

Established, application-specific methods

Clinical and research technologies can stimulate neural tissue using defined methods and protocols. Their effects depend on the method, target, dose, equipment and clinical context.

The existence of neural stimulation does not establish that a remote system can select and impose arbitrary thoughts or complex decisions.

D. Claims of general remote influence

Not established by the reviewed evidence

The public research reviewed here does not establish a general operational technology that can remotely read private thoughts, insert arbitrary mental content or reliably control complex human behaviour through ordinary RF exposure.

Such a claim would require specific technical documentation, controlled experiments, independent replication and evidence of actual deployment.

These categories are related by their broad interest in human perception and neural function, but they are not interchangeable. The existence of a real effect in one category does not validate every claim made about the others.

12. The significance of the Frey effect in the history of technology

The Frey effect is historically significant because it provides a documented example of an unexpected sensory phenomenon emerging from the use of high-power electromagnetic equipment and subsequently becoming the subject of scientific investigation.

Its importance can be understood in three ways.

First, it expanded scientific interest in the interaction between pulsed electromagnetic energy and biological tissue. The initial observation generated a research question, and subsequent experiments examined the conditions under which the response occurred.

Second, it encouraged investigators to examine the physical mechanism rather than rely only on descriptions of subjective experience. The resulting literature considered the relationship between pulse characteristics, energy absorption, tissue motion and auditory perception.

Third, it illustrates the distinction between a demonstrated physical phenomenon and an application inferred from that phenomenon. The ability to produce a sound sensation under particular conditions is not, by itself, evidence of an effective speech-transmission system or a means of controlling human behaviour.

That distinction is not a minor technicality. It is the difference between a scientifically documented observation and an additional claim that requires its own proof.

13. Overall assessment

The public scientific record supports a history of genuine technological development beginning with radar-era observations, continuing through Frey's experiments, and extending into modern RF exposure research and neural-interface engineering.

The record is strongest where it identifies a specific effect, a defined mechanism, a published experimental method and measurable outcomes. It is less conclusive where a claim moves from the existence of a technology to an assertion of unrestricted capability, covert deployment or a particular effect on an individual.

The current public record supports the following conclusions:

- **RF auditory perception is a documented scientific phenomenon under particular conditions.**
- **The proposed mechanisms have been investigated experimentally, and the literature includes both mechanistic hypotheses and challenges to those hypotheses.**
- **Neural measurement, neural stimulation and brain-computer interfaces have developed into real, application-specific technologies.**
- **Defence institutions have publicly documented research programmes in neural interfaces and human-machine interaction.**
- **Modern systematic reviews do not support general claims that short-term RF exposure below guideline limits reliably produces the assessed cognitive-performance or self-reported symptom effects.**
- **The reviewed evidence does not establish unrestricted remote reading or control of human thoughts through RF exposure.**

The most defensible historical conclusion is that the underlying science is real, the technology has advanced substantially in several distinct directions, and the capabilities that can be confirmed today must be defined by the evidence for each individual system.

Principal sources for continued research

The next logical research step is a source-by-source technical chronology: identify the original 1947–1950s radar reports, verify their bibliographic details, follow the citations into Frey's 1961–62 work, then trace subsequent experiments and patent families against independent research and official programme records. That would make it possible to distinguish the earliest observations, the first controlled demonstrations, later technical advances and any documented applications without conflating them.

Yes continue with that and include as much verified scientific research published in

papers or available on the Internet and also includes some Online testimonies that corroborate the technology although possibly not proven because the civilians who are Victims Of the Technology

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Part VI — The Linked Witness-and-Science Register

A source-led forensic framework for electromagnetic auditory effects, anomalous health reports and civilian testimony

The next stage is to turn the historical narrative into a research instrument: two registers that can be examined independently and then cross-referenced. The scientific register asks what has been experimentally demonstrated, what mechanism has been proposed, and what remains uncertain. The testimony register preserves what identifiable people say they experienced, where and when they reported it, and what independent evidence exists. A third layer—the cross-reference register—records whether the two bodies of material genuinely converge or merely use similar language.

This distinction matters because a reported experience, a documented symptom, an experimentally demonstrated physical effect, and proof that a particular technology caused a particular person's experience are four different evidentiary propositions. They may overlap, but one does not automatically establish the next.

For example, Allan H. Frey's 1962 paper is a primary scientific source for reported auditory perceptions produced under particular pulsed-radiofrequency exposure conditions. It does not establish that every unexplained sound, voice-like perception, headache or sensation reported by a civilian has the same cause. Conversely, a civilian's account can be an important record of a person's experience even when the physical mechanism has not been identified.

The register below is designed to preserve both kinds of evidence without discarding testimony simply because causation is unresolved, and without representing a proposed explanation as a scientifically established conclusion.

19.1 Scientific register: primary sources and what they establish

The entries below are an initial source register, not a claim that every relevant paper or programme has been exhaustively identified. Each entry separates the research question from the conclusion that can reasonably be drawn.

S-01. Frey, 1962 — human auditory response to modulated electromagnetic energy

Primary source: Allan H. Frey, "Human Auditory System Response to Modulated Electromagnetic Energy," Journal of Applied Physiology, 17(4), 689–692 (1962).

Frey reported that people could perceive auditory sensations under particular conditions

involving pulsed electromagnetic energy. This work is a foundational source for what is now commonly called the Frey effect or microwave auditory effect.
Its importance is specific: it establishes a historical experimental research question concerning the interaction between pulsed radiofrequency energy and auditory perception. It does not establish that arbitrary speech can be transmitted into a person's mind at a distance, that thoughts can be remotely read, or that a particular unexplained experience was caused by radiofrequency exposure.
Register classification: Primary experimental paper; historically important; limited to the reported experimental conditions and interpretations.

S-02. Frey and Coren, 1979 — examination of a proposed mechanism

Primary source: Frey and Coren, "Holographic Assessment of a Hypothesized Microwave Hearing Mechanism," Science (1979).
This paper is relevant because scientific investigation of a reported effect must also test proposed explanations for how that effect occurs. Demonstrating or reporting a perception is not the same as establishing the underlying mechanism. Research that challenges a proposed mechanism is therefore part of the evidence base, not an inconvenience to be omitted.
Register classification: Mechanism-focused research; relevant to evaluating explanatory hypotheses, not proof of a deployed device.

S-03. Elder and Chou, 2003 — review of auditory responses to pulsed radiofrequency energy

Source: "Auditory Response to Pulsed Radiofrequency Energy," Bioelectromagnetics (2003).
This review examines the microwave auditory phenomenon and proposed mechanisms, including rapid energy absorption and thermoelastic processes. It is useful for distinguishing the established physical interaction being investigated from broader claims about what a system might be capable of transmitting.
Register classification: Scientific review; use it to map mechanisms and experimental findings, not as evidence that a particular person has been exposed.

S-04. National Research Council, 1991 — subliminal stimulation and human performance

Source: In the Mind's Eye: Enhancing Human Performance.
This report is useful for evaluating claims about subliminal material and behavioural enhancement. It helps distinguish the possibility of limited effects under specific experimental conditions from claims that hidden messages can reliably produce broad, complex or involuntary behavioural outcomes.
Register classification: Government-requested scientific review; relevant to subliminal perception and performance claims, but not a direct study of microwave auditory effects.

S-05. National Academies, 2020 — anomalous health incidents among US government personnel

Source: An Assessment of Illness in U.S. Government Employees and Their Families at Overseas Embassies.

This report is central to the witness-and-science register because it considered reported experiences alongside clinical information and possible physical mechanisms. The committee examined reports involving sudden sounds and sensations as well as symptoms such as dizziness, tinnitus, head pressure, balance problems and cognitive difficulties.
The committee considered several possible explanations. It judged directed, pulsed radiofrequency energy plausible for some of the distinctive acute symptoms described, while identifying substantial limitations and uncertainty. The report did not prove that one mechanism explained every case, establish the existence of a particular deployed device, or demonstrate that every reported symptom resulted from an external technological exposure.
Register classification: Expert committee assessment; combines reported accounts and medical information with a discussion of plausible mechanisms and evidentiary limitations.

S-06. ODNI expert panel, 2022 — possible mechanisms and evidentiary gaps

Source: Intelligence Community expert-panel assessment of anomalous health incidents.

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The panel considered whether different classes of external stimuli could account for particular features of reported incidents. Its discussion is relevant to the question of plausibility, but a mechanism being considered plausible is not equivalent to demonstrating that it occurred in a specific case. The report also discusses information gaps and the possibility that different cases might have different explanations.
Register classification: Government expert assessment; mechanism evaluation, not individual case attribution.

S-07. ODNI, March 2023 — updated intelligence-community assessment

Source: Updated Assessment of Anomalous Health Incidents.
This source must be included alongside, not substituted for, the clinical and scientific reports. It addresses intelligence-community judgements about possible foreign involvement and the evidence available to those agencies. Those are different questions from whether a person experienced a symptom, whether a symptom was medically documented, or whether a physical mechanism is possible in principle.
Register classification: Intelligence assessment; relevant to attribution and threat assessment, not a controlled experiment.

S-08. ODNI, January 2025 — further updated assessment

Source: Updated assessment of anomalous health incidents, reflecting the intelligence community’s position as of December 2024.

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This update should be compared with the 2023 assessment and with other public reports, rather than read in isolation. Its judgements concern the likelihood of foreign involvement and the evidence for a responsible actor. Differences in confidence and assessment do not, by themselves, identify a cause for every individual report.
Register classification: Updated intelligence assessment; use for the evolution of attribution judgements, while retaining the stated confidence levels and limitations.

S-09. WHO-commissioned systematic review, 2024 — radiofrequency exposure and

self-reported symptoms

This review is important because it addresses a broader population and a different research question from the anomalous-health-incident investigations. It examined human experimental studies of radiofrequency exposure and self-reported symptoms. The review did not find significant symptom effects from the acute exposures assessed under its included study conditions.

That finding should be reported accurately, but its scope must also be respected: evidence about the exposures, durations and outcomes included in a review does not automatically resolve every possible exposure scenario, every chronic health question, or every individual case.

Register classification: Systematic review; relevant to controlled human exposure evidence and the interpretation of symptom reports.

S-10. WHO-commissioned systematic review, 2024 — radiofrequency exposure and cognition

This review examined human studies of radiofrequency exposure and cognitive performance. Its results do not support a general claim that the radiofrequency exposures studied reliably impair cognitive performance. It is a necessary counterweight when evaluating claims that exposure to radiofrequency energy routinely produces memory impairment, loss of control, or other complex neurological outcomes.

Register classification: Systematic review; relevant to cognition under the study conditions, not a direct investigation of all reported anomalous incidents.

19.2 Testimony register: identifiable accounts and formal records

The testimony register should not require a witness to prove a mechanism before their account can be preserved. Its job is to record what the person actually reported, the setting in which the account was collected, and what independent information can be checked.

For online civilian testimony, I recommend a higher standard of source traceability than simply collecting posts that mention “microwave weapons,” “voices,” “remote influence” or similar terms. An identifiable interview, a published first-person account, a formal hearing transcript or a medical case report can be examined and cited. An anonymous post can still be archived as an online statement, but it ordinarily cannot independently establish the identity of the writer, the circumstances described, the exposure itself or its cause.

W-01. US government personnel reporting sudden directional sounds and sensory symptoms

Primary clinical source: Swanson and colleagues, “Neurological Manifestations Among US Government Personnel Reporting Directional Audible and Sensory Phenomena in Havana, Cuba,” JAMA (2018).

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The paper documents clinical evaluations of personnel who reported unusual auditory and sensory experiences. The reported symptom picture included auditory phenomena alongside symptoms involving balance, vision, sleep, headache and cognitive function. The article is valuable because it moves beyond an anonymous internet description: it is a published

medical account of a defined group of people who sought evaluation.
It must nevertheless be read as a clinical case series. Clinical assessment can document symptoms and identify findings, but it does not necessarily determine the initiating cause. The fact that several people report similar experiences can justify investigating common exposures; similarity alone cannot identify the source of those experiences.
What this entry supports: identifiable reports existed; medical evaluations were undertaken; symptoms and some clinical findings were described in a peer-reviewed publication.
What it does not independently establish: that every person experienced the same exposure, that a particular device caused the symptoms, or that the same explanation applies to unrelated civilian accounts.

W-02. First-person accounts considered by the National Academies

Primary source: National Academies, An Assessment of Illness in U.S. Government Employees and Their Families at Overseas Embassies (2020).
This report is especially relevant to a testimony register because it considered accounts of the circumstances in which symptoms began, including reports of sudden sounds or sensations and, in some cases, a perceived directional quality. The committee considered those descriptions alongside clinical information and competing explanations.
The report's treatment of the accounts is more nuanced than either of two extreme interpretations: that testimony alone proves a technological attack, or that uncertainty about cause makes the testimony worthless. The accounts provided material for investigation, while the committee separately assessed the plausibility and limits of possible mechanisms.
What this entry supports: testimony was part of a formal expert assessment and was considered alongside other available evidence.
What it does not establish: that every reported sensation had a confirmed external source, or that one mechanism explains every affected individual.

W-03. The intelligence-community expert panel's assessment of reported experiences

Primary source: ODNI, 2022 expert-panel material on anomalous health incidents.
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This is a separate record from the National Academies' medical assessment. The panel considered the reported features of the incidents in evaluating possible mechanisms and explicitly discussed gaps in the available information.
For the register, its value is not that it conclusively establishes one explanation. Its value is that it documents how a government expert panel assessed the reports and distinguished between possible physical mechanisms and the limitations of available evidence.
The register should preserve the panel's actual wording and confidence qualifications rather than compressing them into a statement such as "the government proved directed-energy attacks." That stronger claim would go beyond what the source establishes.

W-04. The 2023 intelligence assessment and the question of attribution

Primary source: ODNI, March 2023.
This entry is essential even when it conflicts with a witness's proposed explanation. It records intelligence judgements concerning possible foreign responsibility, not a universal

finding about the truth or falsity of every reported symptom.
The correct comparison is therefore between separate questions:

- Did the person report the experience?
- Was the experience documented by a clinician, investigator or other independent source?
- Is there evidence for a particular physical mechanism?
- Is there evidence connecting that mechanism to the individual incident?
- Is there evidence identifying an actor responsible for the incident?

A source that addresses the last question cannot automatically settle the first three.

W-05. The 2025 intelligence update and the preservation of minority assessments

Primary source: ODNI, updated assessment reflecting the position as of December 2024, published in January 2025.

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The register should record the assessment's majority and minority judgements, their confidence levels, and what each judgement actually says. Where an assessment describes a possibility concerning a small subset of incidents, it must not be rewritten as a finding that the explanation applies to all reported cases. Equally, a low-confidence judgement should not be presented as if it were a confirmed event.
This is particularly important when building a historical record: disagreement between assessments is itself part of the documentary history and should remain visible.

W-06. Congressional oversight and competing interpretations

Source: US House Permanent Select Committee on Intelligence, unclassified report concerning anomalous health incidents.

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A congressional oversight report can provide useful questions, witness material, criticism of agency processes and an alternative interpretation of evidence. It is not interchangeable with a controlled experiment or a clinical study. Its claims should be attributed to the committee and checked against the underlying documents wherever possible.
Register classification: Government oversight record; useful for tracing disputes over evidence, investigative adequacy and interpretation.

19.3 How to add civilian online testimony without overstating corroboration

Civilian accounts are a potentially important part of this research, but they need a clear source policy. Otherwise, the dossier can become impossible to audit: one person's firsthand report gets repeated by another website, a third website cites the repetition, and the same story begins to look like three independent sources.
The following categories make the difference visible.
Category A — Formal first-person testimony
A person describes their experience in a hearing, official inquiry, sworn statement or identifiable investigation. Record the transcript or original document, not merely a summary.
Category B — Identifiable published interview or first-person account
The original interview or article identifies the speaker sufficiently to establish the provenance of the report. Publication verifies that the account was published, not

necessarily that every factual claim within it is independently established.

Category C — Identifiable online self-report

A public post, blog, forum entry or video in which the speaker describes their own experience. Preserve the original URL, date, author or account name as displayed, and the exact words relied upon. Treat identity and circumstances as unverified unless independently established.

Category D — Independently corroborated technical or clinical record

Medical findings, instrument measurements, contemporaneous records, verified environmental data or other evidence that independently supports a specific part of the account. State exactly what it corroborates.

These categories are not a ranking of whose suffering matters. They are a way to distinguish the type of source and the kind of proposition it can support. A person whose account falls into Category C may be describing a deeply distressing experience. The correct record is that the person reported it, with the causal explanation remaining unverified if no independent evidence establishes it.

What counts as independent corroboration?

Suppose two people report hearing an unusual sound at roughly the same time. That is a reason to examine whether their accounts are independent and whether any shared environmental evidence exists. It is not, on its own, proof of a radiofrequency source. If both accounts were copied from the same post, they are not independent witnesses. If a contemporaneous recording exists, its provenance and technical characteristics need to be examined before it can be said to corroborate the sound's source.

Similarly, a clinician's note may independently corroborate that a person reported tinnitus, dizziness or headache on a particular date. It does not necessarily corroborate the person's belief about what caused the symptoms. A measurement can establish that a signal was present under the measured conditions; connecting that signal to a person's symptoms requires additional evidence.

The register should therefore have separate fields for corroboration of the report, corroboration of the symptom, corroboration of an exposure, and corroboration of causation. This prevents the strength of one kind of evidence from silently being transferred to another.

19.4 Identifiable first-person account: Marc Polymeropoulos

W-07 — First-person account of an anomalous health incident reported in Moscow

Former CIA officer Marc Polymeropoulos has publicly described becoming acutely unwell during a trip to Moscow in December 2017. His account includes severe vertigo, tinnitus, nausea and a persistent headache. He has also described the subsequent difficulty of obtaining medical attention and navigating the consequences for his working life. These are identifiable, published first-person accounts rather than anonymous online claims. The CNN transcript and the first-person essay allow a researcher to examine his own description, while the CBS report provides another published record of the account.

For the register, the central distinction is between corroborating that Polymeropoulos publicly reported these symptoms and establishing why they occurred. His testimony

documents his account and its reported consequences. The similarity between his symptoms and symptoms described by other people may be relevant to comparative research, but it does not by itself demonstrate a shared exposure, identify a device or establish responsibility.

This is an important example of why the register should preserve an individual's own words and chronology rather than replacing the account with either an unqualified technological explanation or a dismissive summary.

W-07 — Evidence assessment

Established from the publications

Polymeropoulos made an identifiable public account of acute symptoms, their reported onset during a Moscow trip, and their subsequent effects on his health and career.

Potentially independently verifiable

Contemporaneous medical records, travel records, clinical examinations and other original records, if available and appropriately authenticated.

Not established by the account alone

The identity of a physical source, the presence of radiofrequency or acoustic exposure, the operation of a weapon, or the identity and intent of a responsible actor.

This account is not a civilian case in the strictest sense: Polymeropoulos was a government officer. It is included as a model for documenting identifiable testimony. A separate civilian collection should use the same standards without assuming that government personnel and members of the public necessarily experienced the same circumstances.

19.5 The critical scientific counterpoint: what the microwave auditory effect does—and does not—demonstrate

The witness register should be cross-referenced to the relevant scientific literature, including research that supports the existence of an effect and research that questions whether it explains a wider group of reported incidents.

S-11. Foster and Finch, 1974 — evidence for thermoacoustic auditory stimulation

Source: "Microwave hearing: evidence for thermoacoustic auditory stimulation by pulsed microwaves," Science (1974).

Foster and Finch reported that pulsed microwave energy could generate acoustic transients in water. Their measured transients were sufficient, under the conditions studied, to provide an explanation for the clicks associated with microwave auditory perception. This work is relevant to the physical-mechanism question: it offers a way to connect absorbed electromagnetic energy to an acoustic pressure transient rather than assuming that radio waves directly carry ordinary sound into the brain.

The distinction is important for claims about voices or complex messages. The microwave auditory effect is not synonymous with unrestricted transmission of intelligible speech, and the evidence for perception of clicks, buzzes or other pulse-related sensations cannot simply be extended to claims about remote speech, thought reading or behavioural control.

S-12. Lin, 2007 — review of microwave-pulse hearing, mechanisms and thresholds

Source: James C. Lin and Zhangwei Wang, "Hearing of microwave pulses by humans and

animals: effects, mechanism, and thresholds," Health Physics (2007).
This review discusses human and animal observations, possible mechanisms and exposure thresholds. It describes a thermoelastic account in which absorption of a short pulse produces a pressure wave that travels by bone conduction and stimulates the inner ear. This is a distinct mechanism from a system that directly reads or writes neural information.
The review is useful for specifying the physical phenomenon under discussion, but a literature review does not establish that any given person was exposed to the required pulse conditions. To support an individual exposure claim, the research would need incident-specific evidence about the signal, its parameters, the environment, and the person's exposure.

S-13. Foster, Garrett and Ziskin, 2021 — can the microwave auditory effect be weaponized?

Source: "Can the Microwave Auditory Effect Be 'Weaponized'?" Frontiers in Public Health (2021).
This paper is directly relevant to the boundary between a known physical phenomenon and a proposed operational use. The authors acknowledge the microwave auditory effect and its thermoacoustic explanation, then assess whether it could plausibly explain reported anomalous health incidents. They argue that the available evidence does not support attributing those incidents to microwave exposure and discuss practical and evidentiary uncertainties surrounding weaponization.
The article should not be represented as proving that no technology could ever produce harmful effects. Nor should its discussion of the known auditory effect be represented as proof that a weapon caused reported incidents. It is a scientific argument about plausibility, practical constraints and the state of the public evidence.
That makes it a particularly important entry for an impartial dossier: the existence of a physical phenomenon and the claim that it explains a particular set of human experiences must be assessed separately.

S-14. Lin, 1976 — comparison of proposed transduction mechanisms

Source: "Microwave Auditory Effect—A Comparison of Some Possible Transduction Mechanisms," Journal of Microwave Power (1976).
This paper compares possible mechanisms using mathematical approximations and simplifying assumptions. It is useful as a record of how researchers reasoned about the physical process, rather than merely cataloguing subjective descriptions. As with any theoretical analysis, its conclusions must be understood within the assumptions and experimental evidence on which they rest.
Taken together, these sources provide a defensible scientific foundation for the register: there is a documented research literature on the auditory perception of pulsed microwave energy, and there is continuing debate about the extent to which that phenomenon can account for particular reported health incidents. The literature does not justify collapsing every report of a voice, sound, sensation or neurological symptom into a single technological explanation.

19.6 The cross-reference register: connecting witnesses to research without inventing a connection

The third register is where the dossier becomes analytically useful. It should not simply place a scientific paper beside a testimony and imply that one confirms the other. It should state the precise relationship, if any, between them.

The following is a proposed working matrix.

Cross-reference	Question to investigate	What would count as meaningful support?
Reported sound ↔ microwave auditory research	Does the description match the experimentally reported phenomenon?	A sufficiently specific description, relevant exposure information and a scientifically plausible match. Similar wording alone is weak evidence.
Sudden symptoms ↔ clinical case reports	Were the symptoms independently assessed?	Dated medical records, examination findings or contemporane-ous clinical notes, with appropriate consent and

		privacy protection.
Alleged exposure ↔ physical measurement	Was an external signal present?	Authenticated measurements with calibrated instruments, frequency, power or field-strength information, timing and documented measurement conditions.
Multiple witnesses ↔ shared incident	Are the accounts independent and contemporane-ous?	Separate original statements, independently established timing and location, and records that do not merely repeat a common source.
Patent ↔ alleged device	Does the patent disclose a	The original patent, claims,

	technically relevant concept?	drawings and technical description, followed by evidence of a working implementation if one is alleged.
Government programme ↔ operational use	Was a concept actually built, tested or deployed?	Programme records, contracts, test reports, procurement records or other evidence of the particular stage being claimed.
Online account ↔ published research	Did the research examine the same phenomenon and conditions?	A defensible comparison of the research methods, exposure parameters and reported outcomes. A thematic

resemblance is not causal corroboration.

A further distinction is necessary when patents are involved. A patent can document that an inventor sought protection for a described concept and can provide technical detail about that concept. It does not, by itself, establish that the device was built, worked as claimed, was funded by a military or intelligence agency, or was used against a particular person. Those are separate propositions requiring separate evidence.

Suggested record format

Use the following fields for every entry. The template is designed to be copied into a spreadsheet, document or research database.

Witness record template

One record per original account; create separate linked records for later interviews or republications.

Record ID: W-___

Person or displayed account name:

Original source URL:

Source type: Formal testimony / published interview / online self-report / clinical record / other

Publication date and reported incident date:

Location as reported:

First-person account: Quote only the words necessary for analysis; retain context and link to the original.

Symptoms or sensations reported: Use the witness's own wording first; any standardized symptom categories should be added separately.

Independent corroboration: Identify each record and exactly what it supports.

Alternative explanations considered: Record only explanations actually discussed in the source or supported by relevant evidence.

Proposed mechanism: The witness's explanation, separately identified from any scientific assessment.

Causal status: Established / supported but incomplete / plausible but unverified / undetermined / contradicted by specified evidence.

Confidence and limitations: Explain why the classification was selected.

Linked scientific records: S-, S-

Researcher's notes: Distinguish source facts from interpretation.

Copy witness template

Scientific record template

One record per paper, report, patent, programme document or technical dataset.

Record ID: S-___

Full bibliographic citation:
Original source URL or DOI:
Publication date and source type:
Research question or technical purpose:
Method or evidence examined:
Principal finding:
Exact proposition supported:
What the source does not establish:
Exposure or technical parameters: Record them only if the source provides them.
Limitations and uncertainty:
Related witness records: W-, W-
Related patents or programmes: P-, G-
Researcher's assessment: Separate the source's conclusion from your interpretation.
Copy scientific template

19.7 Evidence-status language for the public-facing dossier

The wording used in the register should remain consistent from one entry to another. I recommend the following distinctions:

- Documented report: A person or source made the stated report, and the original account can be located.
- Independently corroborated detail: A separate source supports a specified part of the account, such as the date of a medical visit or the existence of a recording.
- Experimentally demonstrated phenomenon: A controlled study demonstrated an effect under its stated conditions.
- Mechanistically plausible: A proposed explanation is consistent with some known physical or biological principles, but has not been established for the incident in question.
- Causal link undetermined: The available evidence does not establish what caused the reported experience.
- Attribution unestablished: The evidence does not establish who, if anyone, caused the incident.
- Contradictory evidence identified: A specific source conflicts with a claim; the conflicting material should be linked and explained rather than silently discarded.

These labels should not be used as a hierarchy of human credibility. They classify evidence, not a person's worth, sincerity or right to be heard.

19.8 Priority research sequence

To expand this into a substantial, source-rich dossier, I would proceed in the following order.

- Complete the primary scientific chronology. Retrieve the original papers from Frey, Foster and Finch, Lin, Elder and Chou, and later reviews. Capture each paper's methods, exposure conditions, measured outcomes, proposed mechanism and limitations. Do not rely on abstracts alone where the full text is accessible.
- Build a formal-investigation testimony index. Start with the 2018 JAMA case series, the 2020 National Academies assessment, the 2022 expert-panel assessment and the 2023 and 2025 intelligence assessments. Link the exact passages that describe witness accounts, clinical evidence and uncertainty.

- Add identifiable published first-person accounts. Record the original interview or statement, its date, the person's own description, any independent records mentioned, and whether the proposed mechanism was measured or inferred. Keep the source's claims separate from a journalist's narration.
- Add civilian online accounts under a distinct category. Preserve public URLs and publication dates, avoid reproducing sensitive personal details, and mark identity, incident timing and exposure as unverified where appropriate. Search for independent corroboration rather than treating repetition across websites as multiple independent confirmations.
- Cross-reference patents and programmes. For every technical concept cited, verify the original patent or official programme record. Track separately whether it was proposed, experimentally demonstrated, developed into a prototype, tested, procured or deployed. Do not infer later stages from earlier ones.
- Publish an evidence map and source ledger. Each public-facing claim should link to its original source and carry a concise description of what that source supports and what it leaves unresolved. Update entries when new evidence emerges rather than silently rewriting the historical record.

19.9 Conclusion: preserving testimony and testing explanations

A rigorous register can preserve a person's account without making the person prove a technical explanation before the account is recorded. It can also examine scientific research without allowing a known phenomenon to become a universal explanation for every similar-sounding experience.

The microwave auditory effect is a documented subject of scientific research. Its proposed mechanisms, practical constraints and relevance to reported anomalous health incidents have been discussed in scientific publications and government assessments. Separately, identifiable people have publicly described sudden auditory or sensory experiences and serious subsequent symptoms. Those accounts deserve accurate documentation, and their clinical or documentary corroboration should be recorded wherever available. The connection between an individual report and a specific technological cause, however, must be assessed case by case.

The resulting dossier should therefore make three things easy for any reader to do: find the original testimony, examine the scientific evidence relevant to it, and see precisely where the evidence supports a connection—and where that connection remains unresolved. That is how testimony and science can be preserved together without either being misrepresented.

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The resulting dossier should therefore make three things easy for any reader to do: find the original testimony, examine the scientific evidence relevant to it, and see precisely where the evidence supports a connection—and where that connection remains unresolved. That is how testimony and science can be preserved together without either being misrepresented.

Part VII — Executing the Witness-and-Science Research Register

From a research framework to a source-linked, case-by-case evidence examination

The objective now is to populate the register with actual sources and test the relationships between them, rather than merely describe how the work should be done. The examination below starts with the strongest traceable material already identified: original scientific papers, peer-reviewed clinical reports, the National Academies assessment, intelligence-community documents, and identifiable first-person accounts.

The guiding rule is to record what each source establishes, what it suggests, and what it cannot establish. That makes it possible to preserve reported experiences and investigate proposed technologies without turning an unresolved explanation into a settled finding.

20. The core scientific finding: the microwave auditory effect is a real research subject

The starting point is the original scientific literature—not patent speculation, internet repetition, or an assumption that every unusual auditory experience has the same cause.

20.1 The original Frey research

Allan H. Frey, 1962 — "Human Auditory System Response to Modulated Electromagnetic Energy."

Frey reported auditory perceptions associated with particular conditions of pulsed electromagnetic exposure. This paper is the foundational reference for the phenomenon now widely called the microwave auditory effect or Frey effect.

The evidentiary significance is specific. It establishes that researchers investigated auditory perception associated with pulsed radiofrequency energy and published experimental observations about it. It does not establish that all reported voices, unexplained sounds, pressure sensations, headaches or cognitive symptoms arise from that mechanism.

Register entry S-01

- Source type: Original experimental paper.
- Supported proposition: Auditory perception associated with pulsed electromagnetic energy was reported in scientific literature.
- Further evidence required: Independent replication, the specific exposure conditions, and evidence connecting any particular reported incident to those conditions.
- Not established by this paper: Remote reading of thoughts, unrestricted transmission

of intelligible speech into a person's mind, or the cause of an unrelated individual's symptoms.

20.2 The proposed physical mechanism

Foster and Finch, 1974 — "Microwave hearing: evidence for thermoacoustic auditory stimulation by pulsed microwaves."

This study investigated acoustic transients produced when pulsed microwave energy was absorbed by material. It is relevant to the proposed thermoelastic or thermoacoustic explanation: a short pulse can produce rapid heating and a resulting pressure disturbance, which may stimulate the auditory system through mechanical pathways.

This mechanism is materially different from the proposition that a radio signal simply carries ordinary speech directly into the brain. A pressure transient capable of producing a click or other auditory sensation does not, by itself, demonstrate the ability to transmit arbitrary speech, read thoughts or exert precise behavioural control.

Register entry S-11

- Source type: Experimental mechanism study.
- Supported proposition: A physical mechanism for microwave-associated auditory stimulation has been investigated experimentally.
- Further evidence required: Demonstration that the relevant exposure occurred in a specific incident and that it explains the reported effect.
- Not established: That any particular civilian was exposed, that a device was present, or that a reported voice-like experience was technologically generated.

20.3 Subsequent scientific examination

Elder and Chou, 2003 — "Auditory Response to Pulsed Radiofrequency Energy."

This review considers the reported auditory response and proposed mechanisms. It belongs in the scientific register because it connects earlier observations with subsequent research and identifies the physical questions involved.

Lin, 2007 — "Hearing of microwave pulses by humans and animals: effects, mechanism, and thresholds."

Lin's review examines reported effects, mechanisms and thresholds. For the register, its technical value lies in specifying that exposure characteristics matter. A meaningful comparison between a published experiment and a reported incident needs more than a general similarity of symptoms: it requires attention to the energy source, pulse characteristics, exposure conditions and observed outcome.

20.4 A direct examination of the weaponization claim

Foster, Garrett and Ziskin, 2021 — "Can the Microwave Auditory Effect Be 'Weaponized'?"

-

This paper considers whether the microwave auditory effect plausibly explains reported anomalous health incidents. It accepts the need to examine the established auditory phenomenon but argues against attributing the incidents under discussion to microwave exposure on the public evidence available to the authors.

This is an important counterpoint, not a reason to omit the earlier experimental work. A rigorous dossier needs both the evidence for the phenomenon and the scientific analysis of

whether it explains a broader claim.
Scientific conclusion: There is a documented scientific literature on auditory perception associated with pulsed radiofrequency energy. The existence of that phenomenon does not establish that a particular technology caused a particular person's experience. The latter requires incident-specific evidence.

21. The clinical evidence must be examined as a whole, including findings that differ

An evidence register would be incomplete if it included the early clinical findings but omitted later research that reported different results. The correct method is to preserve each study's population, method and conclusion so the apparent differences can be investigated.

21.1 The 2018 clinical case series

Swanson and colleagues, JAMA, 2018

The authors reported clinical evaluations of 21 US government personnel who described directional auditory or sensory phenomena in Havana. Persistent symptoms included cognitive difficulties, balance problems, visual and auditory symptoms, sleep disturbance and headache. The researchers also reported objective findings in cognitive, vestibular and oculomotor assessments.

These findings are evidence that the evaluated group experienced substantial health problems and that clinical abnormalities were identified in many participants. However, the study was a retrospective clinical case series, not a controlled experiment in which a suspected energy source was measured and its causal effects demonstrated. The authors themselves described the source as undetermined.

Register result: The paper supports the existence of a clinically investigated group with reported sensory events and persistent symptoms. It does not identify a specific source or prove that a particular technology caused the cases.

21.2 The 2019 advanced neuroimaging study

Verma and colleagues, JAMA, 2019

-

This study compared advanced MRI measures from 40 US government personnel with possible exposure to directional phenomena with 48 healthy controls. It reported group-level differences in several measures, including brain tissue volumes, cerebellar microstructure and connectivity in auditory and visuospatial subnetworks. The authors stated that the clinical significance of the differences was uncertain and required further investigation.

That qualification is important. A statistical difference between groups does not automatically show that every participant had the same injury, that the difference was caused by the reported event, or that a particular technology was responsible. Nor should the results be dismissed merely because the imaging differences were not specific to a single cause.

Register result: The paper reports group-level neuroimaging differences in its study population. It does not establish the cause of those differences or prove a specific mechanism of exposure.

21.3 The 2024 National Institutes of Health studies

NIH Clinical Center, 18 March 2024

-

The NIH reported results from detailed clinical assessments and advanced imaging involving more than 80 US government employees and adult family members who had reported anomalous health incidents, compared with healthy control groups. The researchers found no significant evidence of MRI-detectable brain injury or differences in most clinical measures compared with controls. NIH also emphasized that the symptoms reported by participants were real, could be prolonged and disruptive, and could be difficult to treat.

This is a crucial counterpoint to the earlier clinical and imaging papers. The NIH findings do not mean that the people who participated did not experience symptoms. They mean that the particular tests and comparisons used in these studies did not identify the anticipated differences in the areas examined.

The 2018, 2019 and 2024 studies should not be reduced to a simplistic contest in which one paper cancels out the others. Their cohorts, study designs, measures, control groups and analytical approaches differ. The proper next step is to compare those methods directly and determine which results replicated, which did not, and what the differences imply for possible explanations.

Register result: The NIH studies found no significant evidence of MRI-detectable brain injury or differences in most clinical measures in their studied groups. They do not establish that the symptoms were fabricated, and they do not independently identify an alternative cause for every incident.

21.4 Comparison of the clinical and imaging evidence

Source	Reported result	Key limitation
JAMA, 2018	Persistent symptoms and objective clinical abnormalities in a referred group of 21 personnel	Retrospective case series; source and cause undetermined
JAMA, 2019	Group-level differences in several advanced MRI measures among 40	Clinical significance and causal relationship uncertain

	personnel compared with controls	
NIH, 2024	No significant evidence of MRI-detectable brain injury or differences in most clinical measures compared with controls	Applies to the populations, tests and outcomes examined; does not settle every individual case

The scientifically responsible conclusion is that published studies have reported differing findings. The difference itself deserves analysis. It cannot be resolved simply by selecting the paper that best fits a preferred theory.

22. Formal testimony and public records: expanding the witness register

The register can also include testimony preserved in government hearings, where first-person reports, medical summaries and official responses may appear in the same record.

W-08. Congressional hearing on reported incidents in Havana

US Senate hearing: “Attacks on U.S. Diplomats in Cuba: Response and Oversight.”

-

The transcript records official testimony about reports of unusual auditory sensations, ear pain, dizziness, headaches and other symptoms among personnel assigned to Havana. It also describes the medical evaluation process and the effort to determine whether a subset of people had clinically verifiable findings.

This source is valuable because it preserves statements in a formal public record and allows readers to examine the words of the witnesses and officials in context. It should not be described as a hearing that conclusively established a weapon or identified a perpetrator unless a specific finding in the transcript supports that claim.

Evidence classification: Formal public record; supports the fact that the reports and official investigations were discussed and documented. The causal explanation remains a separate question.

W-09. House Homeland Security Subcommittee hearing, May 2024

Hearing: “Silent Weapons: Examining Foreign Anomalous Health Incidents Targeting

Americans in the Homeland and Abroad."
The hearing brought expert witnesses and former government officials before a congressional subcommittee to discuss anomalous health incidents, possible threats, investigative questions and competing interpretations of the evidence. The transcript is the appropriate source for what each witness actually said. The committee's framing, the testimony of an individual witness and a verified scientific finding must be recorded as distinct things.
Evidence classification: Formal oversight record; useful for tracing public testimony and disputes over interpretation. A witness's statement is attributable to that witness; it is not automatically a consensus scientific conclusion.

W-10. Government Accountability Office report on patient care

GAO, 29 July 2024 — "Havana Syndrome: Better Patient Communication and Monitoring of Key DOD Tasks Needed to Better Ensure Timely Treatment."
The GAO documented concerns about access to care and the Department of Defense's processes for assisting people who reported anomalous health incidents. Its report describes the symptom patterns reported by personnel and family members and examines the government's patient-care response.
This is an important source for a witness register because the consequences of a reported incident include not only the initial sensation or symptom but also the experience of seeking evaluation, obtaining records, accessing treatment and navigating administrative processes.
The GAO report should be used for claims it actually examined about care, communication and agency procedures. It should not be used as proof of a particular physical mechanism merely because it documents the seriousness of the reported symptoms or difficulties obtaining care.

23. What this first execution establishes—and what remains to be researched

The initial source set now supports several distinct findings:

- A documented scientific phenomenon: Published research has investigated auditory perception associated with pulsed radiofrequency energy, including proposed thermoacoustic mechanisms.
- Documented reports of unusual sensory experiences: Formal medical and government records contain accounts of people reporting sudden auditory or sensory phenomena.
- Clinical investigation: Some evaluated groups showed persistent symptoms and objective clinical findings; later NIH studies did not find significant differences in most measures or evidence of MRI-detectable brain injury in the populations they examined.
- Unresolved causation: The existence of a reported event or symptom does not by itself establish the physical source, the use of a device, or the identity of a responsible actor.
- A documented administrative dimension: Public records describe investigations, medical assessments, and difficulties relating to care and agency response. These records should be examined independently from theories about the original cause.

These findings justify further research. They do not justify treating all reported experiences as instances of one technology, and they do not justify treating unresolved causation as proof that an account was invented.

24. Civilian testimony: the next evidence gap

The sources assembled so far are strongest on government personnel, their family members, formal investigations and scientific assessments. They do not yet constitute a verified collection of independent civilian accounts demonstrating exposure to a particular technology. It would be misleading to label anonymous online reports as corroboration of that technology without independently verified exposure evidence.

Civilian accounts can nevertheless be added as their own research category. For each one, the register should capture:

- The original public URL and date, not only a repost or a commentary about the account.
- Whether the person is describing a firsthand experience or repeating someone else's account.
- The exact experience described, without rewriting it to make it resemble a published experiment.
- Whether there is independent documentation of the timing, symptoms, location or exposure.
- Whether the proposed technological explanation comes from the witness, a clinician, an investigator or a separate technical analysis.
- Whether the original source can be revisited and the account's provenance checked.

A person's report can be authentic as a report of their experience even if the suggested cause remains unknown. That is precisely why testimony and causation need separate fields.

Current register status

Scientific foundation established

Primary microwave-auditory research, mechanism studies, reviews and competing assessments identified with original links.

Formal testimony and clinical records assembled

JAMA case series, advanced imaging research, NIH findings, government hearings and oversight records are available for cross-reference.

Civilian online testimony requires a dedicated collection

Firsthand public accounts need to be identified and checked individually. The current sources do not establish a verified civilian dataset connecting reported experiences to a particular technological cause.

Case-level causal links remain to be established

The next task is to match each account to independently verifiable clinical, temporal, environmental or technical evidence where available.

25. Research conclusion

The evidence assembled here supports a careful but substantive conclusion: pulsed-radiofrequency auditory effects are a documented scientific research topic; unusual sensory experiences and serious health symptoms have been reported and investigated; and published studies differ in what they found when examining affected groups. Formal investigations have not established one universally applicable cause for all reported incidents.

The next phase should therefore concentrate on building the civilian testimony collection and

tracing each account back to its original source. Where a link to a scientific mechanism is plausible, the dossier should explain why. Where an independent measurement or record corroborates a detail, it should identify that record. Where the connection is uncertain, that uncertainty should remain visible.

The result will be more useful than either an uncritical collection of claims or a document that excludes testimony because its cause has not been determined. It will allow readers to examine the human accounts, the published science and the limits of the evidence side by side.

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From a research framework to a source-linked, case-by-case evidence examination

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The guiding rule is to record what each source establishes, what it suggests, and what it cannot establish. That makes it possible to preserve reported experiences and investigate proposed technologies without turning an unresolved explanation into a settled finding.

20. The core scientific finding: the microwave auditory effect is a real research subject

The starting point is the original scientific literature—not patent speculation, internet repetition, or an assumption that every unusual auditory experience has the same cause.

20.1 The original Frey research

Allan H. Frey, 1962 — "Human Auditory System Response to Modulated Electromagnetic Energy."

Frey reported auditory perceptions associated with particular conditions of pulsed electromagnetic exposure. This paper is the foundational reference for the phenomenon now widely called the microwave auditory effect or Frey effect.

The evidentiary significance is specific. It establishes that researchers investigated auditory perception associated with pulsed radiofrequency energy and published experimental observations about it. It does not establish that all reported voices, unexplained sounds, pressure sensations, headaches or cognitive symptoms arise from that mechanism.

Register entry S-01

- Source type: Original experimental paper.
- Supported proposition: Auditory perception associated with pulsed electromagnetic energy was reported in scientific literature.
- Further evidence required: Independent replication, the specific exposure conditions, and evidence connecting any particular reported incident to those conditions.
- Not established by this paper: Remote reading of thoughts, unrestricted transmission of intelligible speech into a person's mind, or the cause of an unrelated individual's symptoms.

20.2 The proposed physical mechanism

Foster and Finch, 1974 — “Microwave hearing: evidence for thermoacoustic auditory stimulation by pulsed microwaves.”
This study investigated acoustic transients produced when pulsed microwave energy was absorbed by material. It is relevant to the proposed thermoelastic or thermoacoustic explanation: a short pulse can produce rapid heating and a resulting pressure disturbance, which may stimulate the auditory system through mechanical pathways.
This mechanism is materially different from the proposition that a radio signal simply carries ordinary speech directly into the brain. A pressure transient capable of producing a click or other auditory sensation does not, by itself, demonstrate the ability to transmit arbitrary speech, read thoughts or exert precise behavioural control.
Register entry S-11

- Source type: Experimental mechanism study.
- Supported proposition: A physical mechanism for microwave-associated auditory stimulation has been investigated experimentally.
- Further evidence required: Demonstration that the relevant exposure occurred in a specific incident and that it explains the reported effect.
- Not established: That any particular civilian was exposed, that a device was present, or that a reported voice-like experience was technologically generated.

20.3 Subsequent scientific examination
Elder and Chou, 2003 — “Auditory Response to Pulsed Radiofrequency Energy.”
This review considers the reported auditory response and proposed mechanisms. It belongs in the scientific register because it connects earlier observations with subsequent research and identifies the physical questions involved.
Lin, 2007 — “Hearing of microwave pulses by humans and animals: effects, mechanism, and thresholds.”
Lin’s review examines reported effects, mechanisms and thresholds. For the register, its technical value lies in specifying that exposure characteristics matter. A meaningful comparison between a published experiment and a reported incident needs more than a general similarity of symptoms: it requires attention to the energy source, pulse characteristics, exposure conditions and observed outcome.
20.4 A direct examination of the weaponization claim
Foster, Garrett and Ziskin, 2021 — “Can the Microwave Auditory Effect Be ‘Weaponized’?”
-
This paper considers whether the microwave auditory effect plausibly explains reported anomalous health incidents. It accepts the need to examine the established auditory phenomenon but argues against attributing the incidents under discussion to microwave exposure on the public evidence available to the authors.
This is an important counterpoint, not a reason to omit the earlier experimental work. A rigorous dossier needs both the evidence for the phenomenon and the scientific analysis of whether it explains a broader claim.
Scientific conclusion: There is a documented scientific literature on auditory perception associated with pulsed radiofrequency energy. The existence of that phenomenon does not

establish that a particular technology caused a particular person's experience. The latter requires incident-specific evidence.

21. The clinical evidence must be examined as a whole, including findings that differ

An evidence register would be incomplete if it included the early clinical findings but omitted later research that reported different results. The correct method is to preserve each study's population, method and conclusion so the apparent differences can be investigated.

21.1 The 2018 clinical case series

Swanson and colleagues, JAMA, 2018

The authors reported clinical evaluations of 21 US government personnel who described directional auditory or sensory phenomena in Havana. Persistent symptoms included cognitive difficulties, balance problems, visual and auditory symptoms, sleep disturbance and headache. The researchers also reported objective findings in cognitive, vestibular and oculomotor assessments.

These findings are evidence that the evaluated group experienced substantial health problems and that clinical abnormalities were identified in many participants. However, the study was a retrospective clinical case series, not a controlled experiment in which a suspected energy source was measured and its causal effects demonstrated. The authors themselves described the source as undetermined.

Register result: The paper supports the existence of a clinically investigated group with reported sensory events and persistent symptoms. It does not identify a specific source or prove that a particular technology caused the cases.

21.2 The 2019 advanced neuroimaging study

Verma and colleagues, JAMA, 2019

-

This study compared advanced MRI measures from 40 US government personnel with possible exposure to directional phenomena with 48 healthy controls. It reported group-level differences in several measures, including brain tissue volumes, cerebellar microstructure and connectivity in auditory and visuospatial subnetworks. The authors stated that the clinical significance of the differences was uncertain and required further investigation.

That qualification is important. A statistical difference between groups does not automatically show that every participant had the same injury, that the difference was caused by the reported event, or that a particular technology was responsible. Nor should the results be dismissed merely because the imaging differences were not specific to a single cause.

Register result: The paper reports group-level neuroimaging differences in its study population. It does not establish the cause of those differences or prove a specific mechanism of exposure.

21.3 The 2024 National Institutes of Health studies

NIH Clinical Center, 18 March 2024

-

The NIH reported results from detailed clinical assessments and advanced imaging involving

more than 80 US government employees and adult family members who had reported anomalous health incidents, compared with healthy control groups. The researchers found no significant evidence of MRI-detectable brain injury or differences in most clinical measures compared with controls. NIH also emphasized that the symptoms reported by participants were real, could be prolonged and disruptive, and could be difficult to treat.
This is a crucial counterpoint to the earlier clinical and imaging papers. The NIH findings do not mean that the people who participated did not experience symptoms. They mean that the particular tests and comparisons used in these studies did not identify the anticipated differences in the areas examined.
The 2018, 2019 and 2024 studies should not be reduced to a simplistic contest in which one paper cancels out the others. Their cohorts, study designs, measures, control groups and analytical approaches differ. The proper next step is to compare those methods directly and determine which results replicated, which did not, and what the differences imply for possible explanations.
Register result: The NIH studies found no significant evidence of MRI-detectable brain injury or differences in most clinical measures in their studied groups. They do not establish that the symptoms were fabricated, and they do not independently identify an alternative cause for every incident.

21.4 Comparison of the clinical and imaging evidence

Source	Reported result	Key limitation
JAMA, 2018	Persistent symptoms and objective clinical abnormalities in a referred group of 21 personnel	Retrospective case series; source and cause undetermined
JAMA, 2019	Group-level differences in several advanced MRI measures among 40 personnel compared with	Clinical significance and causal relationship uncertain

	controls	
NIH, 2024	No significant evidence of MRI-detectable brain injury or differences in most clinical measures compared with controls	Applies to the populations, tests and outcomes examined; does not settle every individual case

The scientifically responsible conclusion is that published studies have reported differing findings. The difference itself deserves analysis. It cannot be resolved simply by selecting the paper that best fits a preferred theory.

22. Formal testimony and public records: expanding the witness register

The register can also include testimony preserved in government hearings, where first-person reports, medical summaries and official responses may appear in the same record.

W-08. Congressional hearing on reported incidents in Havana

US Senate hearing: "Attacks on U.S. Diplomats in Cuba: Response and Oversight."

-

The transcript records official testimony about reports of unusual auditory sensations, ear pain, dizziness, headaches and other symptoms among personnel assigned to Havana. It also describes the medical evaluation process and the effort to determine whether a subset of people had clinically verifiable findings.

This source is valuable because it preserves statements in a formal public record and allows readers to examine the words of the witnesses and officials in context. It should not be described as a hearing that conclusively established a weapon or identified a perpetrator unless a specific finding in the transcript supports that claim.

Evidence classification: Formal public record; supports the fact that the reports and official investigations were discussed and documented. The causal explanation remains a separate question.

W-09. House Homeland Security Subcommittee hearing, May 2024

Hearing: "Silent Weapons: Examining Foreign Anomalous Health Incidents Targeting Americans in the Homeland and Abroad."

The hearing brought expert witnesses and former government officials before a congressional subcommittee to discuss anomalous health incidents, possible threats,

investigative questions and competing interpretations of the evidence. The transcript is the appropriate source for what each witness actually said. The committee's framing, the testimony of an individual witness and a verified scientific finding must be recorded as distinct things.

Evidence classification: Formal oversight record; useful for tracing public testimony and disputes over interpretation. A witness's statement is attributable to that witness; it is not automatically a consensus scientific conclusion.

W-10. Government Accountability Office report on patient care

GAO, 29 July 2024 — "Havana Syndrome: Better Patient Communication and Monitoring of Key DOD Tasks Needed to Better Ensure Timely Treatment."

The GAO documented concerns about access to care and the Department of Defense's processes for assisting people who reported anomalous health incidents. Its report describes the symptom patterns reported by personnel and family members and examines the government's patient-care response.

This is an important source for a witness register because the consequences of a reported incident include not only the initial sensation or symptom but also the experience of seeking evaluation, obtaining records, accessing treatment and navigating administrative processes.

The GAO report should be used for claims it actually examined about care, communication and agency procedures. It should not be used as proof of a particular physical mechanism merely because it documents the seriousness of the reported symptoms or difficulties obtaining care.

23. What this first execution establishes—and what remains to be researched

The initial source set now supports several distinct findings:

- A documented scientific phenomenon: Published research has investigated auditory perception associated with pulsed radiofrequency energy, including proposed thermoacoustic mechanisms.
- Documented reports of unusual sensory experiences: Formal medical and government records contain accounts of people reporting sudden auditory or sensory phenomena.
- Clinical investigation: Some evaluated groups showed persistent symptoms and objective clinical findings; later NIH studies did not find significant differences in most measures or evidence of MRI-detectable brain injury in the populations they examined.
- Unresolved causation: The existence of a reported event or symptom does not by itself establish the physical source, the use of a device, or the identity of a responsible actor.
- A documented administrative dimension: Public records describe investigations, medical assessments, and difficulties relating to care and agency response. These records should be examined independently from theories about the original cause.

These findings justify further research. They do not justify treating all reported experiences as instances of one technology, and they do not justify treating unresolved causation as proof that an account was invented.

24. Civilian testimony: the next evidence gap

The sources assembled so far are strongest on government personnel, their family members, formal investigations and scientific assessments. They do not yet constitute a verified collection of independent civilian accounts demonstrating exposure to a particular

technology. It would be misleading to label anonymous online reports as corroboration of that technology without independently verified exposure evidence.

Civilian accounts can nevertheless be added as their own research category. For each one, the register should capture:

- The original public URL and date, not only a repost or a commentary about the account.
- Whether the person is describing a firsthand experience or repeating someone else's account.
- The exact experience described, without rewriting it to make it resemble a published experiment.
- Whether there is independent documentation of the timing, symptoms, location or exposure.
- Whether the proposed technological explanation comes from the witness, a clinician, an investigator or a separate technical analysis.
- Whether the original source can be revisited and the account's provenance checked.

A person's report can be authentic as a report of their experience even if the suggested cause remains unknown. That is precisely why testimony and causation need separate fields.

Current register status

Scientific foundation established

Primary microwave-auditory research, mechanism studies, reviews and competing assessments identified with original links.

Formal testimony and clinical records assembled

JAMA case series, advanced imaging research, NIH findings, government hearings and oversight records are available for cross-reference.

Civilian online testimony requires a dedicated collection

Firsthand public accounts need to be identified and checked individually. The current sources do not establish a verified civilian dataset connecting reported experiences to a particular technological cause.

Case-level causal links remain to be established

The next task is to match each account to independently verifiable clinical, temporal, environmental or technical evidence where available.

25. Research conclusion

The evidence assembled here supports a careful but substantive conclusion: pulsed-radiofrequency auditory effects are a documented scientific research topic; unusual sensory experiences and serious health symptoms have been reported and investigated; and published studies differ in what they found when examining affected groups. Formal investigations have not established one universally applicable cause for all reported incidents.

The next phase should therefore concentrate on building the civilian testimony collection and tracing each account back to its original source. Where a link to a scientific mechanism is plausible, the dossier should explain why. Where an independent measurement or record corroborates a detail, it should identify that record. Where the connection is uncertain, that

uncertainty should remain visible.
The result will be more useful than either an uncritical collection of claims or a document that excludes testimony because its cause has not been determined. It will allow readers to examine the human accounts, the published science and the limits of the evidence side by side.

Continue with next phase/ The next phase should concentrate on retrieving the original individual accounts and the records that could independently test their claims. That is where the dossier can move beyond a general comparison of reported symptoms and scientific literature toward a more precise, case-by-case assessment.
The standard remains the same for every witness: preserve what the person says, identify what can be checked, link the relevant scientific literature, and make the status of any causal claim explicit.

Part IX — Original Civilian Accounts and Independent Evidence
Case file W-13: Chris Allen, Havana, April 2014
This phase has identified a particularly useful civilian case because the reporting describes not only the person's recollection but also contemporaneous records reviewed by journalists. It therefore provides a stronger basis for source verification than an online post with no independent documentation.
31. The original account
Witness: Chris Allen, a US tourist from South Carolina.
Original reporting: Josh Lederman, Associated Press, 19 October 2017.
According to the Associated Press report, Allen visited Havana in April 2014 and stayed at the Hotel Capri. He described numbness and tingling beginning shortly after he went to bed, spreading from his toes to other parts of his body. He said the sensations subsided when he got up and returned when he lay down again. The symptoms persisted after he left Cuba, fluctuating over approximately six months. He subsequently underwent medical investigations, including imaging, nerve-conduction testing and blood tests.
The important documentary detail is that the Associated Press said it had reviewed more than 30 pages of Allen's medical records, laboratory results, travel records and contemporaneous emails, including messages sent from Havana. The reporting also said that Allen had described consistent symptoms to multiple physicians well before the public disclosure of the later diplomatic incidents.
That makes this a useful case for the dossier. It provides evidence beyond a retrospective online allegation: there was a reported travel chronology, a documented history of symptoms and a set of records examined by an independent news organisation.
However, the Associated Press also explicitly noted differences between Allen's experience and the later diplomatic cases. His primary symptoms were numbness and tingling rather than the distinctive sounds reported in some diplomatic incidents, and his trip occurred in 2014, before the publicly identified cluster beginning in 2016. The report did not establish a

common cause.

31.1 What the available evidence supports

Contemporaneous chronology

The published account describes a 2014 trip, the reported onset of symptoms in Havana and contemporaneous communications.

Independent documentary review

The Associated Press reported examining medical, laboratory, travel and email records. The underlying records are not all reproduced in the public article, so this is a journalist's account of reviewing them rather than direct public access to the complete file.

Unresolved causal explanation

The published evidence does not establish that Allen was exposed to an acoustic or electromagnetic device, or that his illness shared a cause with the later diplomatic cases.

31.2 The independent tests that could further assess the case

The next research step is to distinguish records that establish symptoms from evidence that could identify their cause.

First, the original medical records and test results would help establish the clinical chronology and what clinicians found. They might narrow possible explanations, but negative tests would not by themselves prove an external exposure.

Second, the original travel records and contemporaneous emails could help verify dates, locations and the timing of symptom reports. They cannot independently identify a physical source.

Third, any surviving environmental measurements, recordings or contemporaneous technical investigations would be relevant to the alleged exposure question. The public reporting cited here does not establish that such measurements exist for Allen's 2014 stay.

Finally, a scientific comparison with the microwave auditory literature would need to account for the fact that Allen's reported symptoms were primarily sensory and neurological, rather than a clearly described microwave-associated auditory perception. Similarity in the broad category of unexplained symptoms is not enough to establish the same mechanism.

Interim assessment: Allen's account is a documented civilian report supported by a journalistic review of contemporaneous records. Its relationship to the later Havana incidents remains unresolved. It should be retained as a distinct case, not labelled a confirmed directed-energy exposure.

32. Case file W-14: Canadian diplomatic families in Havana, 2017–2018

This case adds a second national investigation and an important set of family and dependent reports. It also provides an opportunity to compare early scientific findings with a later government review that reached a different overall conclusion about the likely causes.

Primary government source: Global Affairs Canada, Unexplained Health Incidents in Havana, Cuba, August 2024.

-

The report describes investigations into health incidents experienced by Canadian personnel and their dependants in Havana. The Canadian government explicitly states that it does not

question the authenticity of the symptoms reported. However, following several years of investigation, it concluded that the incidents were not the result of malicious action by a foreign actor and that pre-existing medical conditions, environmental factors and conventional illnesses were likely to have been important contributors to many of the symptoms.

This is a significant source because it distinguishes the reality of reported health problems from the question of their cause. It also records the evolution of the Canadian government's position following further investigation.

32.1 What this case adds to the register

The Canadian report provides an official account of the government's investigation, its coordination with other agencies and its eventual assessment of the likely explanations. It should be read alongside, rather than substituted for, the clinical research conducted on Canadian personnel and family members.

One earlier study, initially circulated as a preprint, examined 26 Canadian diplomats and family members using clinical assessments, cognitive testing, blood tests and brain imaging. It reported abnormalities in some participants and proposed possible explanations that included exposure to cholinesterase-inhibiting substances. The study's conclusions were hypotheses about the observed findings, not proof that a particular chemical or other agent caused every case.

A later peer-reviewed study examined MRI measures in Canadian diplomats and family members, comparing exposed and unexposed groups with healthy controls. It reported differences in white-matter fibre density in particular brain regions. As with the earlier work, those group-level findings do not independently establish a single cause for the reported illnesses.

The Canadian government's 2024 conclusion must be recorded alongside those studies. It represents the government's overall assessment of its investigation; the research papers describe findings from particular samples and methods. Neither kind of source should be misrepresented as answering every individual case conclusively.

Register entry W-14

- Population: Canadian diplomatic personnel and dependants reporting health incidents in Havana.
- Period: Principally 2017–2018.
- Evidence types: Government investigation, reported symptoms, clinical assessments and published imaging research.
- Established: The Canadian government investigated the reported health incidents and acknowledges that the reported symptoms were genuine.
- Scientific findings: Some studies reported group-level clinical or imaging differences in particular samples.
- Government conclusion: The 2024 report rejected malicious foreign-actor causation and considered conventional medical and environmental factors likely to have contributed to many symptoms.
- Remaining distinction: Findings in a group cannot automatically determine the cause of every person's symptoms.

32.2 Why the disagreement must remain visible

A rigorous archive should not hide the earlier studies because a later government report reached a different overall conclusion. Nor should it treat earlier findings as if they overruled the subsequent investigation.

Instead, it should link the sources and identify the different questions each addresses: clinical observations, statistical differences between groups, possible mechanisms, and the government's assessment of the incidents as a whole.

This allows readers to inspect the underlying evidence and understand how the conclusions evolved. It also prevents a common error in contested research: presenting a preliminary finding as a final causal determination.

33. Updated civilian and family testimony register

The following entries now form a more useful starting collection. They differ in the type and strength of evidence available.

Record	Account or population	Indepen-dent evidence publicly described	Causal status
W-11	Heidi and Chris, Virginia	Published interview; reported medical and service-related circum-stances	Alleged directed-energy exposure not indepen-dently established by the interview
W-12	US tourists reporting unusual experiences	Contempo-rary journalism summarising	Undeter-mined

	in Havana	reports; individual source records still need tracing	
W-13	Chris Allen, Havana, April 2014	AP reported reviewing medical, laboratory, travel and contempora-neous email records	Symptoms documented in reporting; technologi-cal cause undeter-mined
W-14	Canadian personnel and dependants, Havana, 2017–2018	Government investigation and published clinical/ imaging studies	Government rejected malicious foreign-actor causation; individual causes may differ

The register should not rank these people by sincerity or suffering. It should distinguish what the sources establish and what remains open to investigation.

The strongest immediate next step is to obtain the underlying records where publicly available, particularly the original medical and contemporaneous documentation referenced in the Chris Allen reporting, and to extract individual accounts from the Canadian

government's report without treating a summary of multiple people as multiple independent firsthand sources.

34. Findings at the end of this phase

The evidence now provides three distinct types of material for the dossier:

- Civilian reporting with a documented chronology: Chris Allen's account includes a journalist's review of contemporaneous medical and travel records, although the proposed technological explanation remains unresolved.
- Civilian and family testimony with an unverified mechanism: The Virginia family's public account merits preservation, while the causal claims require independent technical and medical evidence.
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The next phase should examine the original documents behind these published accounts: public hearing transcripts, released government correspondence, scientific papers in full, available case documentation and any independently authenticated technical measurements. Where those records are unavailable, the register should say so explicitly rather than imply that they have been examined.

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That is the point at which the dossier becomes genuinely auditable: every important conclusion can be traced to an original source, every witness remains distinguishable from the interpretation of their account, and every proposed technological explanation is tested against the evidence actually available.

Part X — From documented symptoms to the physical mechanism: what the science actually establishes

This phase moves from the Canadian investigation to the central scientific question in the register: what would have to be demonstrated before the microwave auditory effect could explain a particular reported incident?

The distinction matters because three different propositions are often blended together. First, pulsed radiofrequency energy can produce auditory sensations under certain experimental conditions. Second, researchers can investigate whether particular forms of energy might produce unpleasant or harmful physiological effects. Third, a specific person

was exposed to a particular source that caused their symptoms. Evidence for the first proposition does not automatically establish the second or third.

39. The Frey effect: a real phenomenon with defined physical conditions

The foundational publication is Allan H. Frey's 1962 paper, Human Auditory System Response to Modulated Electromagnetic Energy. It reported auditory perceptions associated with particular modulated electromagnetic exposures. Later research investigated the phenomenon and proposed a thermoacoustic explanation: absorption of a short radiofrequency pulse produces a minute, rapid temperature change, which generates an acoustic pressure wave that can stimulate auditory structures.

Original paper: .

A useful later review is Foster, Garrett and Ziskin's 2021 paper, Can the Microwave Auditory Effect Be "Weaponized"? The authors accept the underlying auditory phenomenon but examine the engineering and biological constraints involved in trying to use it against a person. They argue that some conceivable systems would be conspicuous, while hypothetical higher-frequency systems raise different technical questions; they also state that they had no evidence linking such systems to the reported Havana incidents.

The scientific significance is not that the paper proves an operational weapon exists. It is that the question can be broken into testable physical requirements: pulse duration, frequency, energy delivered per pulse, distance, beam geometry, tissue absorption and the resulting acoustic pressure. A proposed explanation becomes stronger when those parameters are measured and shown to match the reported event, rather than inferred solely from the similarity of symptoms.

40. What the Canadian monitoring actually established

The Canadian government's August 2024 report gives us a concrete test of the scientific framework. Under Project Spectre, the RCMP-led investigation installed instruments in staff living quarters in 2019 to detect acoustic and radiation surges and measure environmental variables. The report states that the data did not provide relevant, probative information identifying a cause. The equipment was removed in 2022.

The report also states that the RCMP investigation concluded there was no known criminality, no attribution to a foreign actor and no identified pattern linking incidents to location or other variables. The broader Canadian interdepartmental task force likewise found no evidence attributing the reported symptoms to a foreign actor.

These are substantive findings, not merely an absence of comment. They should be included in the dossier alongside the reported symptoms and the steps taken to investigate them. At the same time, the public report does not provide the full technical dataset, calibration records, detection thresholds or a detailed incident-by-incident comparison of monitoring results. Those limitations matter when interpreting the scope of the negative findings.

The appropriate conclusion is therefore precise: the publicly available Canadian investigation did not establish an external attack or identify a common cause. Its published account does not provide enough technical detail to independently reconstruct every monitoring result. Neither limitation proves an alternative explanation.

41. The competing pesticide hypothesis: why the underlying studies matter

The Canadian health investigation illustrates another important evidentiary distinction. According to Global Affairs Canada, the Dalhousie University Brain Repair Centre investigation initially associated the reported health incidents with possible exposure to organophosphate pesticides. In response, Canada commissioned a separate environmental risk assessment by Golder Associates. That assessment analysed samples from soil, food, indoor and outdoor air, water, textiles and other materials. Golder concluded that pesticide-related health risks were at acceptable levels and found no obvious hazards likely to explain the reported health problems. The government's health advisory task force considered the environmental assessment's methodology robust and its conclusions well supported.

The final Canadian report also says the clinical investigation found symptoms associated with mild brain injury in approximately half its participants, with structural and functional evidence supporting injury in some participants. It notes that pre-existing conditions were associated with a greater likelihood of symptoms and that mass fumigation might have contributed.

These findings should not be collapsed into a single yes-or-no verdict on whether the participants were ill. The clinical study assessed health findings; the environmental assessment assessed particular pesticide risks. Evidence of symptoms or injury does not by itself establish the cause, while an environmental assessment that did not identify a likely pesticide hazard does not invalidate the clinical observations.

For the next documentary layer, the underlying clinical reports and the full Golder assessment would be more informative than repeating summaries of their conclusions. The key questions are what was measured, when and where samples were collected, how exposure was estimated, and how the clinical findings were connected—or not connected—to particular environmental conditions.

42. Can the microwave auditory effect explain persistent illness?

The 2021 review by Foster, Garrett and Ziskin considers whether pulsed microwave energy could produce auditory or audiovestibular effects and whether it could be used as a weapon. The authors discuss physical constraints, including the energy required, the effects of pulse characteristics, attenuation through the skull and the practical requirements of a transmitter. They conclude that some auditory or unpleasant vestibular effects may be conceivable under particular conditions, while the exposures they modelled were unlikely to reach thresholds for brain damage. They also state that they had no evidence that such systems played a role in the Havana incidents.

This matters because an auditory sensation, an acute vestibular disturbance and persistent neurological illness are not interchangeable outcomes. Evidence that a physical process can generate a sound-like sensation does not automatically demonstrate that it causes lasting brain injury, cancer, cognitive impairment or a complex pattern of chronic symptoms.

A rigorous investigation would need to match the proposed exposure to the actual event, including the source's characteristics, the energy reaching the person, the timing of symptoms and any independently documented physiological changes. If the proposed

mechanism is intended to explain lasting injury, that additional claim needs its own evidence. The same standard applies in reverse: the fact that one proposed mechanism has not been demonstrated in a particular case does not mean the person's symptoms were imaginary or inconsequential.

43. Evidence register: conclusions at this stage

Question	Current evidence-based position
Can pulsed radiofrequency energy produce auditory sensations under certain conditions?	Yes. This is a documented scientific phenomenon.
Does the phenomenon prove that a particular person was exposed?	No. Exposure must be established independently.
Did the Canadian investigation establish a foreign-directed attack?	No. Its published conclusion was that no such attribution was supported.
Did the Canadian monitoring identify a cause?	No. The report says it produced no relevant, probative information identifying a cause.
Do reported symptoms automatically establish their cause?	No. Clinical observations and causal attribution are separate questions.
Does failure to establish a cause disprove the	No. The Canadian government explicitly

reported symptoms?	acknowledged the authenticity and impact of the reported symptoms.

Sources: and .

44. Next phase: patents, programmes and demonstrated capabilities

The next research phase should examine patents and government research programmes against the same evidentiary standard. For every relevant patent or programme, the register should identify the actual technical disclosure, the applicant or sponsoring organisation, the stated purpose, the development stage, any independent demonstration of capability, and any evidence linking that capability to a reported incident.

A patent can document that an inventor disclosed a proposed method. A government programme can document that an agency funded or investigated a research objective. Neither, on its own, proves that a working device was built, that it operated at the required parameters, or that it was used against a particular person.

The purpose of the next phase is not to discard technical disclosures or to inflate them into proof. It is to establish exactly what each document adds to the historical, scientific and evidentiary record—and where the public record still leaves a question unanswered.

Part XI — Patents, military research and the boundary between disclosure and demonstrated capability

This phase examines a particularly important part of the public record: patents that explicitly describe radiofrequency hearing, a patent that claims methods for covertly producing adverse health effects, and a US defence research programme intended to develop nonsurgical neural interfaces.

These documents deserve examination on their actual terms. They are evidence of what inventors disclosed, what technical problems researchers sought to solve, and what organisations were willing to investigate. Their existence does not, by itself, establish that every proposed capability worked, was deployed, or was used against any particular person.

45. Patent US 6,470,214 B1 — a patent expressly addressing radiofrequency hearing

Primary source: United States patent record

This patent describes a method for encoding speech on a radiofrequency carrier so that the intended auditory sensation may be more intelligible than the clicks or simple tones associated with earlier descriptions of the radiofrequency hearing effect. The patent's abstract discusses carrier-suppressed modulation and preprocessing of the speech signal. Its background also acknowledges a technical problem: ordinary pulsed-carrier modulation can severely distort complex speech waveforms.

That detail is significant. The patent is not merely a passing reference to the Frey effect; it expressly addresses the engineering problem of attempting to transmit intelligible speech through it. The document therefore belongs in any serious history of attempts to develop

radiofrequency auditory applications.
But its evidentiary reach must be stated carefully. A granted patent establishes that a technical disclosure was examined under the patent system and granted protection for its claims. It does not independently establish that every described implementation was experimentally validated, that intelligible speech was reliably produced in real-world conditions, or that the system was deployed operationally.
The patent's statement of government interest is also relevant: it says the invention may be manufactured and used by or for the US government for governmental purposes without payment of royalties. That establishes a government-use provision in the patent record; it does not establish that the government actually manufactured or used the invention.

46. Patent US 12,097,382 B2 — Systems and Methods for Covertly Creating Adverse Health Effects in Subjects

Primary source: United States patent record, granted 24 September 2024
This is a particularly important document for the register because its title and stated purpose are unusually explicit. The patent describes proposed methods involving electromagnetic waves and ultrasonic waves, including pulsed signals intended to interact with neural activity. Its text discusses the possibility of producing adverse health effects without the subject recognising an external influence. The listed inventors are Elwood Norris and Seth Putterman; the record identifies the University of California, San Diego as the current assignee.
The patent's language warrants examination rather than being dismissed or sensationalised. It documents a technical proposal whose stated objective includes covertly producing adverse effects. That makes it relevant to research into proposed remote-stimulation methods, the history of directed-energy concepts and the ethical questions raised by technologies intended to affect people without their awareness.
However, several distinctions are essential.
First, the document is a patent disclosure, not a clinical study or an independently replicated demonstration of every claimed effect. Second, the description includes more than one proposed modality; it should not be treated as evidence that every modality, signal configuration or claimed biological outcome has been demonstrated. Third, the grant does not establish that a working system was deployed against any person.
The public record should therefore preserve the patent's exact title, inventors, claims, dates, ownership history and technical description, while keeping separate any claims about actual use. The patent itself does not identify a particular victim, establish an operational programme targeting civilians, or demonstrate that the reported anomalous health incidents were caused by the described methods.
This is an example of why the register should link directly to original patent documents rather than relying on social-media summaries. The document is significant in its own right; it does not need unsupported claims about deployment added to make it worth examining.

47. DARPA's N3 programme — what the government actually sought to develop

Primary source: US Defense Advanced Research Projects Agency

DARPA’s Next-Generation Nonsurgical Neurotechnology programme, commonly known as N3, aimed to develop high-performance, bidirectional brain–machine interfaces for able-bodied service members. The programme sought to overcome limitations in noninvasive neural interfaces, including the difficulty of obtaining sufficiently precise signals through the skin, skull and brain tissue. Its stated military applications included human interaction with unmanned aerial vehicles, cyber-defence systems and other complex computer systems. DARPA’s page now marks the programme as complete.

The word bidirectional is important. The research goal was not simply to record neural activity. It was to develop systems that could both read neural signals and write information or stimulation back through a neural interface. DARPA described a demanding target involving multiple independent channels and fine spatial and temporal precision. The programme page also acknowledges the physical challenges involved in achieving those targets.

The original 2018 announcement and 2019 funding announcement provide additional context. DARPA announced a four-year effort and subsequently identified six funded research teams. These announcements document a real, officially acknowledged defence research programme—not merely a speculative idea circulating online.

But N3 must not be conflated with the Frey effect or with the covert adverse-health-effects patent. They concern related questions about energy, neural signals and interaction with the human nervous system, but they are not interchangeable technologies.

N3’s publicly stated objective was a neural interface for human–machine interaction. Its existence does not demonstrate a system capable of remotely transmitting intelligible speech into an unsuspecting person’s perception, reading arbitrary private thoughts at a distance, or producing a particular person’s symptoms without detection. Those would be separate capabilities requiring their own evidence.

The programme nevertheless belongs in the register because it documents official investment in advanced nonsurgical neurotechnology and the acknowledged difficulty of achieving high-resolution interaction with neural tissue. The distinction between the programme’s ambitions and demonstrated results must remain visible.

48. Comparative evidence register

The following comparison separates what each document establishes from what it leaves unresolved.

Record	What the document establishes	What it does not establish by itself
US 6,470,214 B1, radiofrequency	An Air Force-assigned patent	Operational deployment,

hearing	disclosure addressing speech modulation through the radiofrequency hearing effect	reliable field performance or use against a named person
US 12,097,382 B2, covert adverse health effects	A granted patent describing proposed electromagnetic and ultrasonic methods for producing adverse effects	Independent demonstration of every claimed effect or evidence that the system was used against a particular person
DARPA N3 programme	A publicly documented defence research effort aimed at high-performance nonsurgical brain–machine interfaces	That all programme targets were achieved or that N3 was a covert remote-targeting system
Frey's 1962	Published	That all reported

research	observations of auditory sensations under particular modulated electromagnetic exposures	internal voices, chronic illnesses or unexplained symptoms share that cause

Sources: , , and .

49. What the combined record means

Taken together, these documents demonstrate that radiofrequency hearing, proposed methods for transmitting speech-like auditory sensations, research into nonsurgical neural interfaces and proposals for covertly inducing adverse effects are all subjects that appear in the public technical record.

That is a meaningful historical and technical finding. It would be inaccurate to suggest that every underlying concept is fictional simply because a particular allegation of its use remains unverified.

The opposite inference would also be unsound. A patent, an acknowledged research programme and a documented physical phenomenon do not collectively prove that a particular individual was targeted, that an undisclosed device exists, or that an unexplained health incident was deliberately caused. The evidence needed to establish those propositions would include, for example, authenticated device records, independently measured exposure data, technically compatible exposure parameters, reliable timing information and a defensible link between exposure and outcome.

For this register, the strongest formulation is therefore that the public record documents genuine research and specific technical proposals, while leaving operational deployment and individual-case attribution as separate questions.

50. Next phase — reconstructing the development timeline

The next stage should place these records in chronological order and examine their relationships without assuming a common programme:

- 1962: Frey's published report of auditory sensations associated with modulated electromagnetic energy.
- 1996–2002: The filing and grant history of the Air Force-assigned radiofrequency-hearing patent.
- 2018–2022: DARPA's announced N3 programme and its publicly described research objectives.
- 2023–2024: The filing and grant of the patents concerning covert adverse health effects.
- 2016 onward: The reported anomalous health incidents, followed by scientific, medical

and government assessments.

The next task is to compare the dates, named inventors, institutional affiliations, patent references and publicly documented experimental results. Any connection between these records should be demonstrated through documentary evidence rather than inferred from overlapping terminology or chronology.

That approach will give the archive a stronger result: not simply a collection of striking documents, but a traceable account of what was proposed, what was researched, what was demonstrated, and what remains unestablished in the public evidence.

Part XII — Reconstructing the technical timeline from original documents

The chronology is useful because it helps distinguish three things that are often merged in public discussion: an experimentally observed physical effect, a later attempt to engineer a practical application, and a separate proposal for influencing biological systems.

The original documents show that these questions developed over decades. The timeline below records the dates that can be established from the sources and identifies what each milestone contributes to the evidence register.

51. 1962 — the scientific phenomenon enters the published record

Allan H. Frey's paper, Human Auditory System Response to Modulated Electromagnetic Energy, was published in the Journal of Applied Physiology in July 1962. Frey reported that people exposed to appropriately modulated electromagnetic energy perceived sounds, and that the effect depended on carrier frequency and modulation. The paper discussed the experimental conditions and power densities associated with the observations.

This is the foundational scientific milestone. It establishes that radiofrequency-induced auditory perception was investigated and reported in a peer-reviewed journal. It does not establish that all experiences described as hearing sounds or voices without an ordinary source arise from this phenomenon.

52. 1996–2003 — the US Air Force patent family

The US patent record provides a more specific development trail than a patent's publication date alone.

- 13 December 1996: The application for US 6,470,214 B1 was filed.
- 22 October 2002: US 6,470,214 B1 was granted and published.
- 1 July 2003: Related patent US 6,587,729 B2 was published.

The listed inventors are James P. O'Loughlin and Diana L. Loree, and the listed assignee is the United States Department of the Air Force. The application describes signal processing intended to make speech intelligible through the radiofrequency hearing effect.

The specification is especially informative because it acknowledges a problem with earlier approaches: the distortions produced by simple pulsed-carrier modulation could make complex speech unintelligible. It then describes an alternative modulation and preprocessing method intended to address that problem. This is evidence of a defined engineering objective and a proposed technical solution—not, by itself, independent evidence of reliable operational performance.

The patent record also shows that the patent later lapsed for non-payment of maintenance

fees. That administrative event is relevant to the patent's legal status; it does not tell us whether any prototype was built or used.

53. 1988–1989 — an earlier patent titled Hearing System

An earlier document adds another milestone to the timeline: , invented by Wayne B. Brunkan. The application was filed on 6 June 1988, and the patent was granted on 31 October 1989. Its claims describe a system using pulsed microwave energy and audio modulation to produce sound perceived in a person's head.

This patent predates the 1996 filing associated with the later US Air Force patent family. It is therefore relevant to the chronology of technical proposals for using the radiofrequency hearing effect to convey audio information.

The distinction between these records matters. The Brunkan patent lists an individual as the original assignee, whereas the later patent family lists the United States Department of the Air Force. Their technical subject matter overlaps, but the documents do not, merely by existing, establish a shared development programme or a chain of operational deployment. The patent also does not establish that the described system was used against an unwilling person. That question requires evidence beyond the specification and grant record.

54. 2018 onward — the DARPA N3 research programme

DARPA's N3 programme is a different branch of the technical history. It sought to develop nonsurgical, bidirectional brain–machine interfaces that could read neural signals and write information or stimulation to neural tissue. The agency identified military applications involving control of unmanned systems and interaction with complex computer systems.

The programme's stated goal was ambitious: precise interaction with multiple small regions of neural tissue, without surgical implantation. DARPA also identified the obstacles, including signal scattering and attenuation through the skin, skull and brain, as well as the challenge of decoding and encoding neural information through noninvasive methods.

The programme page now says the effort is complete. That establishes the programme's administrative status, but the public summary does not, by itself, establish that every stated performance target was achieved. To make that assessment, the next source layer would be the final reports and peer-reviewed publications produced by the funded teams.

N3 should not be treated as a demonstrated remote mind-reading or covert speech-transmission system. Its stated aim was a nonsurgical interface for human–machine interaction. The scientific and engineering challenges it identified are relevant to understanding the state of neural technology, but they do not prove capabilities beyond those documented by the programme.

55. 2023–2026 — the patent on covert adverse health effects

The record for provides a particularly clear example of why patent histories should be recorded precisely.

The timeline is:

- 10 January 2023: Priority date listed in the patent record.
- 25 September 2023: Application filed.
- 11 July 2024: Application published as US 2024/0226592 A1.
- 24 September 2024: Patent granted as US 12,097,382 B2.

- 12 May 2026: The record lists an assignment of Seth Putterman's interest to The Regents of the University of California.

The patent's title and description explicitly concern proposed systems and methods for covertly creating adverse health effects. The disclosure describes electromagnetic and ultrasonic approaches. This is a real technical disclosure in the patent record, and its language is relevant to research into proposals for remotely influencing human subjects.

The 2026 assignment is a further documentary fact: it records an ownership-related transaction. It does not establish that the university manufactured a device, funded a particular deployment, or authorised any use against an individual. Likewise, the patent grant is not an independent scientific finding that all proposed effects are achievable under practical conditions.

For the register, the correct treatment is to preserve the document, quote its relevant claims accurately, record the ownership history and investigate any separately documented experiments or deployments. It should neither be omitted because its stated purpose is troubling nor treated as proof of an unrecorded operation.

56. Chronology and evidentiary weight

The following table brings the main technical records together.

Date	Record	Evidentiary contribution
1962	Frey's published auditory-effect research	Peer-reviewed report of auditory perception under particular electromagnetic exposure conditions
1988–1989	Brunkan's Hearing System patent	A patent disclosure describing a proposed pulsed-microwave audio system

1996–2003	US Air Force patent family	Technical proposals for processing and transmitting speech through the radiofrequency hearing effect
2018 onward	DARPA N3	Officially documented research objectives for nonsurgical, bidirectional neural interfaces
2023–2024	Patent application and grant concerning covert adverse health effects	A patent disclosure describing proposed electromagnetic and ultrasonic methods
2026	Recorded assignment in the latter patent's	A documented ownership-related event, not

	ownership history	evidence of deployment

Sources: , , , , and .

57. What the timeline can—and cannot—tell us

The chronology demonstrates that the microwave auditory effect was not merely a recent internet theory. It entered the scientific literature decades ago, and later inventors documented proposed ways to use it to convey audio information. Separate government research programmes pursued nonsurgical neural interfaces, while another patent described proposed methods for covertly producing adverse effects.

Those are meaningful findings. They establish a history of scientific observation, engineering proposals and officially documented research objectives.

But a chronology is not, by itself, a causal chain. The existence of an earlier patent does not establish that a later system derived from it. Similar terminology does not establish that two organisations worked together. A stated technical objective does not establish successful performance. And the presence of a potentially relevant capability in the public record does not establish that it caused a particular person's reported symptoms.

For the archive, the strongest next step is to follow each record outward: identify its cited scientific papers, related patents, named researchers, publicly documented experiments and any independently authenticated demonstrations. Each claimed connection should have its own documentary support.

58. Next phase — examine what was actually demonstrated

The next phase should focus on the experimental evidence behind the patent claims and programme objectives, particularly whether the public literature demonstrates intelligible speech transmission under controlled conditions, what exposure parameters were used, what limitations the investigators reported, and whether any independently replicated results support broader claims.

That investigation should also distinguish laboratory demonstrations from practical systems operating at a distance. A successful experiment under controlled conditions may establish a physical or engineering capability without establishing that the same capability is practical, covert, reliable or operationally deployed in an unrelated setting.

The aim is to move beyond the existence of documents to the quality of the demonstrations they describe. That is the point at which the register can begin assessing not only what was proposed, but how much of the proposed capability the public scientific record actually supports.

Part XIII — Experimental demonstrations: what has actually been made to happen?

The previous phase established that patents and research programmes document proposed capabilities. This phase examines a harder question: what do the experiments and published technical analyses demonstrate, and what remains a prediction rather than an observed result?

The distinction is crucial to the archive. A patent can describe a method for transmitting speech through the radiofrequency hearing effect, but that is not the same as a controlled experiment demonstrating intelligible speech transmission at a specified distance. A mathematical model can predict an acoustic response in tissue, but a prediction is not a measurement of injury in a living person.

59. The microwave auditory effect is not merely a patent claim

The scientific basis begins with experimental reports, not patent language. Frey's 1962 paper reported auditory perceptions under particular modulated electromagnetic exposures. Later research examined the mechanism and the conditions under which pulsed radiofrequency energy can generate acoustic transients inside the head.

The 2021 article by Kenneth Foster, David Garrett and Marvin Ziskin provides a detailed review of this mechanism and the relevant engineering constraints. It explains that short radiofrequency pulses can produce small, rapid temperature changes in absorbing tissue, generating pressure waves that may stimulate the auditory system. The authors distinguish this established physical phenomenon from the separate question of whether it could practically be used to harm or harass a person.

That distinction gives the evidence register a firm starting point: the underlying auditory effect has scientific support; a claim that a particular individual was exposed to it still requires independent evidence of that exposure.

60. The speech-transmission patent contains an important qualification

The full text of is more revealing than the title alone.

Its background acknowledges that earlier pulsed-radiofrequency methods could produce sensations resembling simple tones, but that complex speech was severely distorted. It says listeners who were told beforehand that a message had been transmitted might recognise it as speech, while listeners who did not know the content could find it unintelligible. The patent then proposes signal preprocessing and carrier-suppressed modulation to address that limitation.

This gives us two separate findings:

- The patent documents a recognised technical problem: turning the auditory effect into intelligible speech rather than a click, tone or distorted sound.
- The patent claims a method intended to overcome that problem, but the grant alone does not establish how reliably it performed, at what distances, or under what independently replicated experimental conditions.

That is precisely the kind of distinction that makes the register more useful than a list of patent titles. The limitation is recorded alongside the proposed advance, rather than being lost in either an exaggerated claim of capability or a blanket dismissal.

61. Modelling is evidence, but it is not a field demonstration

Foster, Garrett and Ziskin's 2021 article models the pressure waves that could be produced by pulsed radiofrequency energy absorbed in tissue. The authors discuss reported thresholds for auditory perception, estimate the acoustic effects under selected exposure assumptions, and consider whether adverse audiovestibular effects might be possible. They conclude that the practical use of the microwave auditory effect as a weapon appears

unlikely for reasons including the equipment and exposure requirements, while acknowledging uncertainty about classified high-power systems and poorly established thresholds for adverse effects.

A later 2023 commentary by Foster examines a related theoretical possibility involving pulsed infrared laser energy and thermoelastic sound generation. It explicitly describes the proposed non-lethal device as hypothetical and says further analysis is required. This is a different modality from radiofrequency transmission; it must not be cited as proof that a microwave device was used in any reported incident.

The correct evidentiary labels are therefore:

- Experimentally reported: auditory perceptions associated with particular electromagnetic exposures.
- Patent-disclosed: a proposed method for encoding speech for the radiofrequency hearing effect.
- Modelled: predicted acoustic pressures under specified assumptions.
- Hypothetical: a proposed system whose practical operation and use against a person have not been demonstrated by that paper.
- Not established by these sources: a causal connection between those technical proposals and a particular anomalous health incident.

62. The question of weaponisation remains separate

The 2021 review also refers to a JASON assessment commissioned by the US Department of State, An Analysis of Data and Hypotheses Related to the Embassy Incidents (JSR-21-01, 2021). Foster's later commentary summarises that assessment as finding no evidence of microwave use in the embassy incidents.

That conclusion is relevant, but it should be attributed accurately: it is an assessment of the available evidence about those incidents, not a claim that the microwave auditory effect does not exist. Equally, the existence of the physical effect is not evidence that it caused those incidents.

For the archive, the original JASON report is the next document to obtain and examine directly, including its scope, evidence reviewed, assumptions and conclusions. The commentary is a useful pointer to the report, but it is not a substitute for the full assessment.

63. Experimental-evidence register

Evidence item	Status	Main limitation
Frey's 1962 research	Published experimental report	Does not establish the cause of unrelated reported experiences

US 6,470,214 B1	Granted patent with a specific speech-encoding proposal	Patent status does not establish reliable, independently replicated field performance
Foster et al. (2021)	Scientific review and physical modelling	Predictions depend on assumptions and do not establish actual exposure in a reported incident
Foster (2023)	Commentary proposing a separate infrared mechanism	The proposed weapon is hypothetical; it is not evidence of an operational device
JASON assessment, 2021	Government-commissioned technical assessment, cited by later authors	The original report needs direct examination before detailed claims about its

methods can be made

64. Next phase — the original JASON assessment and the government intelligence record

The next phase should compare the technical literature with the official investigations of anomalous health incidents. In particular, it should examine the original JASON assessment alongside the 2020 National Academies report and the ODNI assessments published in 2023 and 2025.

The aim is to determine what evidence each assessment considered, whether it found evidence of a specific exposure mechanism, how it treated clinical findings and alternative explanations, and how its conclusions changed—or remained consistent—over time.

That comparison should preserve an important distinction throughout: a technology can be physically real, a proposed application can be technically documented, and a person's symptoms can be serious, without the available evidence establishing that the technology caused those symptoms. The task is to examine each link in that chain, not to assume it or dismiss it.

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The 2021 article by Kenneth Foster, David Garrett and Marvin Ziskin provides a detailed review of this mechanism and the relevant engineering constraints. It explains that short radiofrequency pulses can produce small, rapid temperature changes in absorbing tissue, generating pressure waves that may stimulate the auditory system. The authors distinguish this established physical phenomenon from the separate question of whether it could practically be used to harm or harass a person.

That distinction gives the evidence register a firm starting point: the underlying auditory effect has scientific support; a claim that a particular individual was exposed to it still

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That is precisely the kind of distinction that makes the register more useful than a list of patent titles. The limitation is recorded alongside the proposed advance, rather than being lost in either an exaggerated claim of capability or a blanket dismissal.

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A later 2023 commentary by Foster examines a related theoretical possibility involving pulsed infrared laser energy and thermoelastic sound generation. It explicitly describes the proposed non-lethal device as hypothetical and says further analysis is required. This is a different modality from radiofrequency transmission; it must not be cited as proof that a microwave device was used in any reported incident.

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The 2021 review also refers to a JASON assessment commissioned by the US Department of State, An Analysis of Data and Hypotheses Related to the Embassy Incidents (JSR-21-01, 2021). Foster's later commentary summarises that assessment as finding no evidence of microwave use in the embassy incidents.

That conclusion is relevant, but it should be attributed accurately: it is an assessment of the available evidence about those incidents, not a claim that the microwave auditory effect does not exist. Equally, the existence of the physical effect is not evidence that it caused those incidents.

For the archive, the original JASON report is the next document to obtain and examine directly, including its scope, evidence reviewed, assumptions and conclusions. The commentary is a useful pointer to the report, but it is not a substitute for the full assessment.

63. Experimental-evidence register

Evidence item	Status	Main limitation
Frey's 1962 research	Published experimental	Does not establish the

	report	cause of unrelated reported experiences
US 6,470,214 B1	Granted patent with a specific speech-encoding proposal	Patent status does not establish reliable, independently replicated field performance
Foster et al. (2021)	Scientific review and physical modelling	Predictions depend on assumptions and do not establish actual exposure in a reported incident
Foster (2023)	Commentary proposing a separate infrared mechanism	The proposed weapon is hypothetical; it is not evidence of an operational device
JASON assessment,	Government-commissioned	The original report needs

2021	technical assessment, cited by later authors	direct examination before detailed claims about its methods can be made

64. Next phase — the original JASON assessment and the government intelligence record

The next phase should compare the technical literature with the official investigations of anomalous health incidents. In particular, it should examine the original JASON assessment alongside the 2020 National Academies report and the ODNI assessments published in 2023 and 2025.

The aim is to determine what evidence each assessment considered, whether it found evidence of a specific exposure mechanism, how it treated clinical findings and alternative explanations, and how its conclusions changed—or remained consistent—over time.

That comparison should preserve an important distinction throughout: a technology can be physically real, a proposed application can be technically documented, and a person's symptoms can be serious, without the available evidence establishing that the technology caused those symptoms. The task is to examine each link in that chain, not to assume it or dismiss it.

Part XVI — Physical evidence: what would establish an external exposure?

The next phase moves from clinical findings to the physical evidence needed to determine whether an external signal was present during a reported incident. This is a critical step because symptoms, patent descriptions and scientific mechanisms cannot substitute for measurements of an actual event.

The question is not simply whether radiofrequency energy exists or whether a device could theoretically generate it. The question is whether there is reliable evidence that a particular signal occurred, whether it could have reached the person under the reported conditions, and whether its characteristics fit the proposed explanation.

A strong evidence register should preserve the distinction between a reported exposure, a measured signal, an identified source and a demonstrated causal effect. Each is a separate proposition requiring its own evidence.

79. What would count as a reliable measurement?

A claim of external radiofrequency exposure becomes substantially more testable when it is supported by contemporaneous instrument data rather than symptoms alone.

The National Academies report describes the distinctive acute experiences reported by some

personnel: sudden sounds or sensations, sometimes perceived as directional or restricted to particular locations. It also notes that the committee did not have access to the complete underlying clinical records.

For a physical exposure claim, a rigorous investigation would ideally preserve the following evidence:

- Instrument identity and calibration. The make, model, measurement range, calibration history and known limitations of the equipment.
- Original measurements. Unaltered data files, timestamps, metadata, instrument settings and records of any gaps in collection.
- Environmental context. Measurements of relevant radiofrequency and acoustic conditions, with enough context to distinguish background activity from an unusual event.
- Independent replication. Where feasible, a separate investigator's measurements using appropriate equipment and a documented protocol.
- Incident correlation. A reliable timeline connecting any recorded signal with the reported event, while distinguishing contemporaneous observations from memories recorded later.
- Alternative explanations. Consideration of ordinary radiofrequency sources, acoustic effects, equipment artefacts and other environmental or medical factors.

No single item is sufficient in isolation. A recording might be authentic but not identify its source. A measured signal might be unusual but unrelated to the reported symptoms. A symptom timeline might be well documented without establishing the physical mechanism that produced it.

The aim is to build a chain of evidence in which each conclusion is supported by the kind of measurement or record capable of establishing it.

80. Why the absence of a recording is not the same as a negative measurement

There is an important distinction between an instrument that was operating during an incident and an instrument that was never present, was not recording the relevant parameter, or did not cover the relevant time period.

If appropriate equipment operated continuously during a reported incident and recorded no signal above its documented detection threshold, that negative result would be evidence against signals within the instrument's measurable range under those conditions. It would not necessarily exclude every possible signal or mechanism.

If no suitable instrument was operating, the absence of a recording tells us much less about whether an exposure occurred.

This distinction is relevant to the Canadian monitoring programme discussed earlier. The Canadian government report says that monitoring equipment was installed in staff living quarters, but that it did not yield relevant, probative information identifying a cause. The public report does not provide all the raw data, calibration records and detection thresholds needed to independently assess every possible signal.

The defensible conclusion is that the published investigation did not identify a cause through its monitoring. The available public summary is insufficient to reconstruct the full technical performance of the equipment.

81. A physical signal does not automatically establish biological causation

James C. Lin's review of microwave auditory effects describes the proposed thermoelastic mechanism and discusses the physical parameters relevant to auditory perception. The 2021 review by Foster, Garrett and Ziskin similarly distinguishes the underlying phenomenon from the question of whether it could be practically weaponised.

This literature establishes why exposure measurements need to be interpreted in context. Radiofrequency energy is not a single uniform exposure: the frequency, pulse structure, energy absorbed by tissue and circumstances of exposure can all matter.

Even if an unusual signal were detected, the investigation would still need to establish whether its characteristics were compatible with the proposed mechanism. If lasting injury were alleged, evidence of the exposure would need to be considered alongside the timing, clinical findings and plausible competing causes.

The same logic applies to any other proposed mechanism. A measurement that is compatible with a hypothesis is not automatically proof of it. A strong causal conclusion requires evidence that distinguishes the proposed explanation from reasonable alternatives.

82. Physical-evidence status register

Evidentiary question	Current status in the public record reviewed here
Is the microwave auditory effect documented in scientific literature?	Yes.
Does the effect provide a possible physical mechanism for certain auditory sensations?	Yes, under particular exposure conditions.
Has the Canadian public report identified a signal that explains the reported incidents?	No.
Does the absence of a	No. The strength of a

publicly identified signal prove that no external exposure occurred?	negative result depends on what was measured, when and with what equipment.
Do patents and theoretical models establish exposure in a particular incident?	No.
Has the public record established a specific external source as the cause of all reported anomalous health incidents?	No.

Sources: , , and .

83. Next phase — applying the same standard to individual case files

The next phase should bring the physical, clinical and documentary evidence together in individual case files. Rather than asking whether a person's experience resembles a known phenomenon, each file should separately record:

- What the person reported and when the account was first recorded.
- What contemporaneous documents support the timeline.
- Which symptoms or clinical findings were independently documented.
- Whether any external signal was measured during the relevant period.
- Whether the proposed mechanism fits the available physical and clinical evidence.
- What alternative explanations were considered and what evidence supports or weakens each one.
- Whether the identity of a responsible actor is supported by independent evidence.

This method gives testimony its proper place in the record without requiring a witness to provide technical proof before their account can be preserved. It also prevents a patent, a plausible mechanism or a general scientific finding from being presented as proof of an individual event.

The objective is a documented, auditable register that makes clear what is known, what is reported, what has been measured, and what remains unresolved.

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